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Carrillo Rincón et al.

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(54) **INDUCIBLE PROMOTERS**

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Worcester, MA (US)

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(*) Notice: Subject to any disclaimer, the term of this
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19, 2022.

(51) **Int. Cl.**

C12N 15/63 (2006.01)

C12N 15/11 (2006.01)

(52) **U.S. Cl.**

CPC **C12N 15/635** (2013.01); **C12N 15/111**
(2013.01); **C12N 2830/36** (2013.01); **C12N**
2840/002 (2013.01)

(58) **Field of Classification Search**

CPC C12N 15/635; C12P 23/00

USPC 435/320.1, 91.1

See application file for complete search history.

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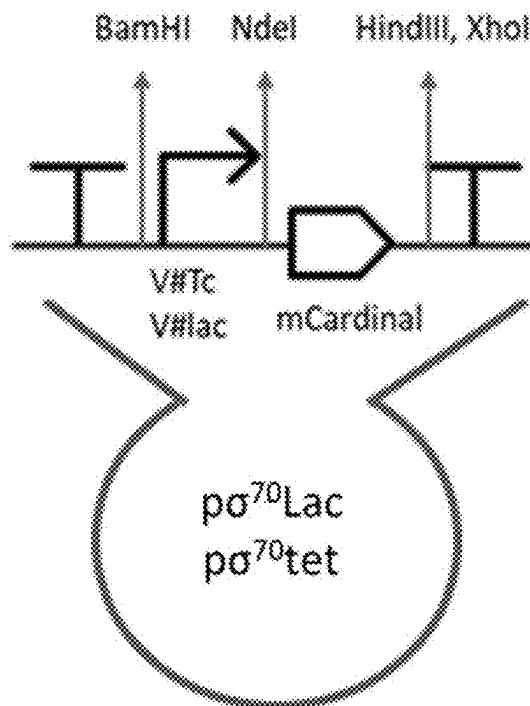
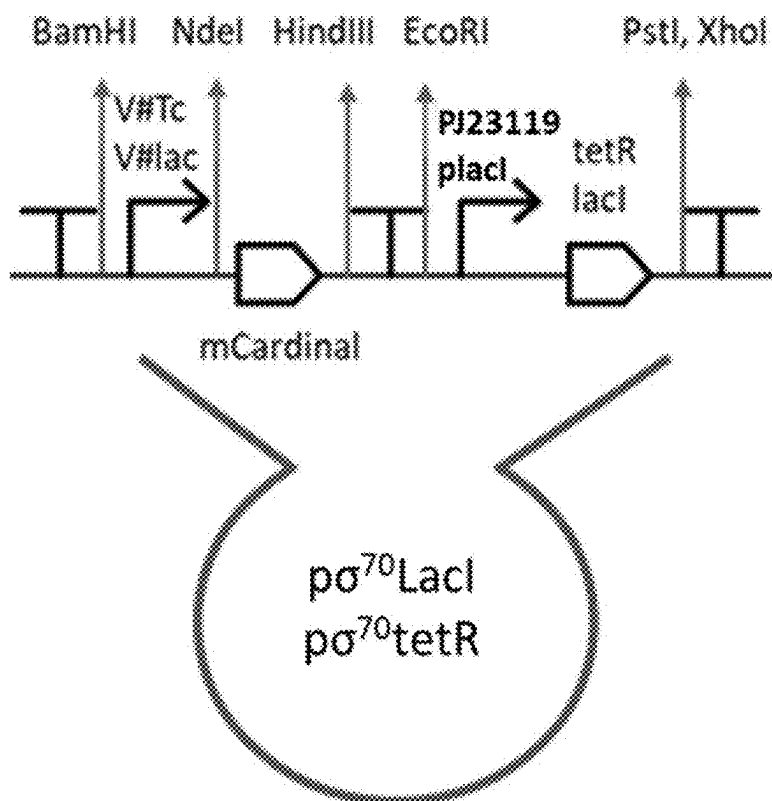
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(57) **ABSTRACT**

Provided herein are nucleic acid constructs that comprise an
inducible promoter. Dual expression systems are provided
comprising two nucleic acid constructs or a single nucleic
acid construct with two inducible promoters. Also provided
are methods of expressing transcripts by transforming a
nucleic acid construct described herein into a prokaryotic
cell and contacting the prokaryotic cell with an inducer.
Methods of producing a carotenoid are also disclosed herein.

16 Claims, 34 Drawing Sheets

Specification includes a Sequence Listing.

**FIG. 1A****FIG. 1B**

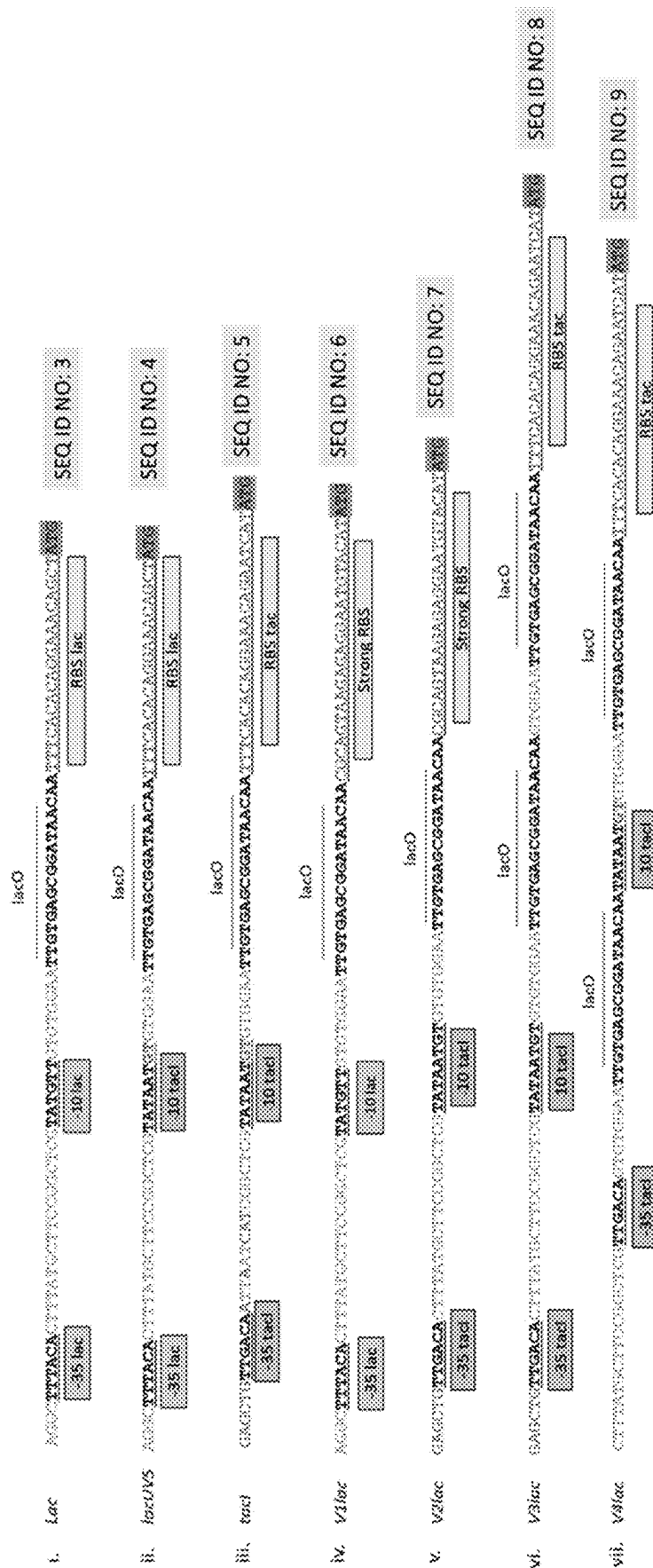


FIG. 2A

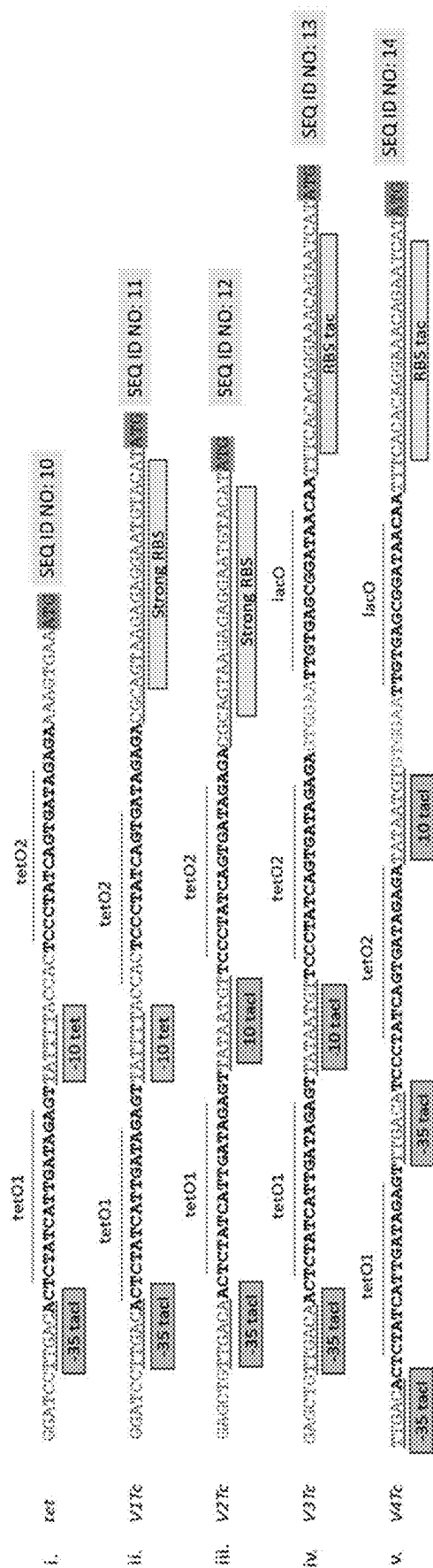


FIG. 2B

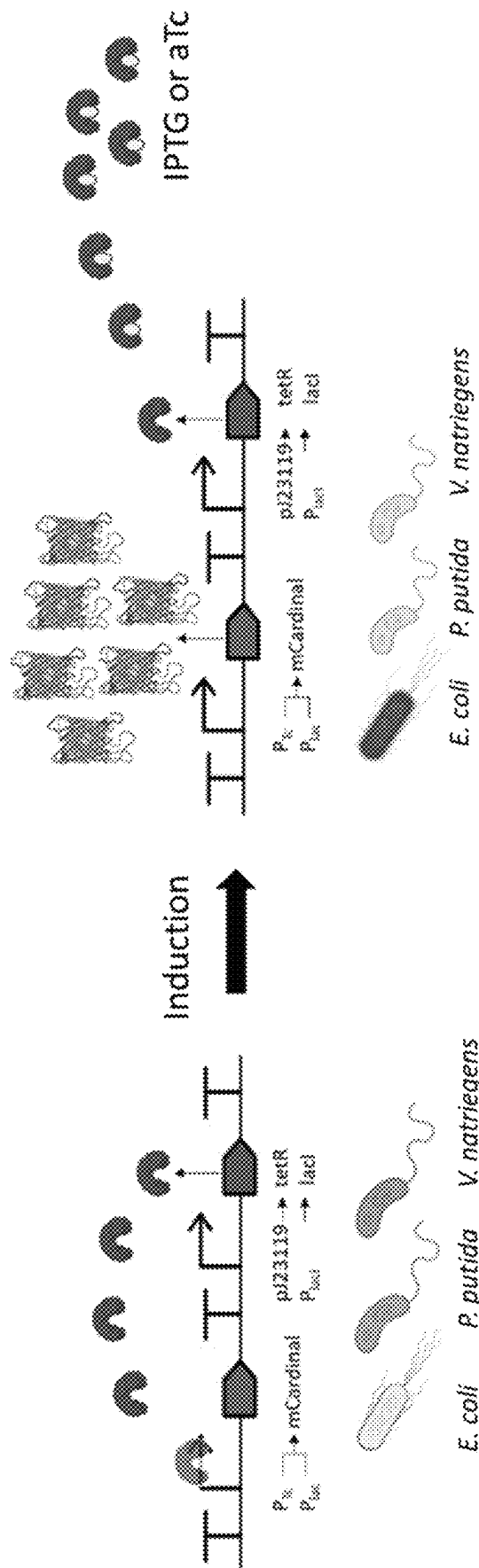
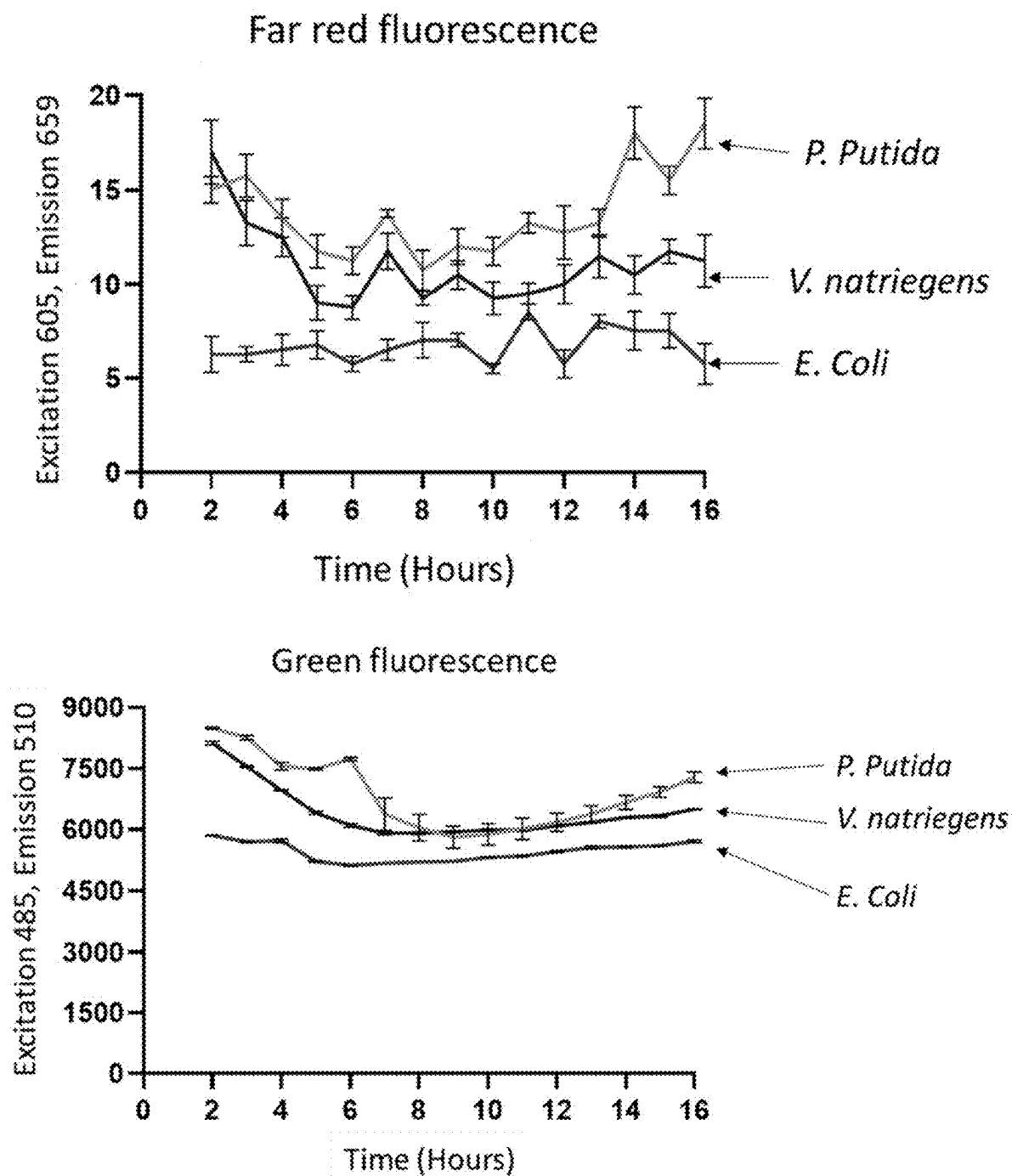
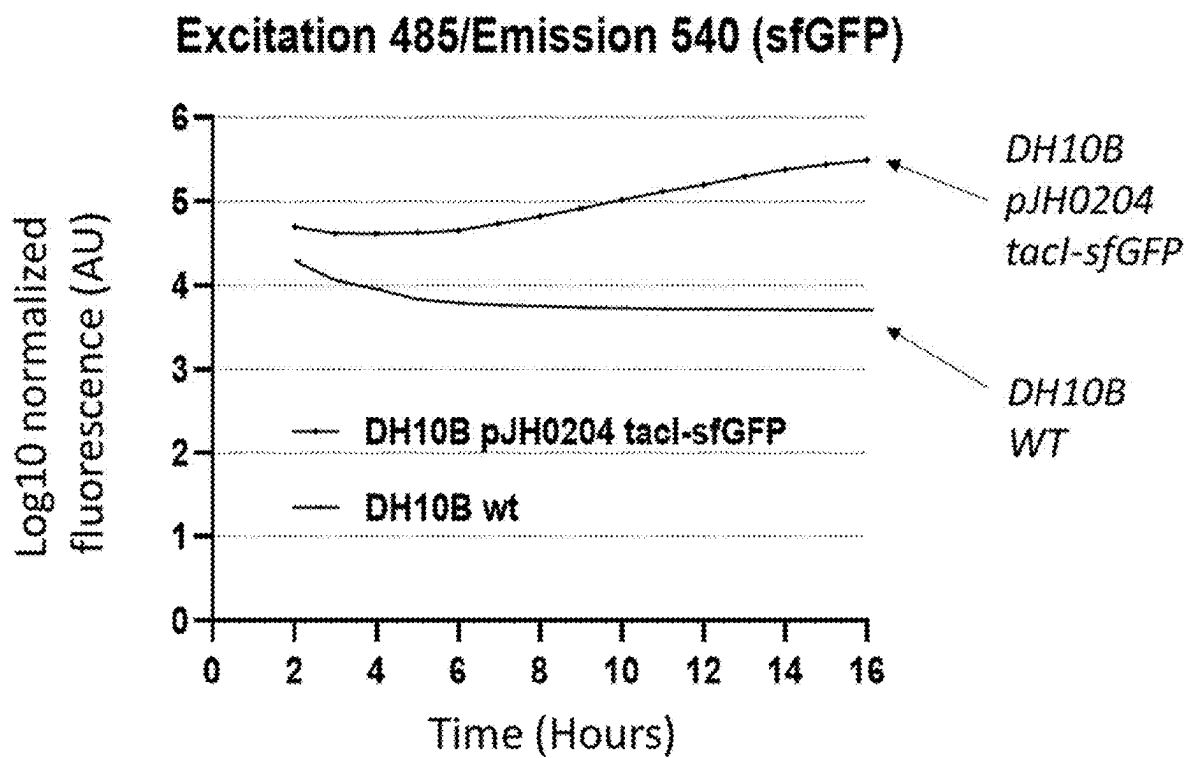
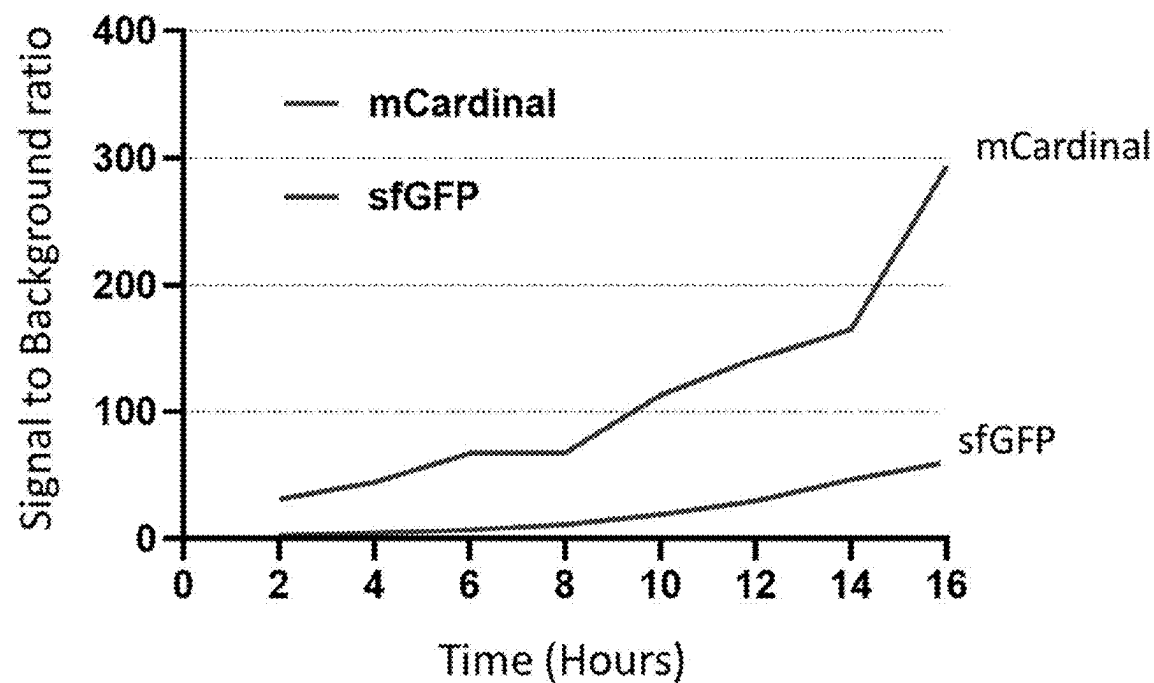


FIG. 2C

**FIG. 3A**

**FIG. 3B**

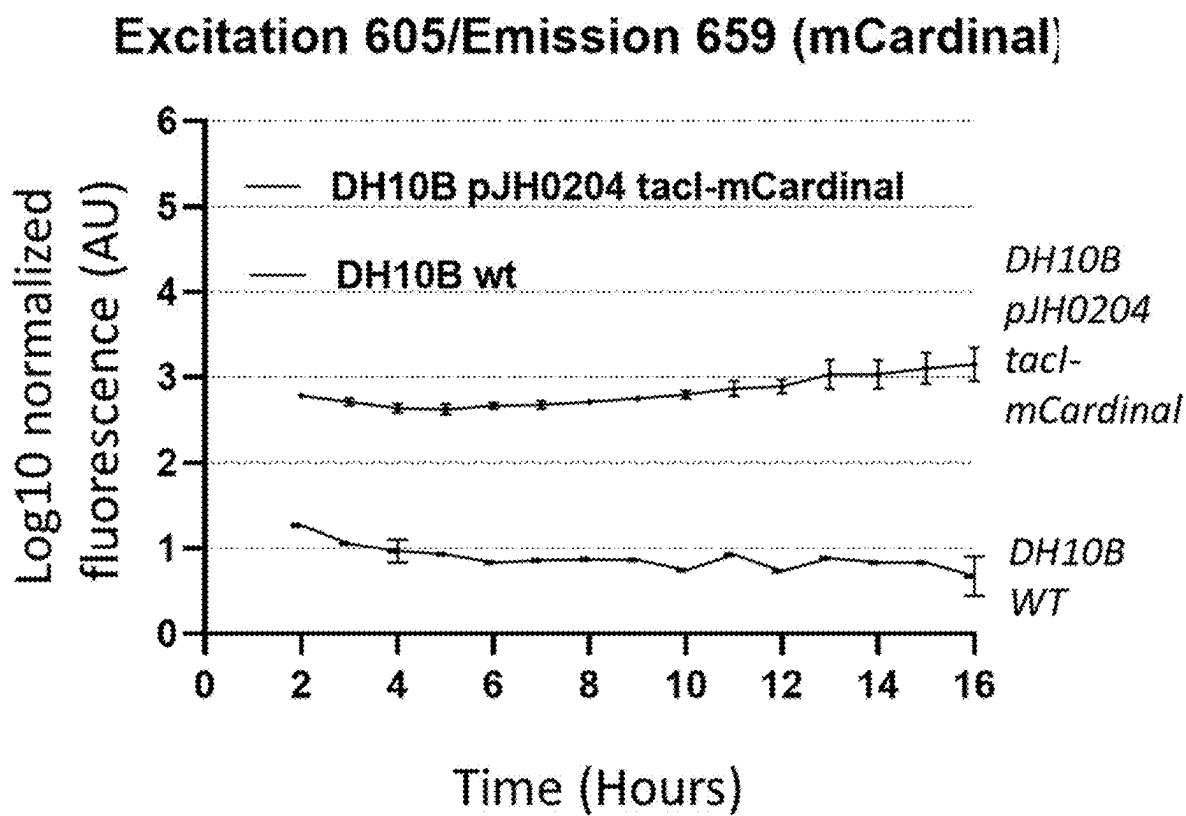


FIG. 3B (Continued)

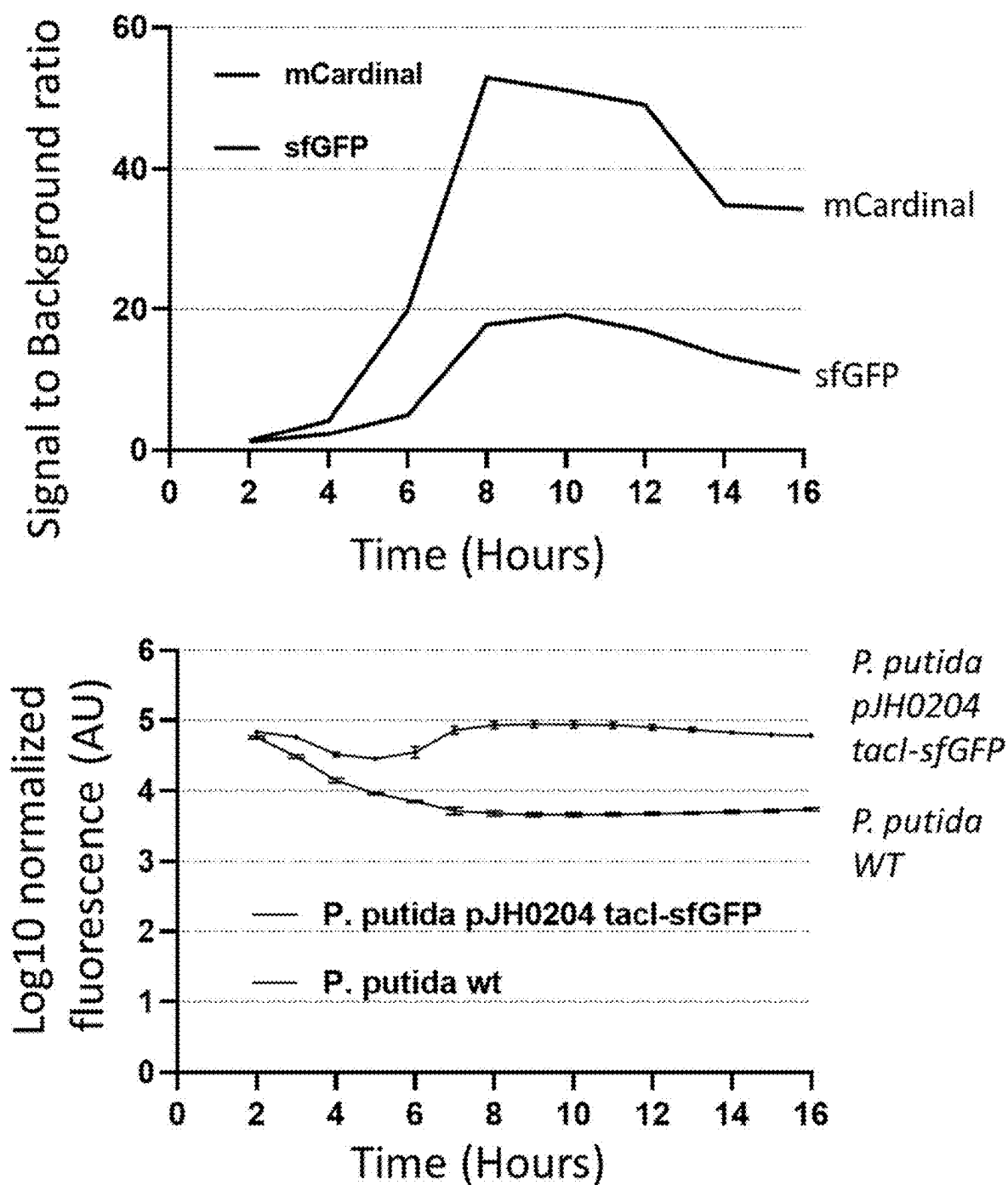
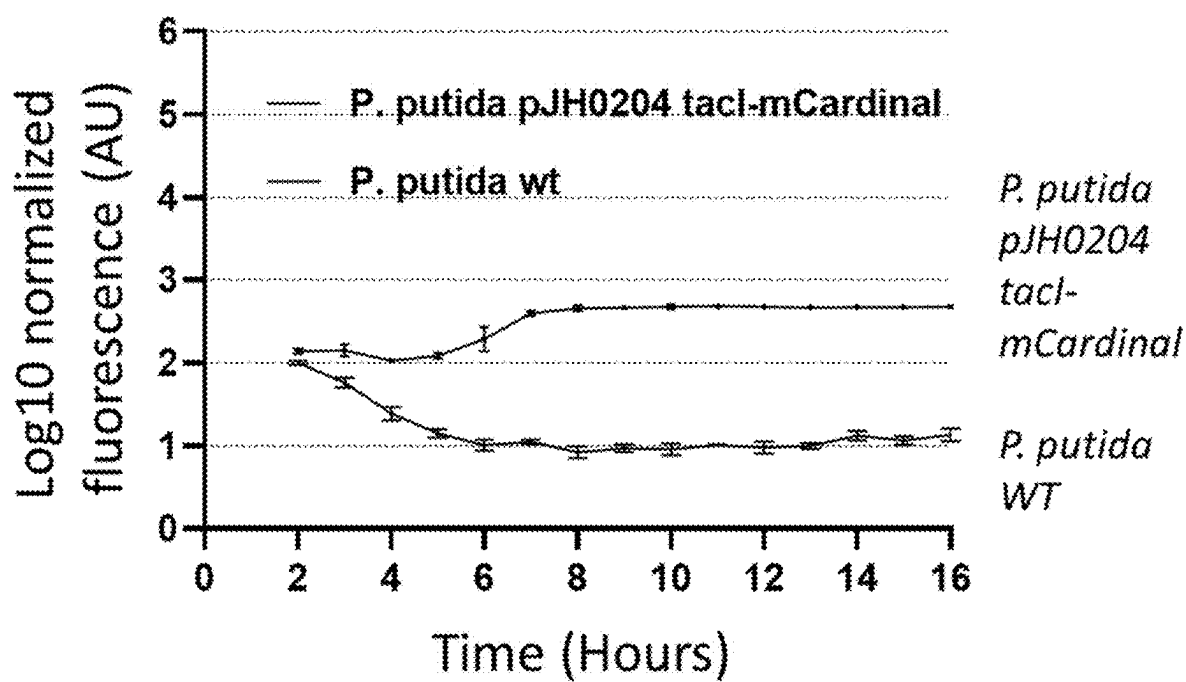


FIG. 3C

*FIG. 3C (Continued)*

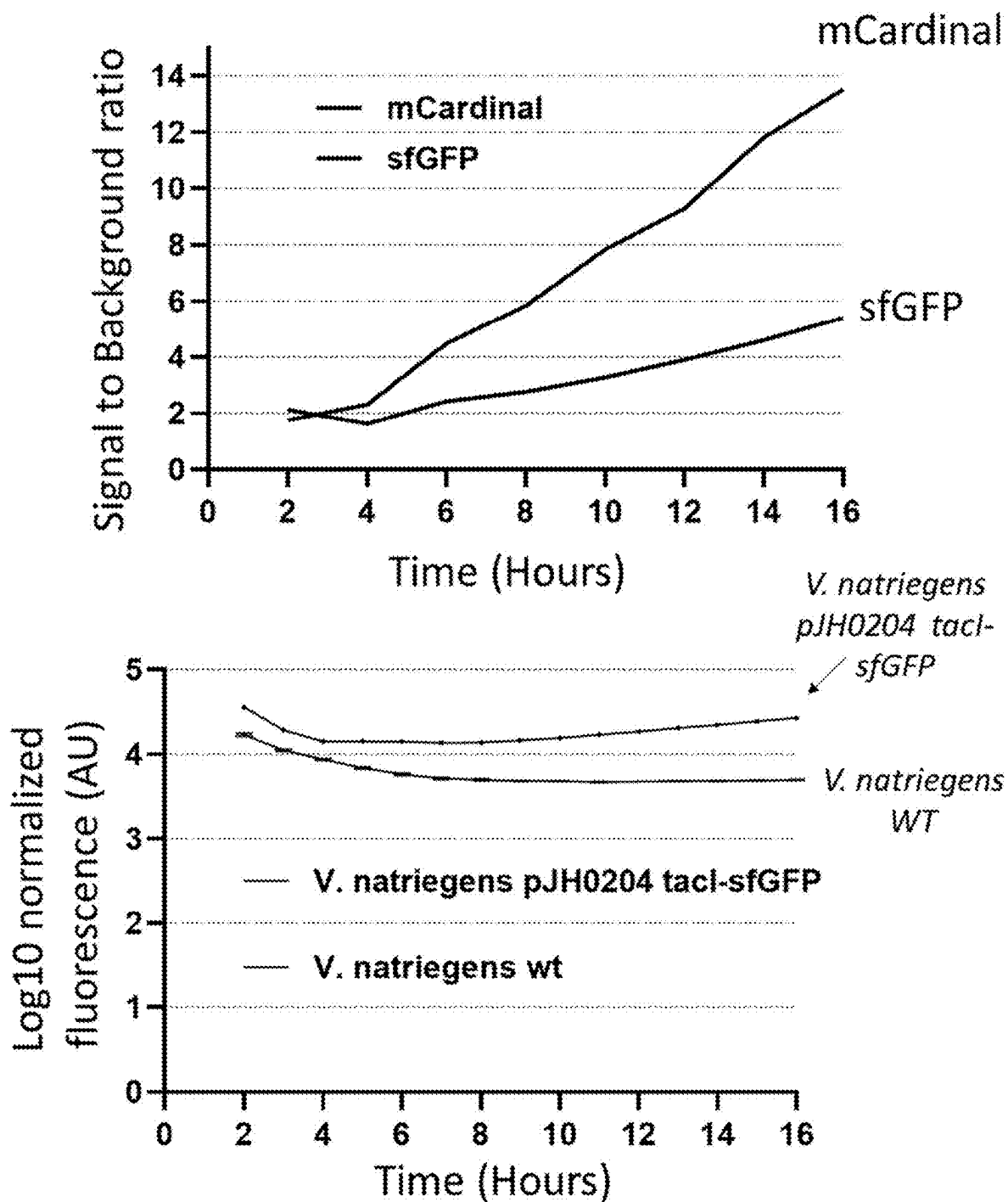
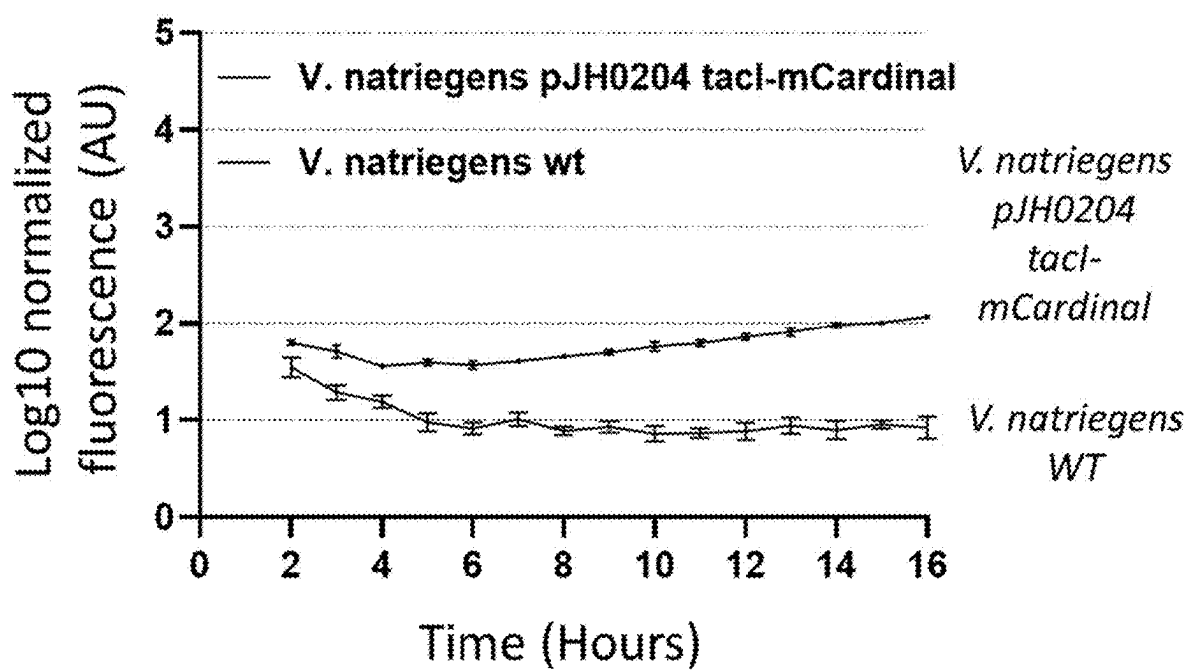
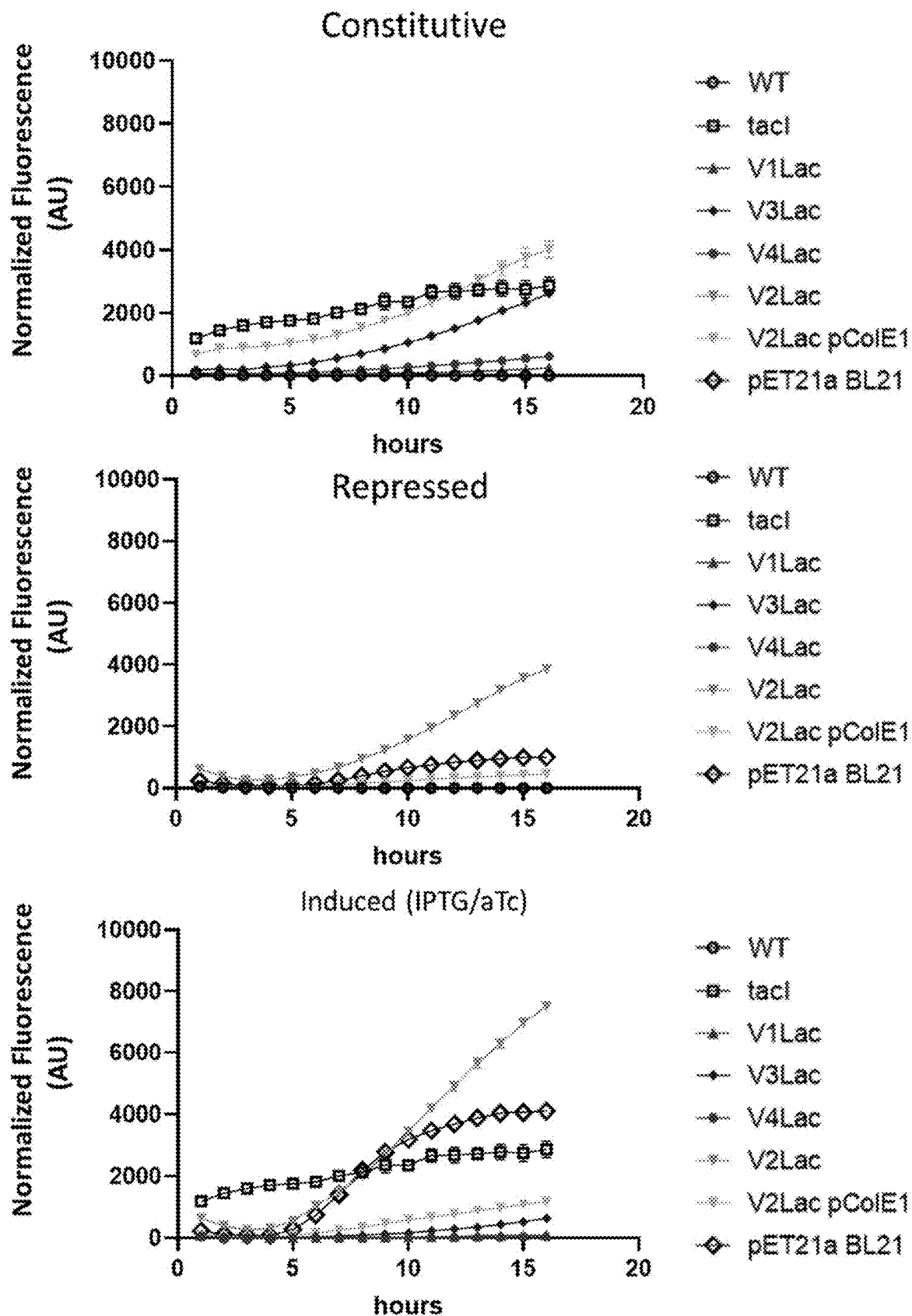
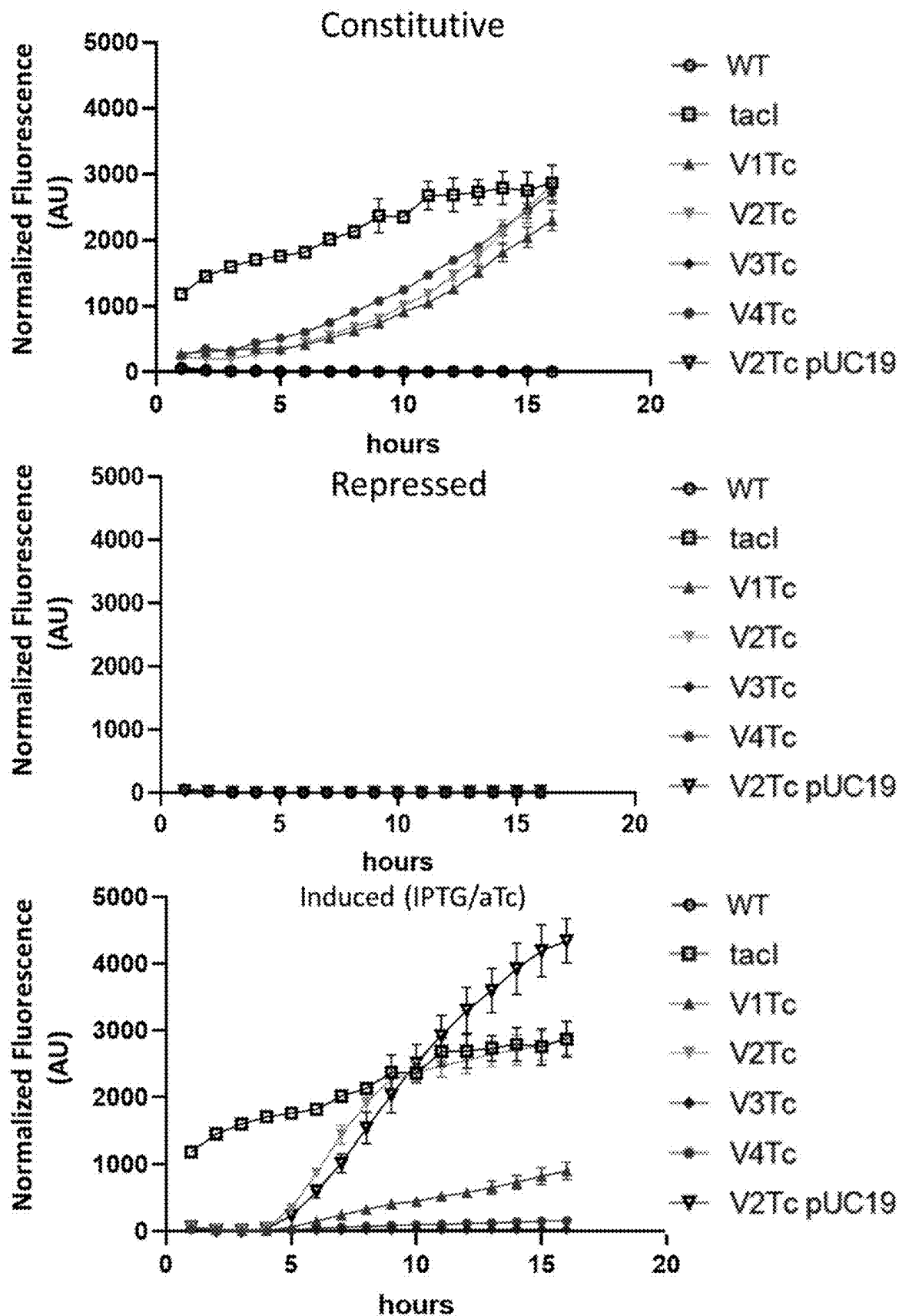


FIG. 3D

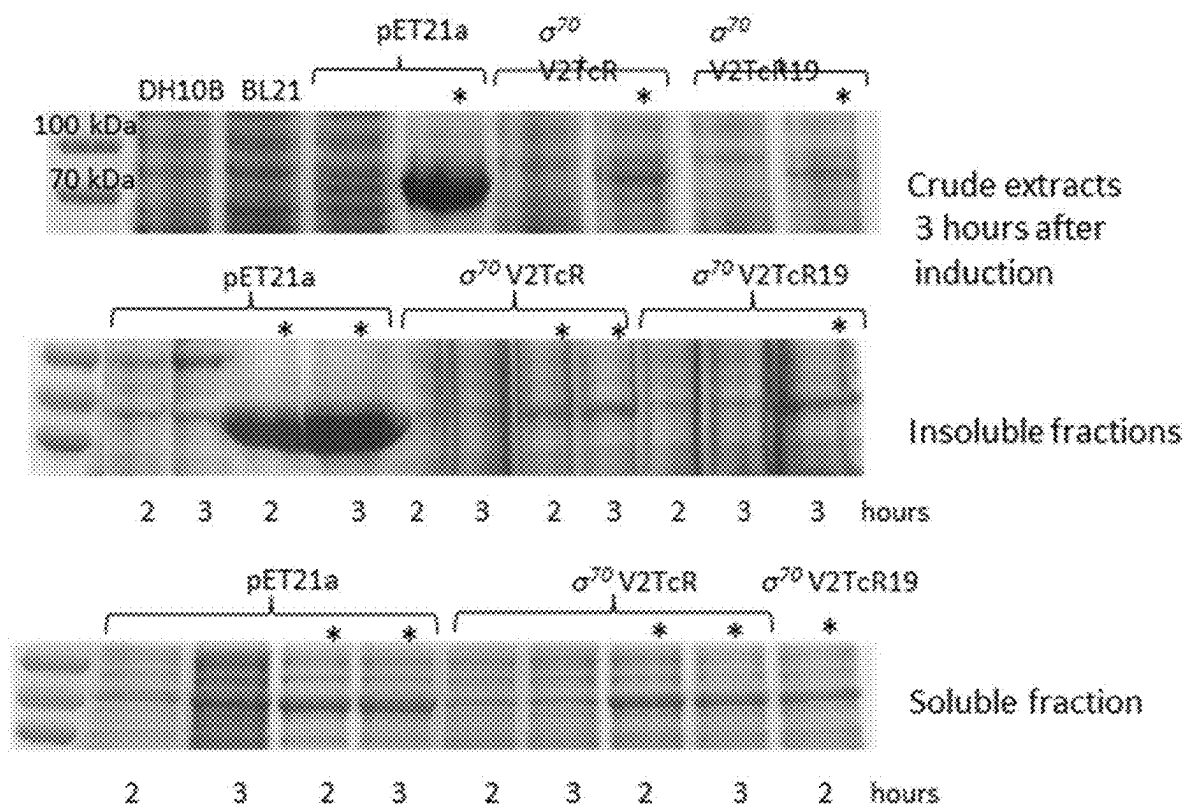
*FIG. 3D (Continued)*

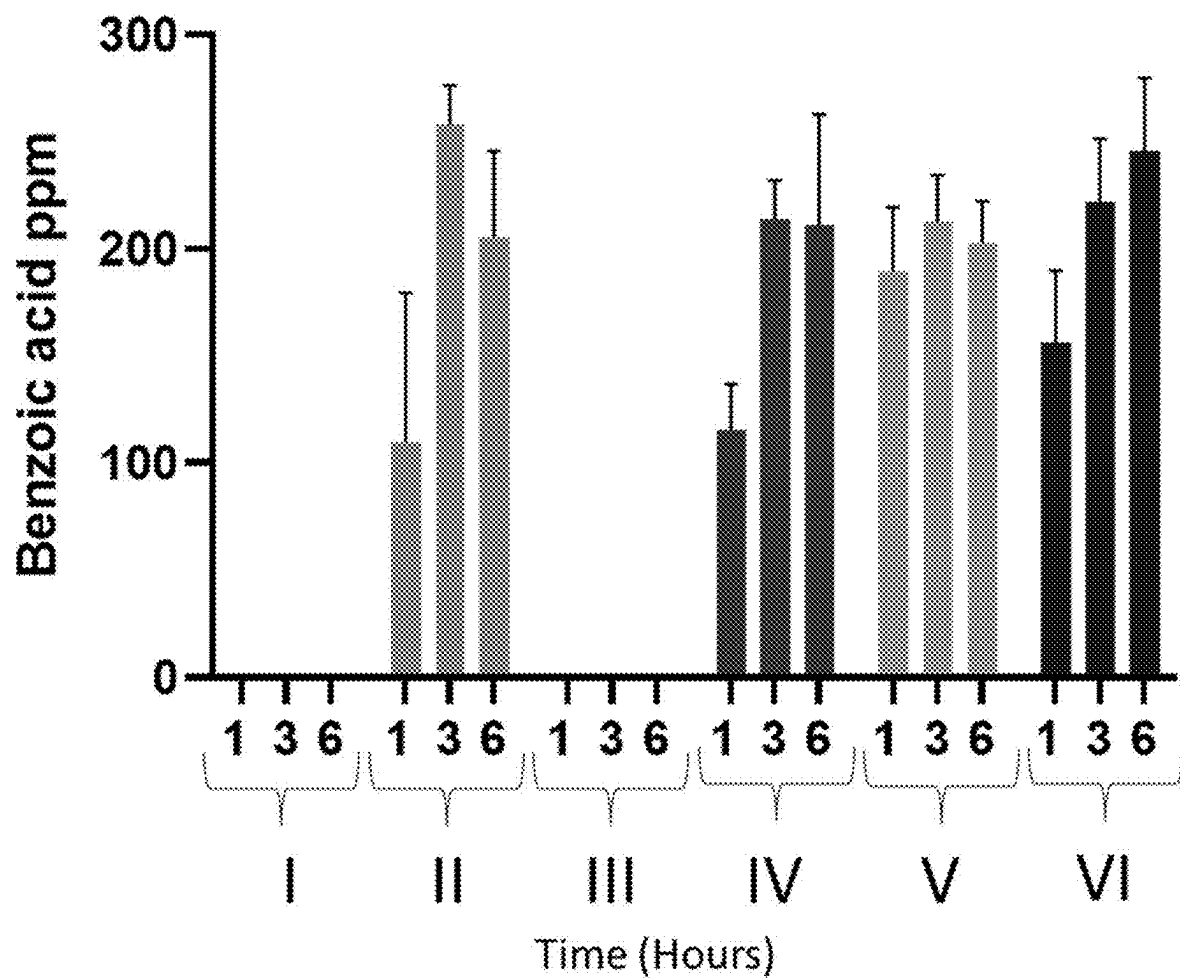
**FIG. 4A**

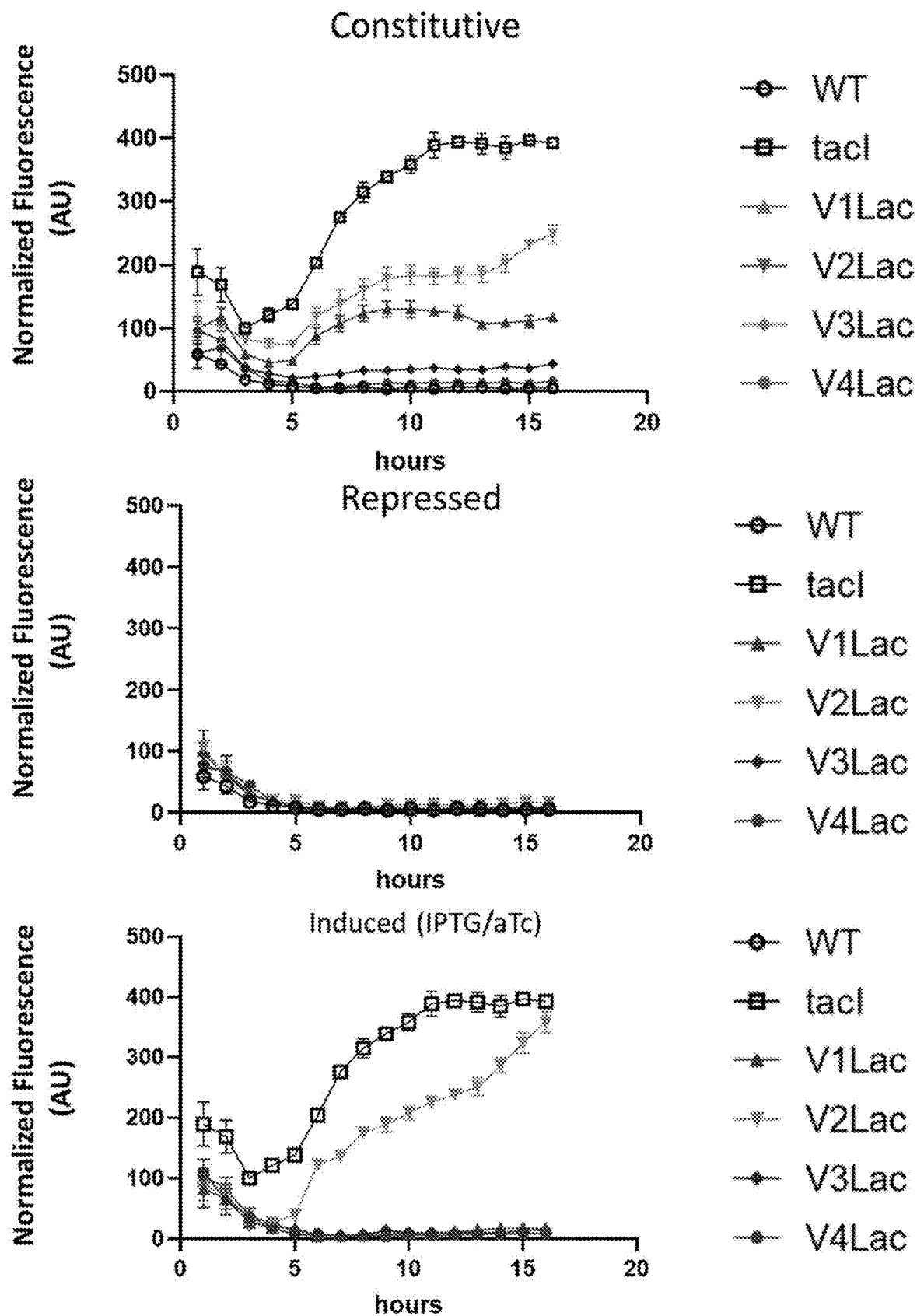
**FIG. 4B**

[illegible]

FIG. 4C

**FIG. 5A**

**FIG. 5B**



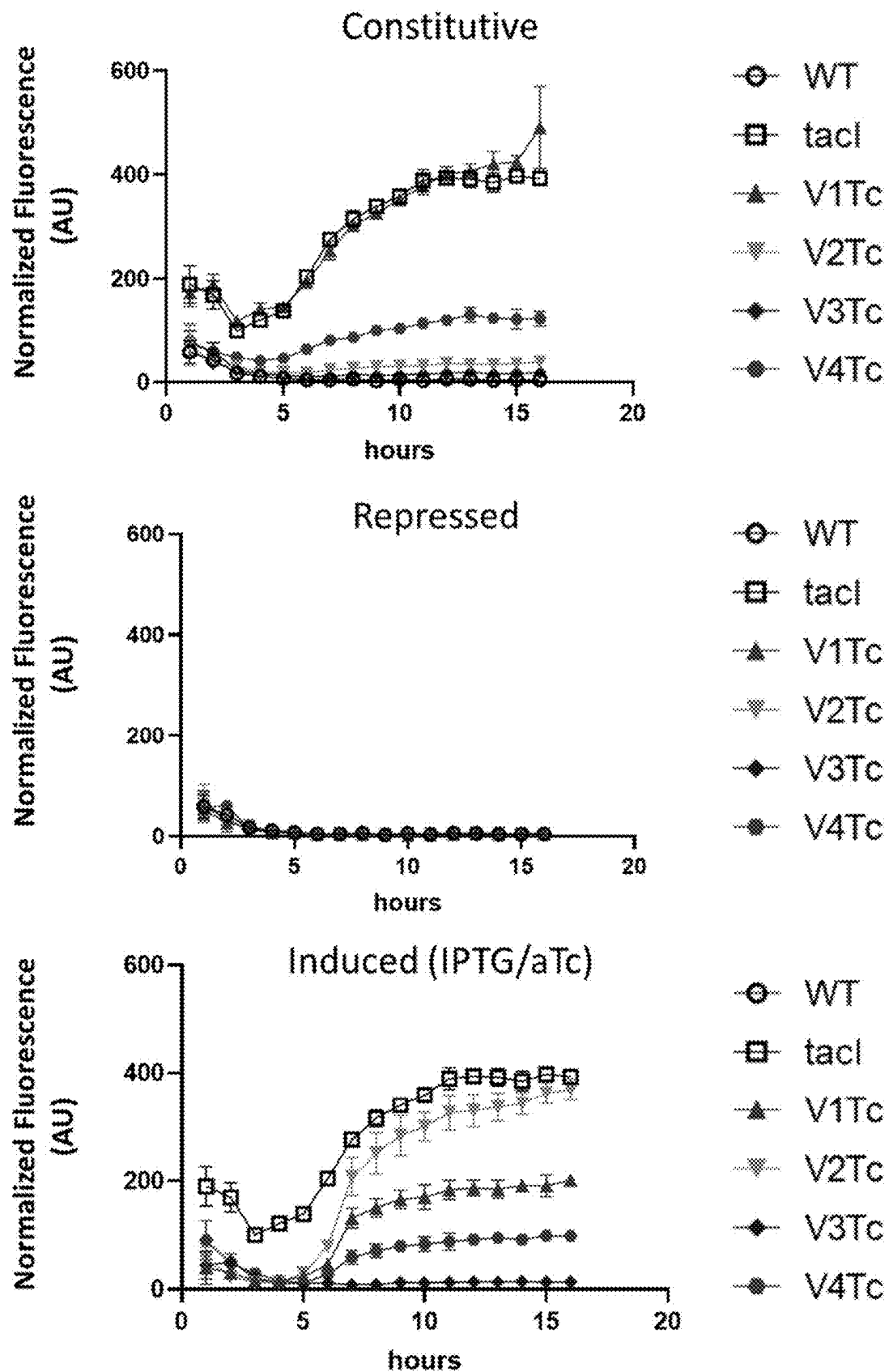
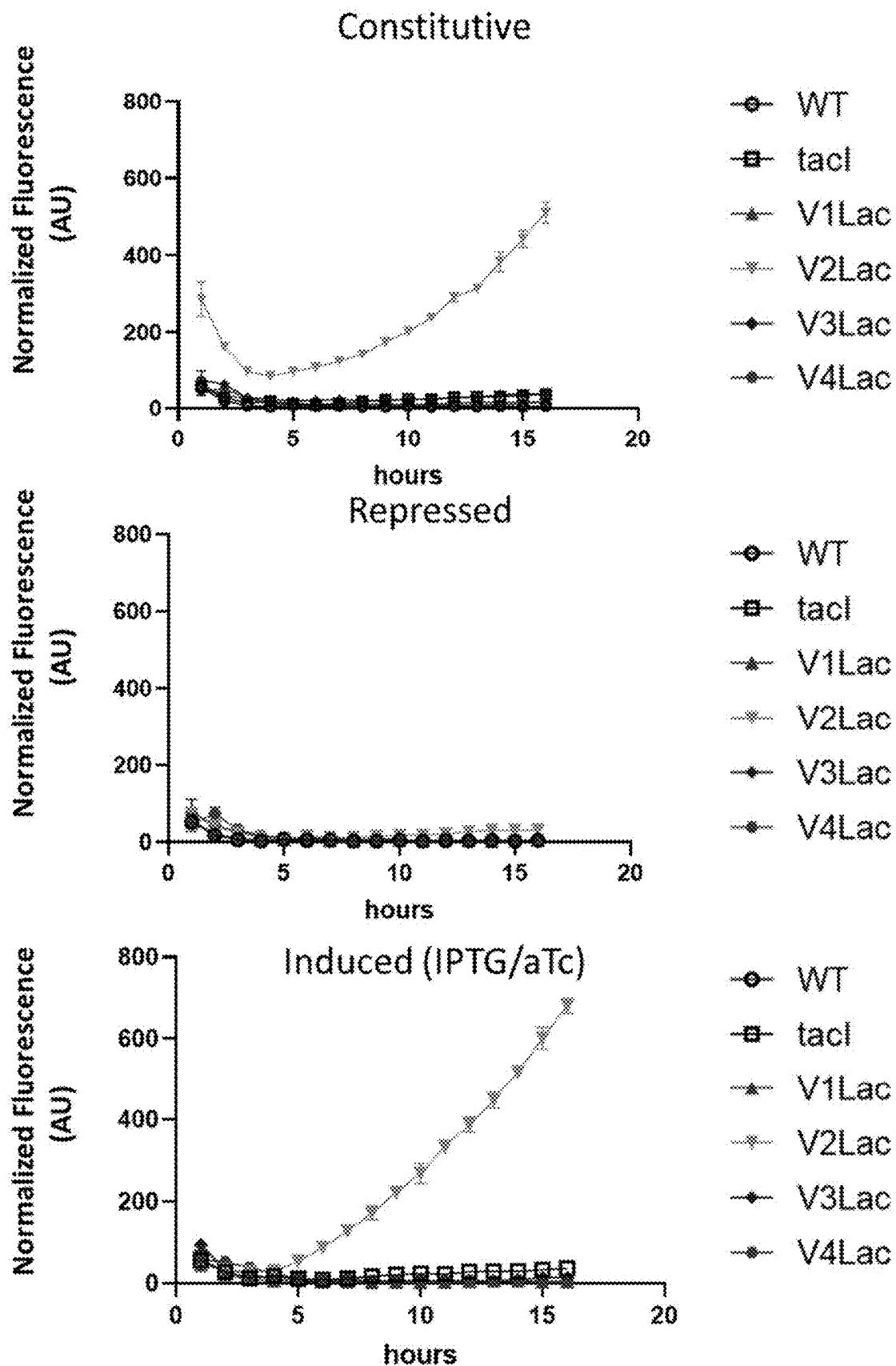
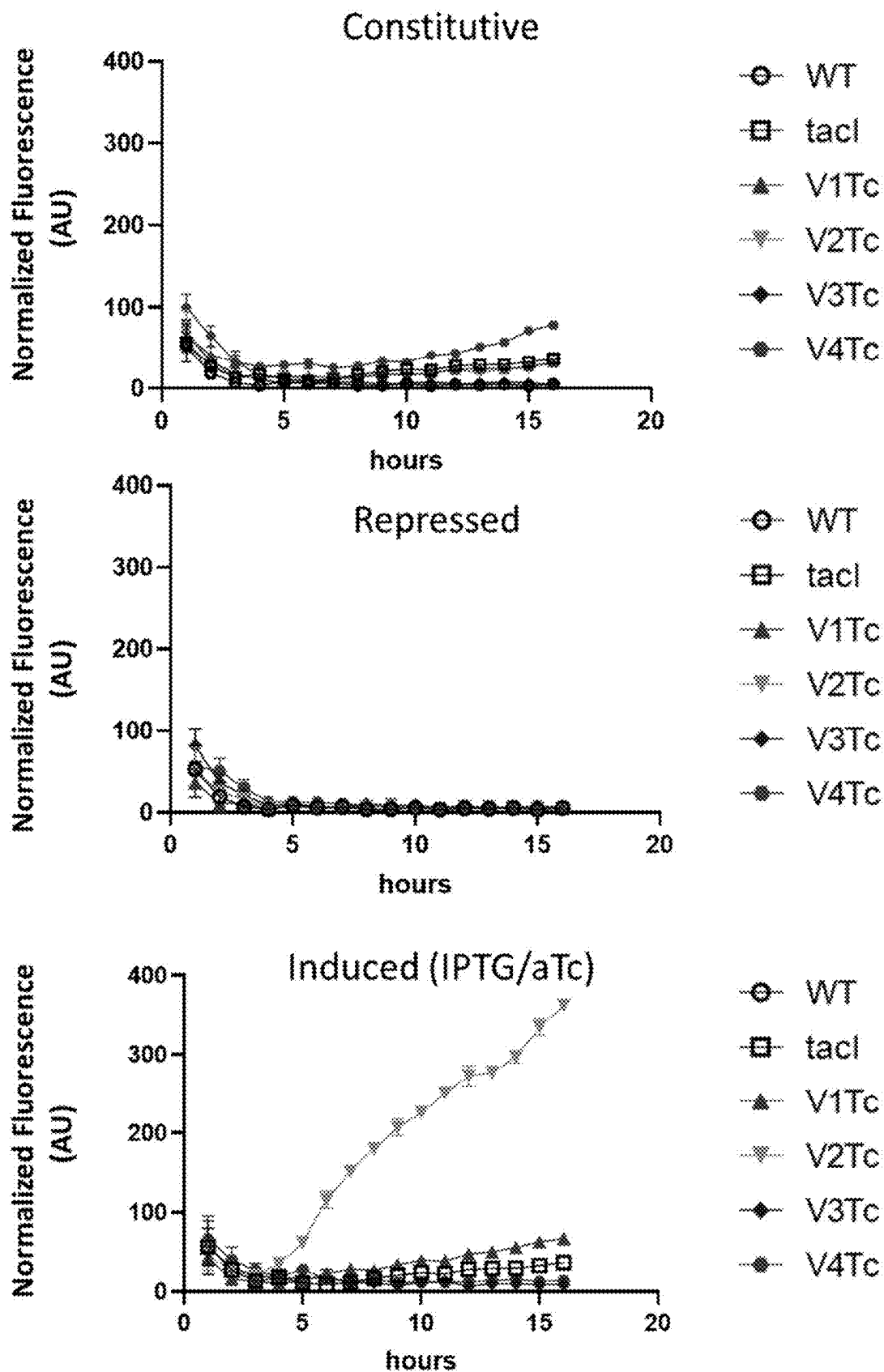


FIG. 6B

	aTc				aTc				aTc				IPTG				IPTG				IPTG			
hours	WT	lacI	V1Tc	V2Tc	V2Tc	V2Tc	V3Tc	V3Tc	V3Tc	V4Tc	V4Tc	V4Tc	V1Lac	V1Lac	V1Lac	V2Lac	V2Lac	V2Lac	V3Lac	V3Lac	V3Lac	V4Lac	V4Lac	
4	0%	100%	119%	0%	3%	6%	0%	0%	0%	0%	28%	0%	4%	30%	6%	59%	8%	13%	15%	6%	6%	4%	13%	
8	0%	100%	96%	0%	46%	7%	0%	79%	0%	1%	26%	0%	21%	38%	0%	50%	0%	55%	9%	0%	0%	1%	0%	
12	0%	100%	102%	0%	46%	8%	0%	83%	0%	2%	29%	0%	22%	30%	0%	46%	1%	60%	7%	0%	1%	2%	0%	
16	0%	100%	125%	0%	51%	9%	0%	94%	3%	0%	31%	0%	24%	29%	1%	3%	63%	3%	91%	10%	1%	2%	3%	

FIG. 6C





hours	aTc				aTc				aTc				IPTG				IPTG				IPTG			
	lacI	M1Tc	V1Tc	tetR	M2Tc	V2Tc	M3Tc	tetR	V3Tc	M4Tc	V4Tc	tetR	M1Lac	V1Lac	tetR	M2Lac	V2Lac	tetR	M3Lac	V3Lac	tetR	M4Lac	V4Lac	
0	100%	86%	0%	48%	58%	22%	224%	51%	31%	99%	165%	107%	42%	23%	24%	574%	67%	187%	135%	80%	36%	77%	94%	150%
4	0%	100%	81%	179%	66%	20%	1346%	27%	49%	58%	182%	38%	86%	47%	0%	1053%	73%	1288%	117%	21%	7%	14%	19%	20%
8	0%	100%	80%	185%	81%	5%	1174%	2%	10%	7%	168%	16%	31%	31%	0%	3%	1255%	76%	1678%	95%	0%	8%	14%	0%
12	0%	100%	86%	198%	86%	0%	1155%	0%	0%	12%	237%	4%	28%	39%	1%	0%	1635%	85%	2186%	101%	0%	28%	35%	4%

FIG. 7C

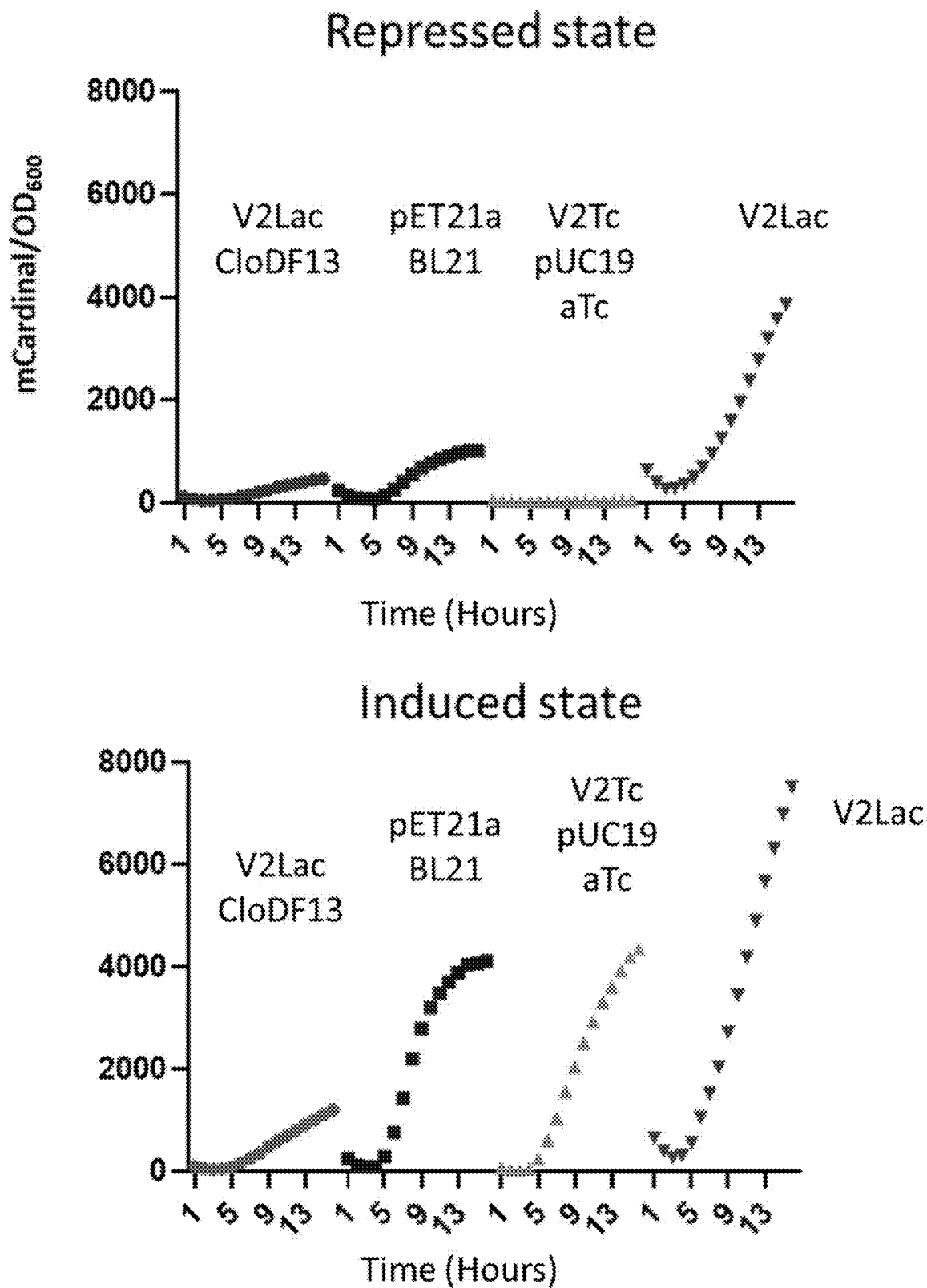
**FIG. 8**

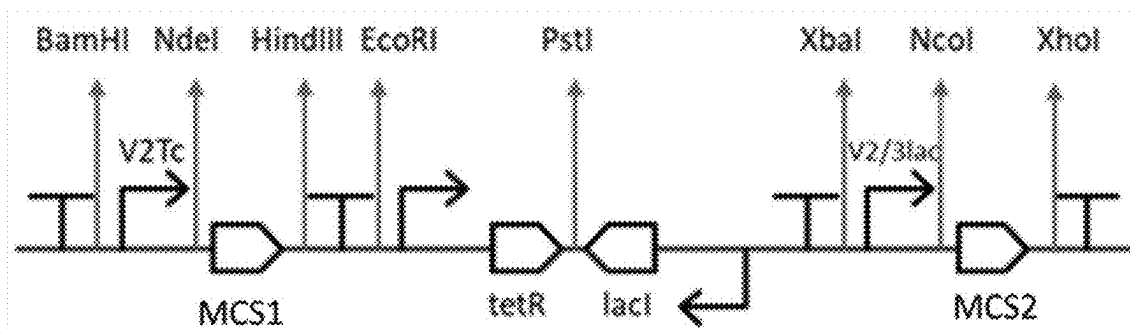
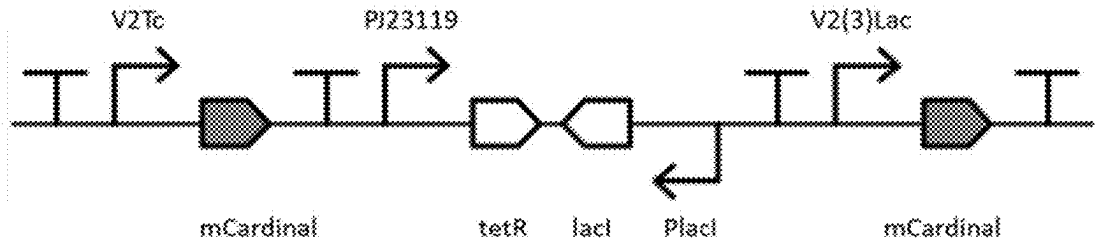
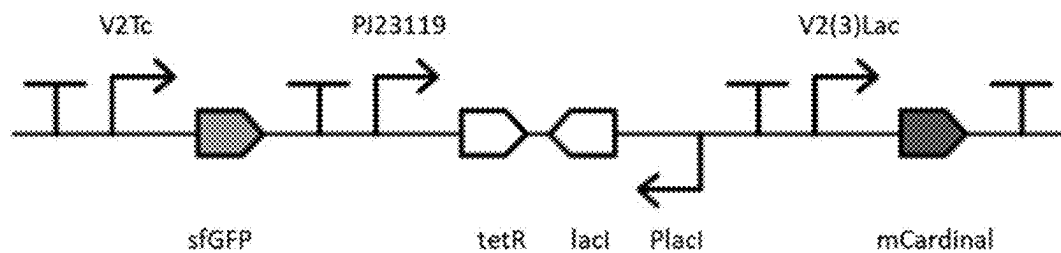
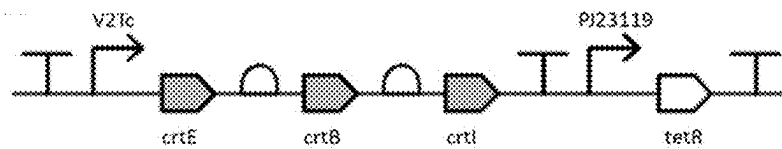
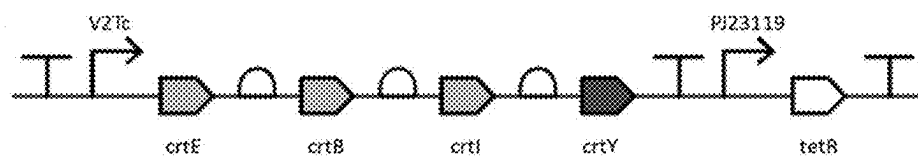
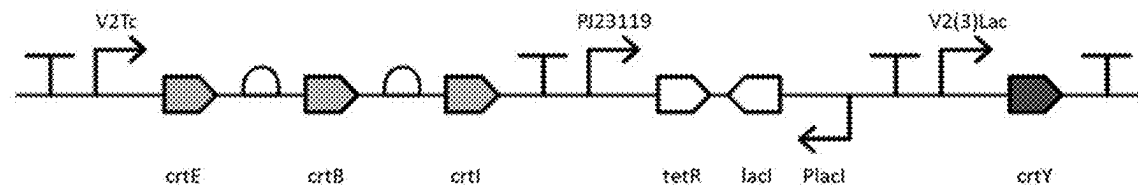
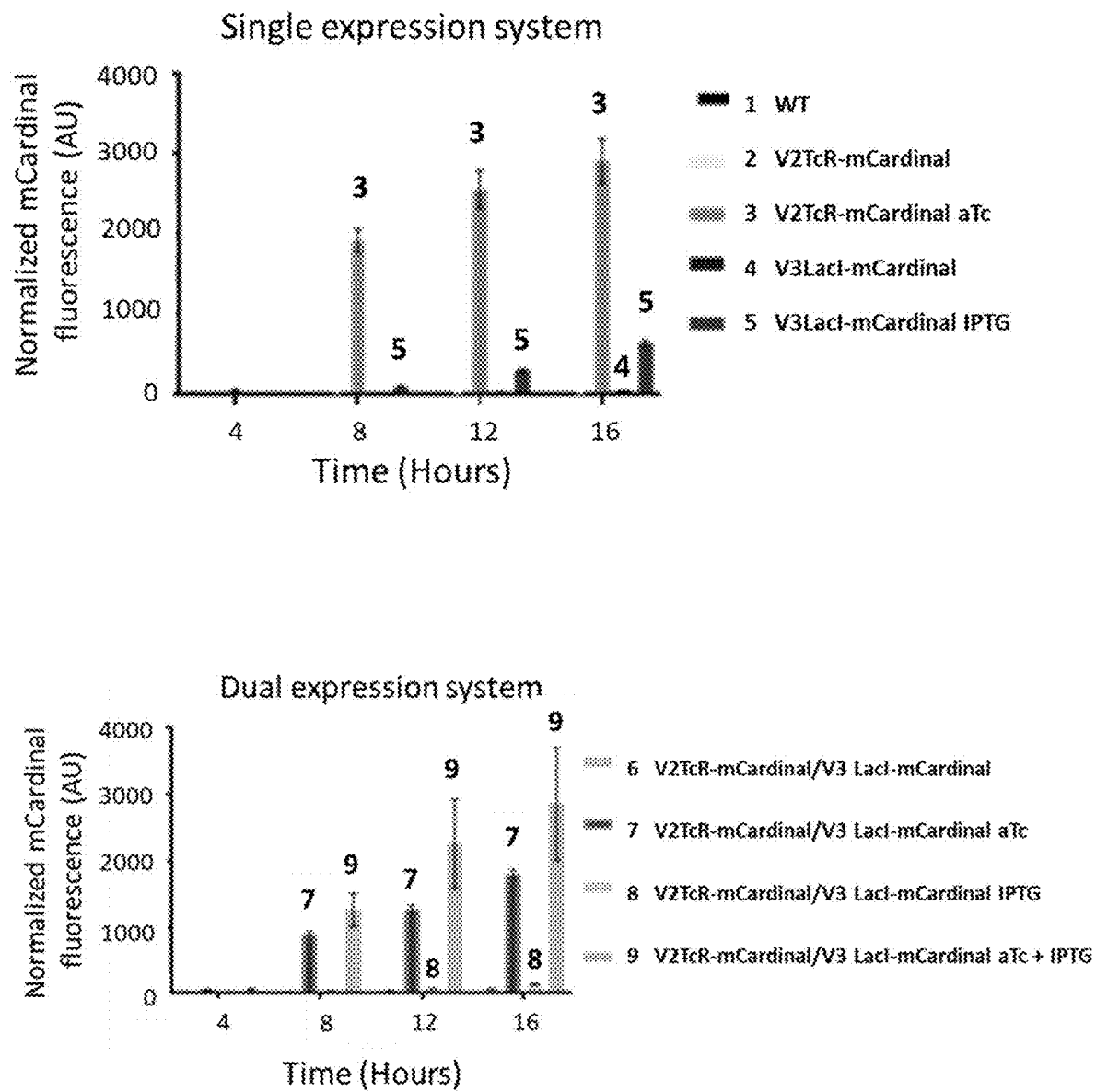
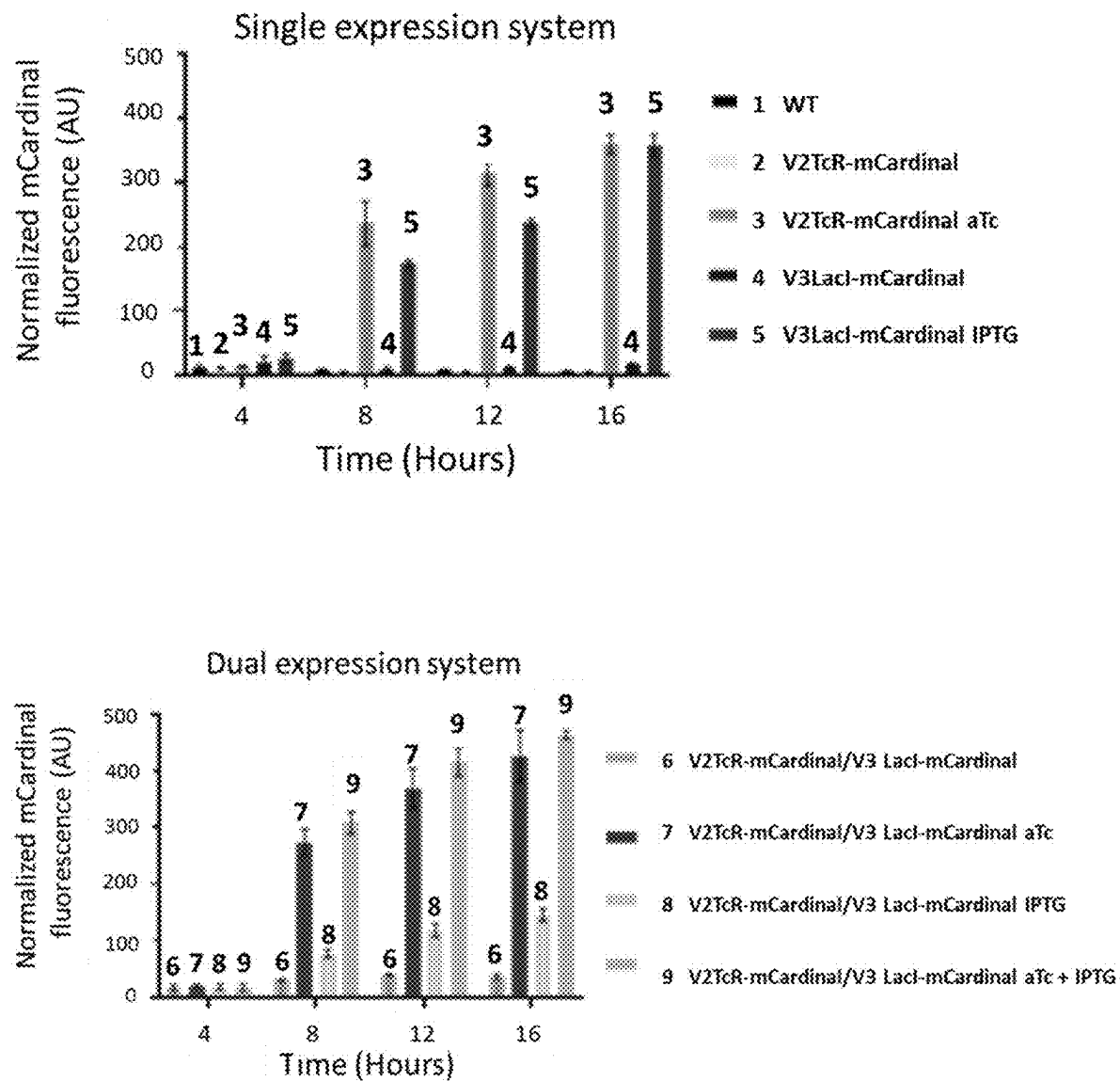
FIG. 9A**FIG. 9B****FIG. 9C**

FIG. 9D**FIG. 9E****FIG. 9F**

**FIG. 10A**

**FIG. 10B**

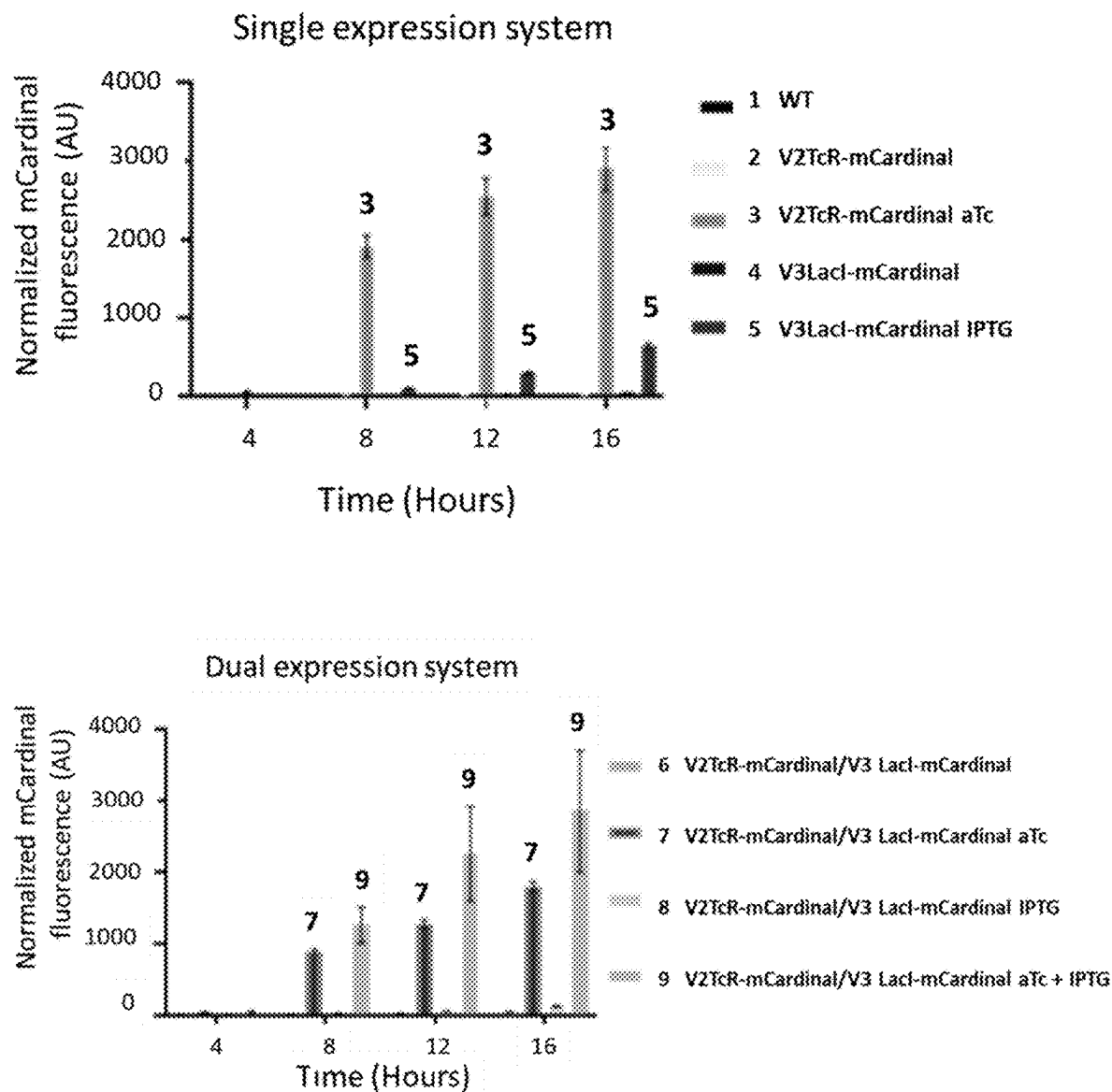
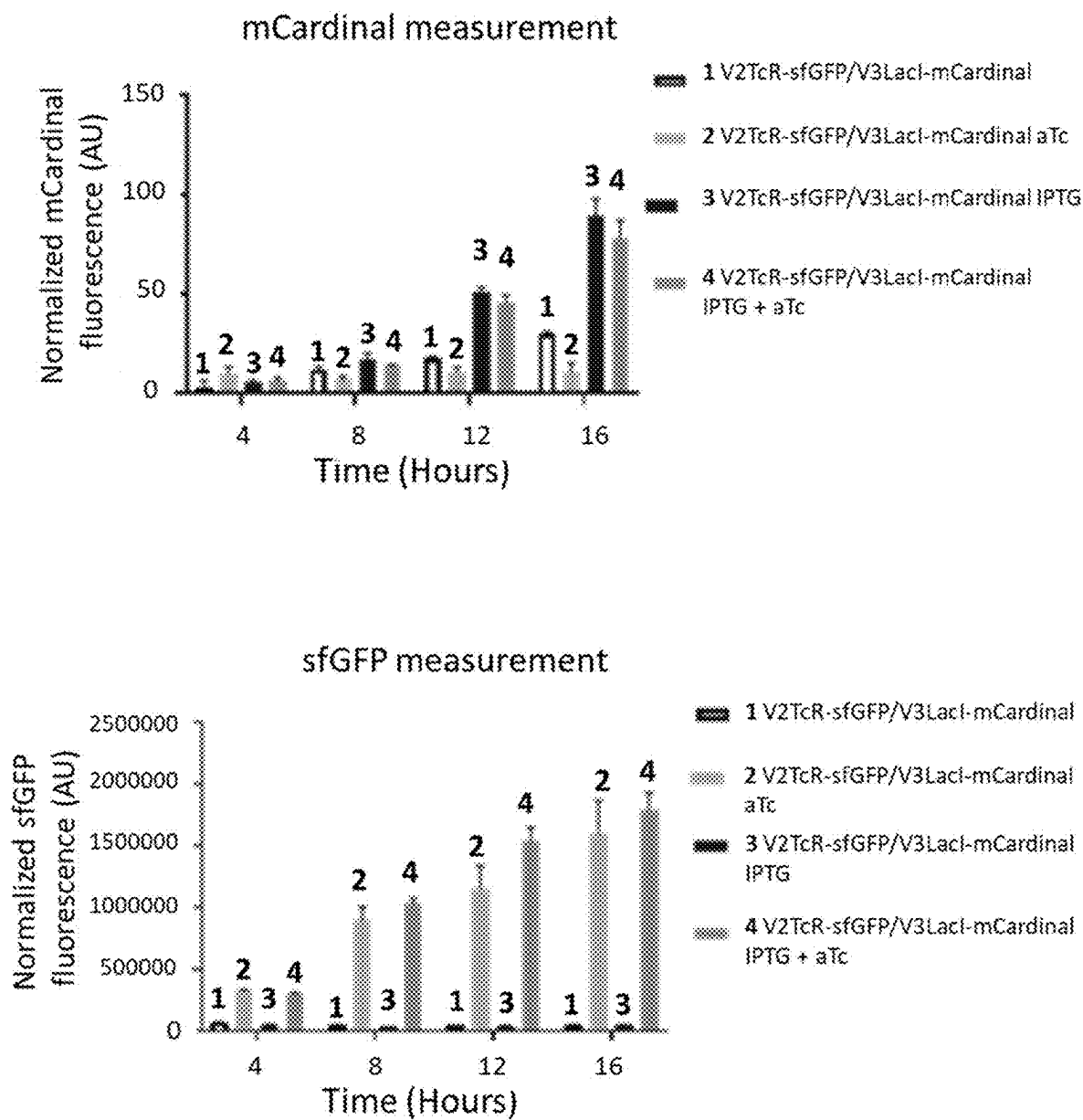
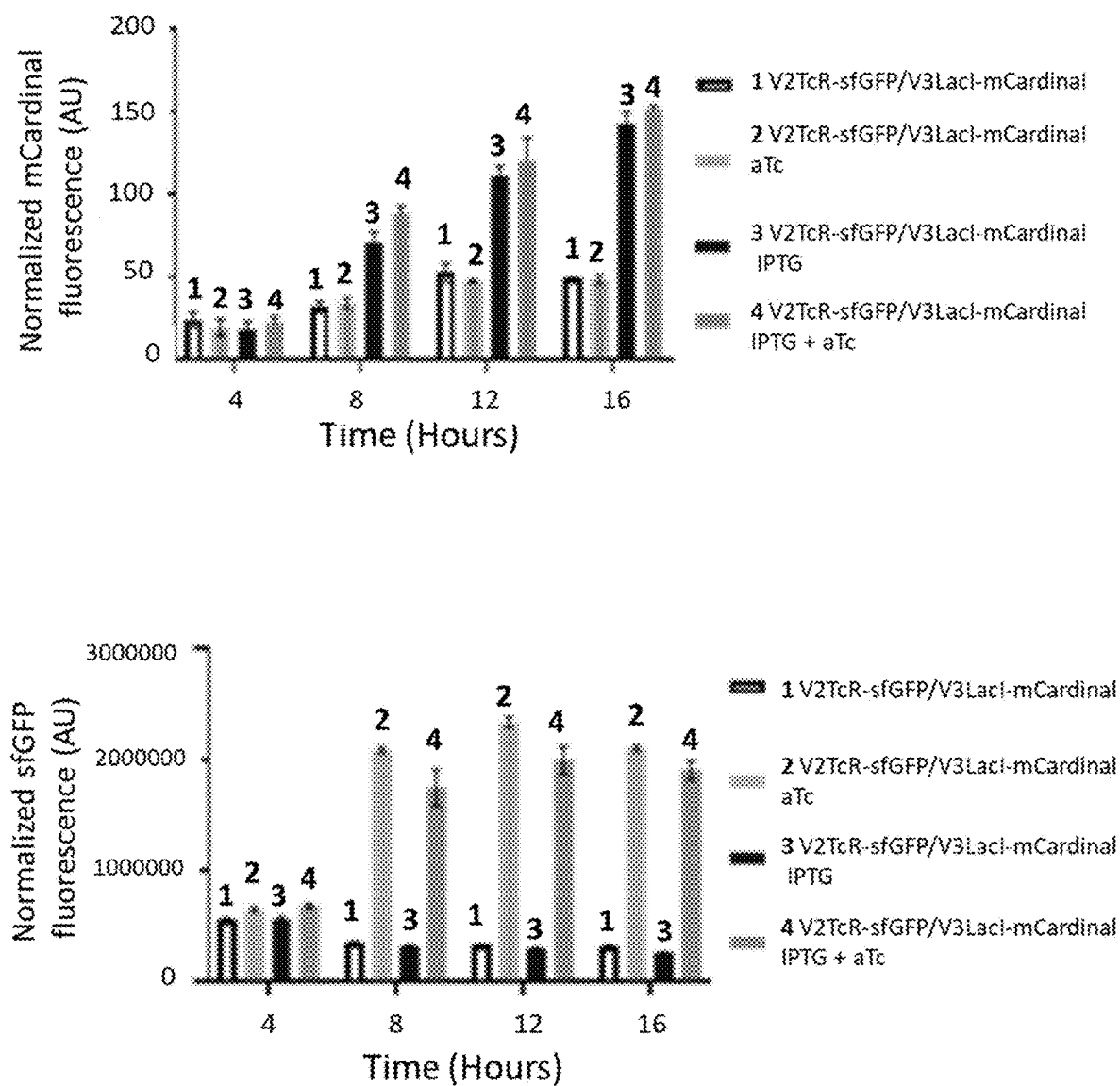
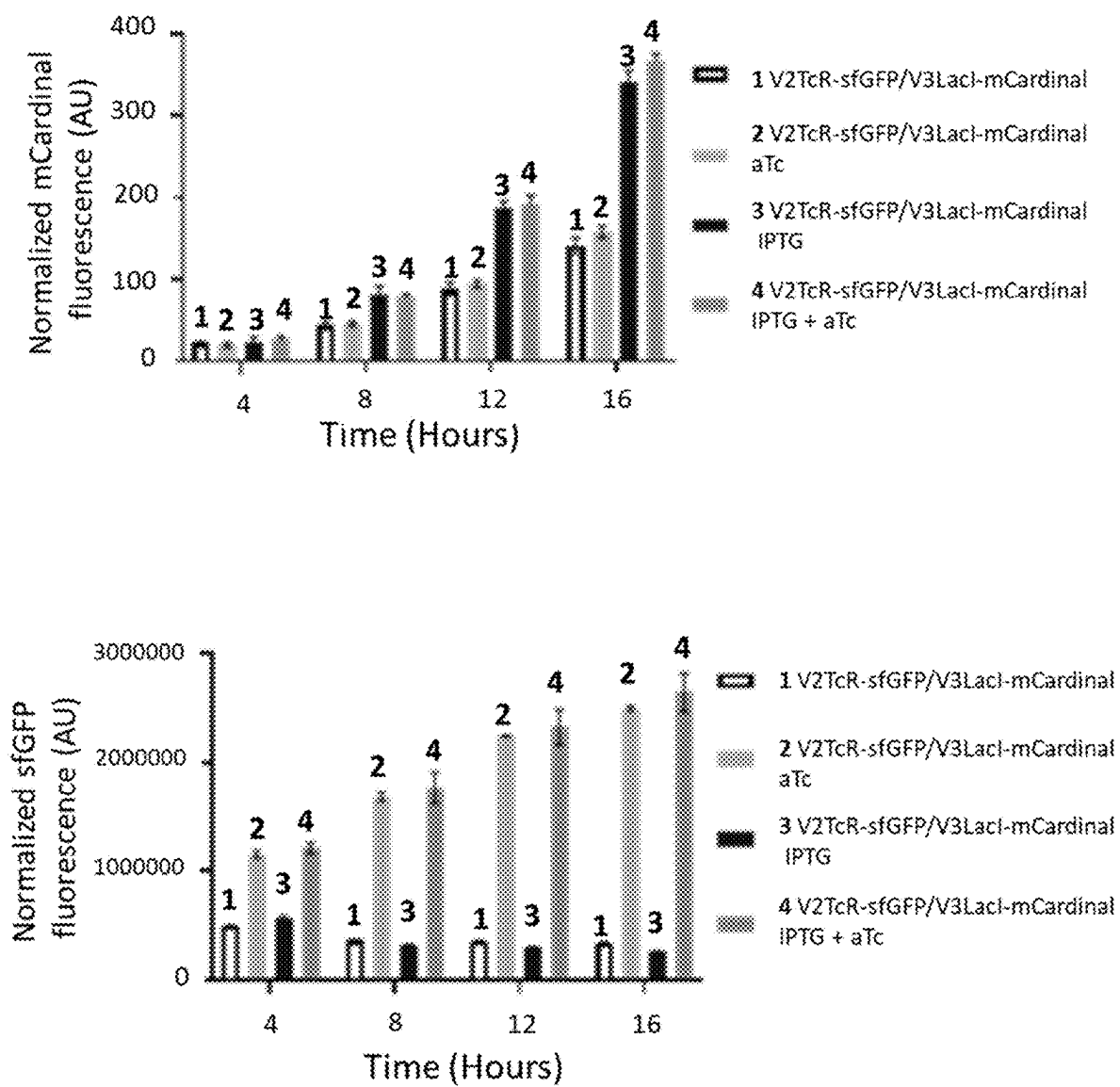


FIG. 10C

**FIG. 11A**

**FIG. 11B**

**FIG. 11C**

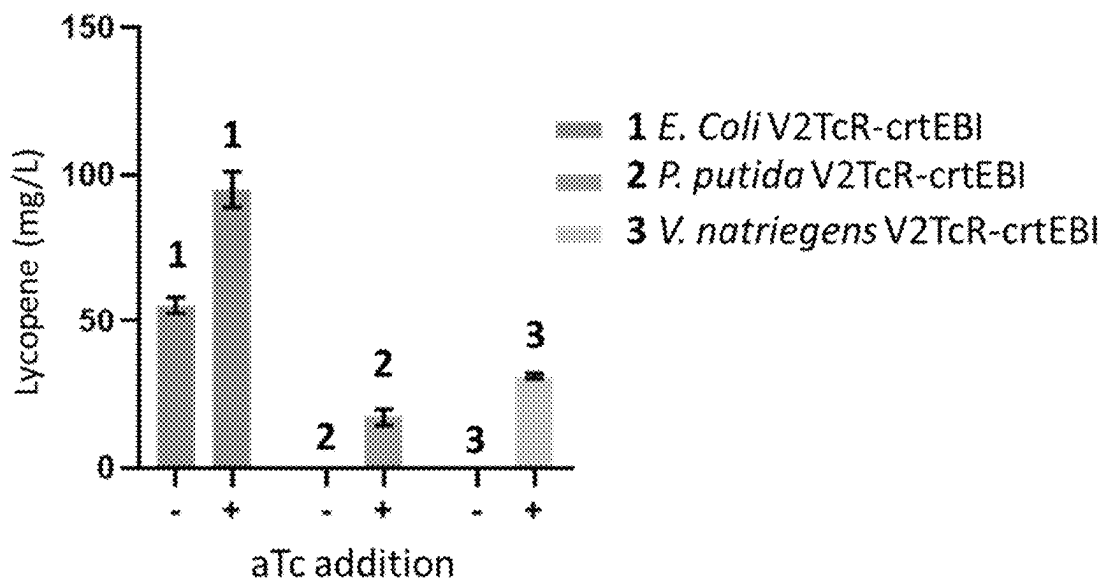


FIG. 12A

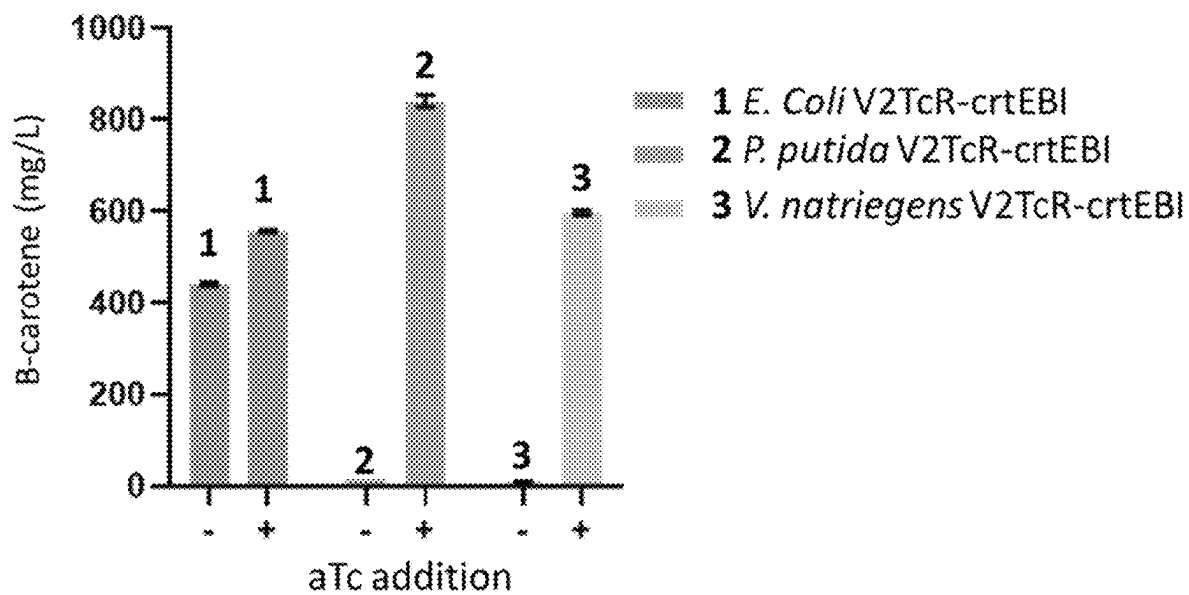


FIG. 12B

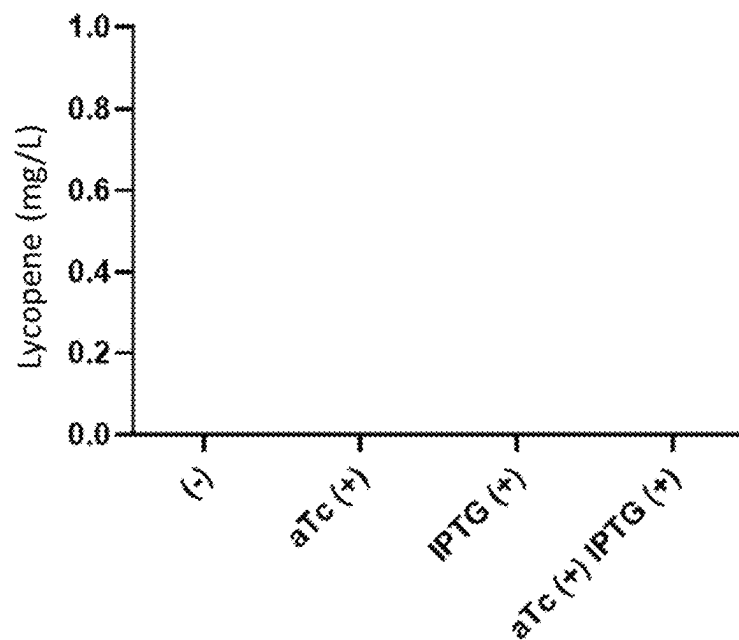


FIG. 13A

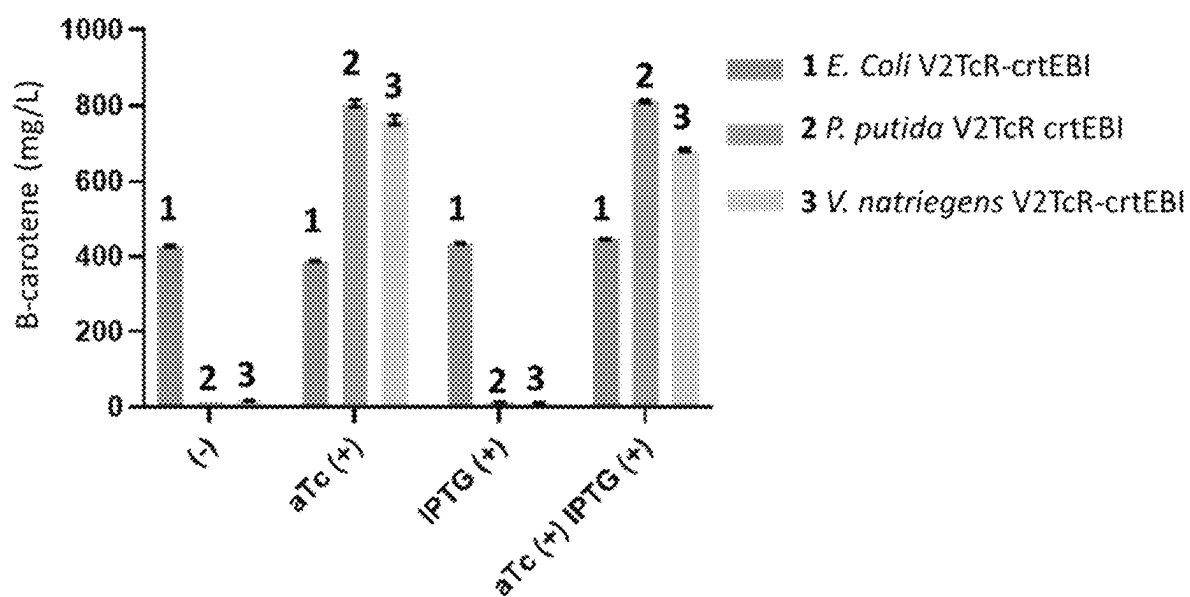
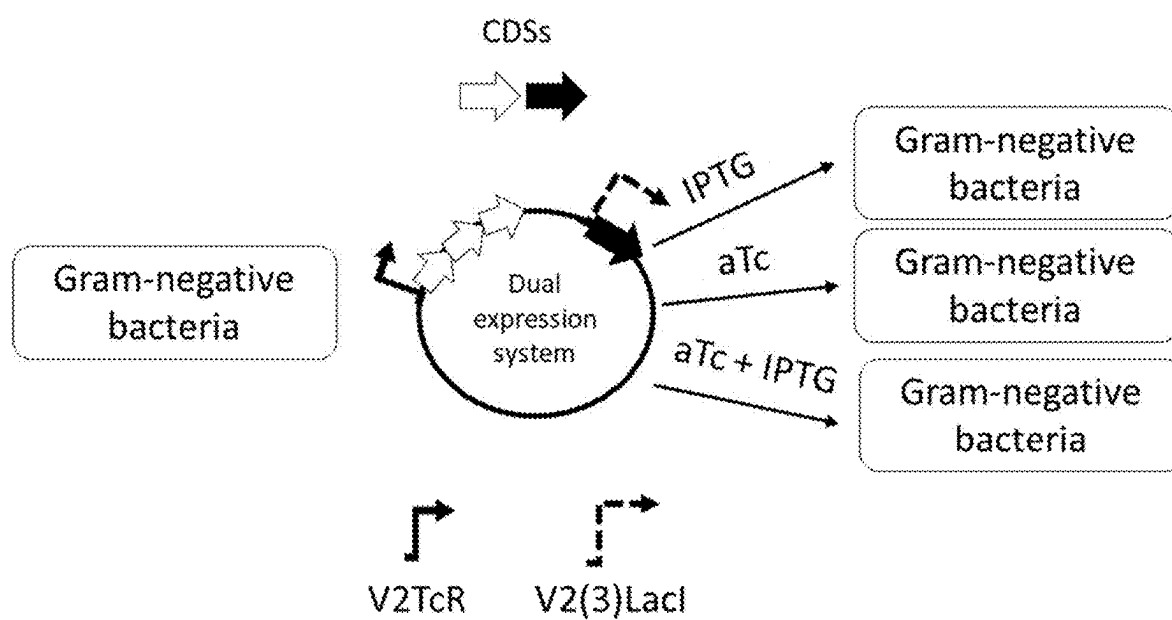


FIG. 13B

**FIG. 14**

INDUCIBLE PROMOTERS**CROSS REFERENCE TO RELATED APPLICATIONS**

This application claims the benefit of, and priority to, U.S. Provisional Patent Application No. 63/332,507 filed Apr. 19, 2022, the entirety of which is incorporated herein by reference.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in XML file format and is hereby incorporated by reference in its entirety. Said XML copy, created on Aug. 4, 2023, is named 110697-015901_US_SL.xml and is 133,847 bytes in size.

BACKGROUND

Inducible promoters are ubiquitous biotechnology tools for manufacturing proteins, providing molecular models of biosynthesis pathways, and as synthetic switches for a variety of environmental, physiological, and cellular tools. Inducible promoters have a consistent architecture including two key elements: the operator region recognized by transcriptional regulatory proteins and consensus sequences that recruit the sigma (σ) subunits of RNA polymerase to initiate transcription of the inducible gene. Despite their widespread use, leaky transcription in the “OFF” state remains a challenge for inducible promoters. Therefore, improved inducible promoters, cellular systems, and methods of generating proteins are needed to enable advances in protein production and various biotechnology applications.

BRIEF SUMMARY

Provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TXTXXTGT sequence at position -10, relative to a transcriptional start site of the promoter; (b) a TXGXCX-sequence at position -35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence, wherein X is any nucleobase. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TATAAT sequence at position -10, relative to a transcriptional start site of the promoter; (b) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. In some embodiments, the nucleic acid construct further comprises a transgene. In some embodiments, the modified inducible promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is inserted into a prokaryotic cell in the presence of an inducer; and wherein: (i) the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct; and (ii) the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct. Further provided

herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (b) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. In some embodiments, the nucleic acid construct further comprises a transgene. In some embodiments, the modified inducible promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is inserted into a prokaryotic cell in the presence of an inducer; and wherein: (i) the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct; and (ii) the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct. In some embodiments, a strong promoter increases the amount of expression of a transgene provided herein relative to a comparable inducible promoter that does not comprise TATAAT, TTGACA, TATAATGT, TXTXXTGT, or TXGXCX. In some embodiments, the strong promoter increases the amount of expression of a transgene provided herein by at least 10% relative to the expression of a transgene expressed by a comparable inducible promoter that does not comprise TATAAT, TTGACA, TATAATGT, TXTXXTGT, or TXGXCX.

Provided herein are methods of expressing a transgene in a cell, the methods comprising: (a) transforming a nucleic acid construct into the cell, wherein the nucleic acid construct comprises a modified inducible promoter, wherein the modified inducible promoter comprises: (i) a TATAAT sequence or a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; (iii) a nucleic acid sequence encoding a bacterial ribosome binding sequence; and (iv) a transgene; and (b) contacting the cell with an inducer, thereby expressing the transgene. In some embodiments, when cloned into a comparable nucleic acid construct that comprises a strong promoter, the transgene inhibits growth of the cell prior to the contacting step (b) with the inducer, thereby preventing expression of the transgene via the comparable nucleic acid. In some embodiments, the cell is a prokaryotic cell. In some embodiments, the cell is a eukaryotic cell.

Provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: (a) a first modified inducible promoter sequence, wherein the first modified inducible promoter comprises: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the first modified inducible promoter sequence; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the first modified inducible promoter sequence; and (iii) a nucleic acid sequence that encodes a first bacterial ribosome binding sequence; and (b) a second modified inducible promoter sequence, wherein the second modified inducible promoter sequence comprises: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the second modified inducible promoter sequence; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the second modified inducible promoter sequence; and (iii) a nucleic acid sequence that encodes a second bacterial ribosome binding sequence. In some

embodiments, the nucleic acid constructs further comprise a transgene. In some embodiments, wherein the first modified inducible promoter sequence, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is in the presence of an inducer.

A composition comprising two or more of a nucleic acid construct provided herein.

Provided herein are isolated prokaryotic cells, wherein the isolated prokaryotic cells comprise a nucleic acid construct provided herein. Provided herein are isolated eukaryotic cells, wherein the isolated eukaryotic cells comprise a nucleic acid construct provided herein. Provided herein are cell-free systems, wherein the cell-free systems comprise a nucleic acid construct provided herein.

Provided herein are non-naturally occurring organisms, wherein the non-naturally occurring organisms comprise: a nucleic acid construct comprising: (a) a first inducible promoter sequence comprising: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (iii) a nucleic acid sequence that encodes a bacterial ribosome binding sequence; (b) one or more of a biosynthesis pathway transgene; and (c) a second modified inducible promoter sequence comprising: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the second modified inducible promoter sequence; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the second modified inducible promoter sequence; and (iii) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. In some embodiments, the non-naturally occurring organisms comprise prokaryotic organisms. In some embodiments, the prokaryotic organisms comprise a population of bacteria. In some embodiments, the biosynthesis pathway transgene comprises a carotenoid synthesis gene.

Provided herein are compositions comprising a non-naturally occurring organism provided herein and an inducer.

Further provided herein are methods of expressing a transgene in a cell that does not comprise a T7 RNA polymerase, wherein the methods comprise: (a) transforming a nucleic acid construct into the cell, wherein the nucleic acid construct comprises a modified inducible promoter, wherein the modified inducible promoter comprises: (i) a TATAAT sequence or a TATAATGT-sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA-sequence at position -35, relative to a transcriptional start site of the promoter; and (iii) a bacterial ribosome binding sequence; and a transgene; and (b) contacting the cell with an inducer, thereby expressing a protein encoded by the transgene. In some embodiments, the expression of the protein in the absence of the inducer is less than an amount of expression of the protein in the absence of the inducer of the transgene operatively coupled to a T7 promoter in a comparable nucleic acid construct. In some embodiments, the expression of the protein encoded by the transgene in the presence of the inducer is greater than an amount of expression of the protein in the presence of the inducer of the transgene operatively coupled to the T7 promoter in the comparable nucleic acid construct. In some embodiments, the transgene is toxic to a prokaryotic cell that does not express a modified inducible promoter. In some embodiments, the cell is a prokaryotic cell. In some embodiments, the cell is a eukaryotic cell.

Further provided herein are methods of expressing a transgene in a cell-free system, wherein the methods com-

prise: (a) transforming a nucleic acid construct into the cell-free system, wherein the nucleic acid construct comprises a modified inducible promoter, wherein the modified inducible promoter comprises: (i) a TATAAT sequence or a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (iii) a bacterial ribosome binding sequence; and a transgene; and (b) contacting the cell-free system with an inducer, thereby expressing a protein encoded by the transgene. In some embodiments, the cell-free system further comprises: an RNA polymerase, ribonucleotides, and/or a buffer.

Provided herein are methods for producing a protein, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the protein. Further provided herein are methods for producing a carotenoid, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the protein. Provided herein are methods for producing an organic compound, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the organic compound. Further provided herein are methods for producing a carotenoid, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the carotenoid. In some embodiments, the organic compound or the protein comprises a polyketide, a terpene, a non-ribosomal peptide, or an enzyme protein. Provided herein are methods for producing benzoic acid, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the benzoic acid. Further provided herein is a composition comprising benzoic acid made by a method provided herein. Further provided herein is a composition comprising a carotenoid made by a method provided herein.

Provided herein are kits, wherein the kits comprise a nucleic acid construct provided herein, a cell provided herein, a cell-free system provided herein, or a non-naturally occurring organism provided herein, packaging and materials, therefore.

BRIEF DESCRIPTION OF THE DRAWINGS

Novel features of exemplary embodiments are set forth with particularity in the appended claims. A better understanding of the features and advantages will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the disclosed systems and methods are utilized, and the accompanying drawings of which:

FIGS. 1A-1B show schematic diagrams of exemplary plasmids produced in accordance with the teachings of the present disclosure. FIG. 1A shows a schematic of a constitutive version of the synthetic lac and tet expression systems. FIG. 1B shows a schematic of inducible versions of the synthetic lac and tet expression systems in pColE1 backbone. Plasmids contain the Kanamycin selection marker and the attP sites specific for Bxb1 integrase. # indicates the version of the promoter (e.g. V1Tc, V2lac, etc).

FIGS. 2A-2C show the architecture of exemplary synthetic lac and tet promoters. The sequences of the operators

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lacO and tetO are overlined, in bold text. The -35 and -10 sequences of the promoters are underlined. Boxes represent the native -35 and -10 sequences, and consensus σ^{70} -35 and -10 sequences. Conserved regions of the original lac promoter and the original tacI promoter are shown. FIG. 2A shows the original lac (i) and synthetic lac (ii-vii) promoters. SEQ ID NOS: 3-9 are shown. FIG. 2B shows the original tet (i) and synthetic tet (ii-v) promoters. SEQ ID NOS: 10-14 are shown. FIG. 2C shows schematic representation of the function of synthetic lac and tet promoters. Each transcriptional unit is insulated by terminators. LacI/TetR regulators bind to the operators lacO/tetO. Addition of the inducers (yellow ovals) IPTG or anhydrotetracycline (aTc) remove the repressor allowing transcription from the promoters.

FIGS. 3A-3D show a graphical comparison of sfGFP and mCardinal emissions in Gram-negative bacteria. FIG. 3A shows absolute values of fluorescence measured in arbitrary units with far red wavelengths (excitation 604, emission 659, left panel), and green wavelengths (excitation 485, emission 510) *E. coli* DH10B, *P. putida* and *V. natriegens*. Note that these strains are wild type and do not contain any fluorescent protein genes. N=3. Error bars +/-SD. FIGS. 3B-3D show graphs of the fluorescence signal-to-background ratio of recombinant strains expressing sfGFP and mCardinal from the constitutive tacI promoter (these represent "signal") vs. wild type strains of (FIG. 3B) *E. coli* (FIG. 3C) *P. putida* & (FIG. 3D) *V. natriegens* (these wild type strains represent "background"). Signal-to-background measurement ratios are then plotted over time (left panels) The fluorescence values (in arbitrary units, AU) measured to create the signal-to-background ratios are shown in the middle (sfGFP) and right (mCardinal) panels. N=3. Error bars +/-SD.

FIGS. 4A-4C show graphs of the time course of mCardinal production in *E. coli* DH10B with synthetic lac and tet promoters. FIG. 4A shows the synthetic lac promoters, and FIG. 4B shows the synthetic tet promoters, with and without the transcriptional regulators tetR and lacI and induced after 3 hours with aTc or IPTG, respectively. FIG. 4C shows direct comparison of each promoter under evaluation against the strong constitutive promoter tacI. The recombinant DH10B carrying tacI-mCardinal was normalized to 100% mCardinal production and the wild type DH10B strain normalized to 0% mCardinal production. For all samples, the fluorescence mean of mCardinal signal (excitation 605, emission 659) was normalized by the cell density (OD600). N=4. Error bars +/-SD.

FIGS. 5A-5B show an SDS-Page gel and graph of the expression profile of CocE in *E. coli*. FIG. 5A shows gel electrophoresis of total crude extracts (top), the insoluble fraction analysis (middle panel), and the soluble fraction (bottom). The expected size of CocE is 63 kDa. *E. coli* expressing the pET and V2TcR expression systems, uninduced and induced (induced samples indicated with an asterisk). FIG. 5B shows a graph of benzoic acid production by CocE present in the soluble fraction, using cocaine as substrate. N=3. Error bars +/-SD. Key for X axis Groups: I: DH10B p σ^{70} V2TcR-CocE; II: DH10B p σ^{70} V2TcR-CocE aTc; III: DH10B p σ^{70} V2TcR19-CocE; IV: DH10B p σ^{70} V2TcR19-CocE aTc; V: BL21 pET21-cocE; VI: BL21 pET21-cocE IPTG.

FIGS. 6A-6C show graphs of the time course of mCardinal production in *P. putida*. Synthetic lac (FIG. 6A) and tet (FIG. 6B) promoters in their constitutive, repressed and induced states. All constructs were integrated in a single copy into the same genomic locus. FIG. 6C shows direct comparison of each promoter under evaluation against the

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constitutive promoter tacI in *P. putida*. *P. putida* with a single-copy integration of tacI-mCardinal was normalized to 100% mCardinal production and the wild type *P. putida* strain normalized to 0% mCardinal production. For all samples, the fluorescence mean of mCardinal signal (excitation 605, emission 659) was normalized by the cell density (OD600). N=4. Error bars +/-SD.

FIGS. 7A-7C show graphs of the time courses of mCardinal production in *V. natriegens*. Synthetic lac (FIG. 7A) and tet (FIG. 7B) promoters in their constitutive, repressed and induced states are represented. FIG. 7C shows direct comparison of each promoter under evaluation against the constitutive promoter tacI in *V. natriegens*. *V. natriegens* carrying tacI-mCardinal was normalized to 100% mCardinal production and the wild type *V. natriegens* strain normalized to 0% mCardinal production. For all samples, the fluorescence mean of mCardinal signal (excitation 605, emission 659) was normalized by the cell density (OD600). N=4. Error bars +/-SD.

FIG. 8 shows a graphical comparison of the pET21a expression system in *E. coli* BL21 versus the synthetic V2Lac/lacI and V2TcR/tetR promoters in *E. coli* DH10B. Cultures were induced after 3 hours of cultivation and mCardinal mean was normalized by the cell density (OD₆₀₀). N=4. Error bars +/-SD.

FIGS. 9A-9F shows a schematic representation of an exemplary dual expression system. FIG. 9A shows the design of the V2TcR-V2(3)LacI system. FIG. 9B shows a dual expression system controlling expression of mCardinal. FIG. 9C shows a dual expression system controlling expression of sfGFP and mCardinal. FIG. 9D shows a V2TcR expression system controlling the LYC operon. FIG. 9E shows a V2TcR expression system controlling expression of crtEBIY. FIG. 9F shows a dual expression system controlling expression of crtEBI and crtY. Exemplary sequences tested include SEQ ID NOS: 69-73 and 76-79.

FIGS. 10A-10C show graphs of the time course of mCardinal production in *E. coli* (FIG. 10A), *P. putida* (FIG. 10B) & *V. natriegens* (FIG. 10C). Left graphs show the recombinant strains containing the V2TcR-mCardinal (SEQ ID NO: 56) and V2(3)LacI-mCardinal expression systems and the right graphs show the recombinant strains containing the dual V2TcR-mCardinal/V2(3)LacI-mCardinal expression system. For all samples, the fluorescence mean of mCardinal signal (excitation 605, emission 659) was normalized by the cell density (OD600). N=3. Error bars +/-SD.

FIGS. 11A-11C shows graphs of the time course of the reporter systems mCardinal and sfGFP in *E. coli* (FIG. 11A), *P. putida* (FIG. 11B), and *V. natriegens* (FIG. 11C) containing the dual expression system V2TcR-sfGFP/V2(3)LacI-mCardinal. Left graphs measure mCardinal production and right graphs measure sfGFP. For all samples, the fluorescence mean of mCardinal signal (excitation 605, emission 659) and sfGFP signal (excitation 485, emission 510) was normalized by the cell density (OD600). N=3. Error bars +/-SD.

FIGS. 12A-12B shows graphs of the production of lycopene and β -carotene with the V2TcR expression system in *E. coli*, *P. putida*, and *V. natriegens*. Cultures were induced with aTc at OD600. 0.7 and cultured for 4 hours. FIG. 12A shows Lycopene levels that were extracted and measured by UHPLC. FIG. 12B shows β -carotene levels that were extracted and measured by UHPLC. N=3. Error bars +/-SD.

FIGS. 13A-13B shows graphs of the production of lycopene and beta (β)-carotene by the dual expression system in *E. coli*, *P. putida*, and *V. natriegens*. Cultures were induced with aTc, IPTG or both inducers at OD600. 0.7 and cultured

for 4 hours. FIG. 13A shows lycopene levels that were extracted and measured by UHPLC. FIG. 13B shows β -carotene levels that were extracted and measured by UHPLC. N=3. Error bars $\pm$ SD.

FIG. 14 shows a schematic of an exemplary dual expression system. Exemplary sequences include SEQ ID NO: 54 and SEQ ID NO: 55.

DETAILED DESCRIPTION

Overview

Disclosed herein are nucleic acid constructs that comprise a modified inducible promoter. Such constructs can be utilized to express a transgene operatively coupled to the promoter upon transformation into a cell or a cell-free system. As described herein, the modified inducible promoter does not substantially express the transgene in the absence of an inducer and in the presence of a specific transcriptional regulator. In some cases, a transgene is toxic to the cell when expressed utilizing the modified inducible promoter described herein. In some cases, a transgene is not toxic to the cell when expressed utilizing the modified inducible promoter described herein. Expression of toxic genes is one of many advantages to the modified inducible promoter systems provided herein. In addition, non-toxic gene product yields (e.g., proteins or organic compounds) are also higher as compared to inducible promoters that are not modified.

In contrast, many inducible promoters produce significant expression of the transgene even in the absence of an inducer. Such promoters, known as “leaky” promoters, cannot be used to express transgenes that are toxic to a cell, as the leaky expression of the transgene inhibits growth of the cell culture and thus prevents overexpression or avoids formation of inclusion bodies.

Further, nucleic acid constructs provided herein comprising a modified inducible promoter that generates the expression of a protein encoded by a transgene in the presence of an inducer that is comparable or to a greater extent than the expression of a protein encoded by the transgene when expressed using a different promoter (e.g., a T7 promoter such as a pET vector). In some embodiments, expression of the transgene can be carried out in various prokaryotic cells, eukaryotic cells, or cell-free systems without the need for T7 lysogenization.

Definitions

The terminology used herein is for the purpose of describing particular cases only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Furthermore, to the extent that the terms “including”, “includes”, “having”, “has”, “with”, or variants thereof are used in either the detailed description and/or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising”.

The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, such as plus or minus 10%. Where ranges and/or subranges of values are provided, the ranges and/or subranges include the endpoints of the ranges and/or subranges.

The term “substantially” as used herein refers to a value approaching 100% of a given value. For example, an expression system described herein that does not “substantially” express a transgene in the absence of an inducer can indicate that less than 10% of the transgene (e.g. less than 5%, less

than 1%, less than 0.1%, or less than 0.01%) is expressed, relative to an amount of transgene expressed in the presence of the inducer.

As used herein, the term “operably linked” indicates that a promoter is in a correct functional location and/or orientation in relation to a nucleic acid sequence it regulates to control transcriptional initiation and/or expression of that sequence.

Modified Inducible Promoters

Inducible promoters often have a consistent architecture including two elements: (1) the operator region recognized by the transcriptional regulator proteins (e.g., lacI and tetR, and the -10 and -35 consensus sequences that recruits the sigma (σ) subunits of RNA polymerase to initiate transcription. Improvements to the lac promoter can be made to increase its strength and improve its regulation. For example, the -10 and -35 boxes of the original lac promoter (FIG. 1Ai) can be updated to generate improved variants such as the lacUV5 (FIG. 1Aii) and tacl (FIG. 1Aiii) promoters. In the lacUV5 promoter, the -10 box of the original lac promoter can be replaced by the Pribnow box (e.g., TATAAT) or any one of TATAATGT, TXTXXT, TATXXTGT, TATAXTGT, TATAATGT or TATAATGT. The lacUV5 and trp promoters shown in the working examples can be combined to create the tacl promoter, which increased transcription 11-fold compared to its predecessor with the incorporation of the -35 consensus box TTGACA. The tet promoter (FIG. 1Bi) shares the highly conserved -35 hexamer with tac promoter, but does not contain the Pribnow box.

Despite their ubiquitous use in biology, there remain problems with the current lac-based inducible expression systems. Leaky transcription in the OFF state remains a consistent challenge. Tight transcriptional control is indispensable to produce high yields of challenging recombinant proteins, including toxic genes and proteins that are impossible for a heterologous host to process or fold. For the pET system, target proteins are driven indirectly by controlling expression of the T7 polymerase under the lacUV5 promoter, and then driving the target gene transcription by the T7 promoter. Still, low level T7 transcription in the uninduced state leads to leaky transcription of the target gene. To combat this problem further, *E. coli* strains such as pLysS and pLysE with integration of the T7 lysozyme that inhibits low level T7 activity are used to obtain tighter transcriptional control. However, a host strain with T7 RNA polymerase under control of the lacUV5 promoter, and integration of T7 lysozyme, is required to tightly regulate target gene expression. Expression with the tet system has historically provided tighter transcriptional control than lac derived promoters, does not require specialized strains (such as the BL21), and fully induction is achieved by anhydrotetracycline (aTc) at concentrations that do not cause growth defect due the high affinity of aTc to TetR, and its imperceptible antibiotic activity. Remarkably, at the uninduced state just one mRNA molecule per three cells is produced, and up to 5000-fold induction has been reported, however, the yields of recombinant protein obtained by the tet expression system remains low compared to the pET expression system when using *E. coli* as heterologous host.

The repertoire of organisms used both in academic and industrial settings is rapidly expanding. To address challenges related to complex protein expression in *E. coli*, other chassis organisms such as *Pseudomonas putida* and *Vibrio natriegens* have been employed to produce challenging proteins (e.g., carotenoids) in a variety of biotechnological processes. Both lac and tet expressions systems can be

adapted to *P. putida* and *V. natriegens*, though in some instances with lower total protein yield than achieved in pET. However, previous systems for inducible expression were not configured for use across various cell types (e.g., prokaryotic cells). Accordingly, disclosed herein are universal expression systems than can be directly ported between different gram-negative species, yield high quantities of recombinant protein comparable to expression via a pET in *E. coli*, and maintain tight transcriptional repression in the uninduced state. The expression system of the current disclosure is a significant improvement over the pET system in providing tight OFF state control while achieving similar yields of recombinant protein, and with the advantage of direct portability to alternative host species. Further, additional modifications can be performed to enable the expression of the transgene to be turned OFF after induction, thus allowing for reversible control of expression.

A modified, inducible promoter of the current disclosure comprises modified architecture of the lac and tet expression systems to improve their strength, control, and portability. In some embodiments, the genetic architecture of the lac and tet expression systems were modified in three ways: (1) addition of the consensus -10 and -35 sequence boxes to be strongly targeted by σ^{70} , (2) incorporation of a nucleic acid sequence that encodes a strong ribosome binding site recognized by a broad spectrum of gram-negative bacteria, and (3) independent control of the transcriptional regulators by appropriately-tuned constitutive promoters.

The nucleic acid constructs provided herein can comprise DNA, RNA, an artificial nucleic acid analog, or any combination thereof. In some embodiments, the nucleic acid constructs provided herein comprise, a nucleic acid modification (e.g., a chemical modification), a nucleobase substitution, or a nucleotide substitution. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises one or more nucleobase substitutions. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TXTXXT, where X is any nucleobase. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TXTXXTGT, where X is any nucleobase. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TATXXTGT, where X is any nucleobase. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TATXXTGT, where X is any nucleobase. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TATAATGT, where X is any nucleobase. In some embodiments, the sequence at position -10, relative to the transcriptional start site of the promoter comprises a sequence of TATAATGT, where X is any nucleobase.

In some embodiments, the sequence at position -35, relative to the transcriptional start site of the promoter comprises one or more nucleobase substitutions. In some embodiments, the sequence at position -35, relative to the transcriptional start site of the promoter comprises a sequence of TXGXCX, where X is any nucleobase. In some embodiments, the sequence at position -35, relative to the transcriptional start site of the promoter comprises a sequence of TTGXCX, where X is any nucleobase. In some embodiments, the sequence at position -35, relative to the transcriptional start site of the promoter comprises a sequence of TTGACX, where X is any nucleobase. In some

embodiments, the sequence at position -35, relative to the transcriptional start site of the promoter comprises a sequence of TTGACA, where X is any nucleobase.

In some embodiments, a nucleic acid construct provided herein comprises a nucleotide analogue. Nucleotide analogues include nucleotides having modifications in the chemical structure of the base, sugar and/or phosphate, including, but not limited to, 5-position pyrimidine modifications, 8-position purine modifications, modifications at cytosine exocyclic amines, substitution of 5-bromo-uracil, and the like; and 2'-position sugar modifications, including but not limited to, sugar-modified ribonucleotides in which the 2'-OH is replaced by a group selected from H, OR, R, halo, SH, SR, NH₂, NHR, NR₂, or CN. shRNAs also can comprise non-natural elements such as non-natural bases, e.g., inosine and xanthine, sugars, e.g., 2'-methoxy ribose, or non-natural phosphodiester linkages, e.g., methylphosphonates, phosphorothioates and peptides.

The nucleic acid construct provided herein can comprise a sequence encoding a ribosome binding site. Ribosome binding sites (RBSs) are nucleic acid sequences that promote efficient and accurate translation of mRNAs for protein synthesis, and are also provided for use in the inducible promoters provided herein to permit modulation of the efficiency and rates of synthesis of the proteins encoded by the system. An RBS affects the translation rate of an open reading frame in two main ways—i) the rate at which ribosomes are recruited to the mRNA and initiate translation is dependent on the sequence of the RBS, and ii) the RBS can also affect the stability of the mRNA, thereby affecting the number of proteins made over the lifetime of the mRNA. Accordingly, one or more nucleic acid sequence encoding a ribosome binding site (RBS) or an RBS mRNA can be added to the nucleic acid constructs described herein to control expression of proteins.

In some embodiments, a nucleic acid construct provided herein further comprises a terminator sequence. Terminators are sequences that usually occur at the end of a gene or operon and cause transcription to stop, and are also provided for use in the modules and engineered systems described herein to regulate transcription and prevent transcription from occurring in an unregulated fashion, i.e., a terminator sequence prevents activation of downstream modules by upstream promoters. A terminator or termination signal can include the DNA sequences involved in specific termination of an RNA transcript by an RNA polymerase. Thus, in certain embodiments a terminator that ends the production of an RNA transcript is contemplated. A terminator can be necessary for use in vivo to achieve desirable message levels.

The most commonly used type of terminator is a forward terminator. When placed downstream of a nucleic acid sequence that is usually transcribed, a forward transcriptional terminator will cause transcription to abort. In some embodiments, bidirectional transcriptional terminators are provided. Such terminators will usually cause transcription to terminate on both the forward and reverse strand. Finally, in some embodiments, reverse transcriptional terminators can be used to terminate transcription on the reverse strand only.

In some embodiments, a nucleic acid construct provided herein comprises additional regulatory elements that increase or decrease transgene expression in a cell or cell-free system depending on the absence or presence of a particular inducer or set of inducers. In some embodiments, the regulatory element is an enhancer. Additional non-limiting examples of regulatory elements include: lasR

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activator (e.g., from *P. aeruginosa*), cinR activator, toxicity-gene activator (e.g., ToxR, from *Vibrio cholerae*), lacI (e.g., a wild-type, derivative, or variant thereof), lacI repressor, tetracycline repressor (e.g., TetR from transposon Tn10), mnt repressor, TP901, heat shock proteins, and any derivative or variant thereof.

Thus, in some embodiments, a nucleic acid construct provided herein can be part of a synthetic gene network. Synthetic gene networks can include an engineered composition that comprises at least one nucleic acid construct provided herein and can perform a function including, but not limited to, sensing the presence or absence of an analyte or inducer, a logic function, or a regulatory function. In some embodiments of a synthetic gene network comprising at least two nucleic acid constructs, the nucleic acid constructs can interact with each other directly or indirectly. A synthetic gene network can comprise a nucleic acid encoding a transgene operably linked to a modified inducible promoter provided herein.

The nucleic acid constructs provided herein can be used to visualize chemical, analyte, or protein production in the presence and absence of an inducer provided herein. In some embodiments, a nucleic acid construct provided herein further comprises a reporter gene. In some embodiments, the reporter gene is mCardinal or a green fluorescent protein (e.g., superfolder GFP or sfGFP). A reporter gene encoding any fluorescent protein can be applicable to the nucleic acid constructs and methods of use provided herein. Additional examples of genes encoding fluorescent proteins that can be used in accordance with the compositions and methods described herein include, without limitation, enhance yellow fluorescent protein (EYFP), engineered cyan fluorescent protein (ECFP), mOrange, mCherry, Venus YFP, Cerulean, mBanana, orange fluorescent protein (OFP), derivatives, or variants thereof.

In some embodiments, the reporter gene encodes for a colorimetric protein enzyme. Colorimetric enzymes can cleave a substrate (e.g., a chemical) to yield a color-changing product. In some embodiments, the protein tag is chitinase (which cleaves colorless 4-Nitrophenyl N,N'-diacetyl-beta-D-chitobioside substrate to yield a yellow p-nitrophenol product). In some embodiments, the reporter gene is LacZ (which encodes beta-galactosidase) or a fragment thereof. When LacZ is expressed, the enzyme cleaves the yellow chlorophenol Red-beta-D-galactopyranoside (CPRG) substrate to produce the purple chlorophenol red product.

In some embodiments, the reporter gene comprises a catalytic nucleic acid. Examples of catalytic nucleic acids include, but are not limited to, a ribozyme, an RNA-cleaving deoxyribozyme, a group I ribozyme, RNase P, a Hepatitis delta ribozyme, and DNA-zymes.

In some embodiments, the reporter gene comprises an antigen for which a specific antibody or antibody fragment is available. In some embodiments, a reporter gene comprises an antibody, which when expressed, binds to a complementary antigen.

Methods of Expression

Provided herein are methods for expressing a transgene or a protein encoded by a transgene sequence using a nucleic acid construct provided herein having a modified inducible promoter as described herein. In some embodiments, a transgene can be expressed in a host cell that is toxic to the host cell or a non-naturally occurring organism (e.g., a bacterium). Such transgenes can be difficult to express utilizing a pET or other vector containing a leaky promoter, because the leaky expression upon transformation can

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inhibit growth of the host cell and thereby either prevent expression or express the transgene as an inclusion body. Indeed, tight OFF state control can allow the host cell to reach mid-log phase growth prior to induction, thus allowing for improved expression of the toxic transgene in the prokaryotic cell relative to expression via a vector having a leaky promoter.

The nucleic acid constructs provided herein can comprise a transgene encoding a protein or a fragment thereof. In some embodiments, the transgene or a protein encoded by the transgene is not toxic to the host cell. A protein and/or peptide or fragment thereof can be any protein of interest, for example, but not limited to: biological proteins; mutated proteins; therapeutic proteins; truncated proteins, and the like. Proteins can also be selected from a group comprising: mutated proteins, genetically engineered proteins, peptides, synthetic peptides, recombinant proteins, chimeric proteins, antibodies, midibodies, tribodies, humanized proteins, humanized antibodies, chimeric antibodies, modified proteins and fragments thereof. A transgene provided herein can comprise a gene that is, for example, part of a biosynthesis pathway for the production of a protein, an organic compound, or a molecule of interest. The cells, systems, and organisms provided herein provide for a facile method of manufacturing such proteins.

The modified inducible promoters provided herein can be introduced to any cell type or any system that can transcribe and translate the protein encoded by the transgene downstream of the promoter. In some embodiments, a cell provided herein is a prokaryotic cell or a eukaryotic cell. In some embodiments, the system is a cell-free system that can be used to produce the protein or organic compound of interest. A cell-free system is a composition comprising a set of reagents capable of providing for or supporting a biosynthetic reaction (e.g., transcription reaction, translation reaction, or both) in vitro in the absence of cells. For example, to provide for a transcription reaction, a cell-free system comprises promoter-containing DNA, RNA polymerase, ribonucleotides, and a buffer system. Cell-free systems can be prepared using enzymes, coenzymes, and other subcellular components either isolated or purified from eukaryotic or prokaryotic cells, including recombinant cells, or prepared as extracts or fractions of such cells. A cell-free system can be derived from a variety of sources, including, but not limited to, eukaryotic and prokaryotic cells, such as bacteria including, but not limited to, *E. coli*, *P. putida*, *V. natriegens*, thermophilic bacteria and the like, wheat germ, rabbit reticulocytes, mouse L cells, Ehrlich's ascitic cancer cells, HeLa cells, CHO cells, or budding yeast. In some embodiments, the cell-free system comprises an RNA polymerase. In some embodiments, the cell-free system comprises components sufficient for the translation reaction. In some embodiments, the cell-free system comprise ribosomes, aminoacyl transfer RNAs, translation factors, and a buffer system. The components can also comprise amino acids or amino acids and aminoacyl tRNA synthetases. Components of translation factors are disclosed, for example, in Shimizu and Ueda, "Pure Technology," Cell-Free Protein Production: Methods and Protocols, Methods in Molecular Biology, Endo et al. (Eds), Humana 2010, the contents of which is incorporated herein by reference in its entirety. Exemplary translation factors include, but are not limited to, factors responsible for protein biosynthesis are initiation factors (IF1, IF2, and IF3), elongation factors (EF-G, EF-Tu, and EF-Ts), and release factors (RF1, RF2, and RF3), as well as RRF for termination.

Prior to induction, the modified inducible promoter provided herein does not substantially express the transgene in the absence of an inducer (e.g. isopropyl β -D-1-thiogalactopyranoside (IPTG), anhydrotetracycline, and the like). In some embodiments, the amount of expressed transgene produced in the absence of inducer is less than 10%, less than 9%, less than 8%, less than 7%, less than 6%, less than 5%, less than 4%, less than 3%, less than 2%, less than 1%, less than 0.9%, less than 0.8%, less than 0.7%, less than 0.6%, less than 0.5%, less than 0.4%, less than 0.3%, less than 0.2%, less than 0.1%, less than 0.09%, less than 0.08%, less than 0.07%, less than 0.06%, less than 0.05%, less than 0.04%, less than 0.03%, less than 0.02%, or less than 0.01% of an amount of expressed transgene produced in the presence of the inducer. In some embodiments, the amount of expressed transgene produced in the absence of inducer is less than 10%, less than 9%, less than 8%, less than 7%, less than 6%, less than 5%, less than 4%, less than 3%, less than 2%, less than 1%, less than 0.9%, less than 0.8%, less than 0.7%, less than 0.6%, less than 0.5%, less than 0.4%, less than 0.3%, less than 0.2%, less than 0.1%, less than 0.09%, less than 0.08%, less than 0.07%, less than 0.06%, less than 0.05%, less than 0.04%, less than 0.03%, less than 0.02%, or less than 0.01% of an amount of expressed transgene produced by a comparable nucleic acid having a weak promoter (e.g. lacI) promoter in the absence of the inducer.

Expression of the transgene is performed by contacting the host cell or cell-free system provided herein with an inducer (e.g. isopropyl β -D-1-thiogalactopyranoside (IPTG), anhydrotetracycline, and the like). Non-limiting examples of inducers include: a chemical, a compound, and organic compound, a protein, an analyte, tetracycline and derivatives thereof, metallothionein, ecdysone, cocaine, hormones, steroids, and antibiotics (e.g., rapamycin, kanamycin). Exemplary environmental inducers include exposure to heat (i.e., thermal pulses or constant heat exposure), light (e.g., photoirradiation within the defined range of wavelengths), various steroidal compounds, divalent cations (including Cu^{2+} and Zn^{2+}), galactose, tetracycline, IPTG (isopropyl- β -D-thiogalactoside), as well as other naturally occurring and synthetic inducing agents and gratuitous inducers.

In some embodiments, the amount of expressed transgene produced in the presence of inducer is at least equal to the amount of expressed transgene produced by a comparable nucleic acid having a lacUV5 promoter and a T7 polymerase in the presence of the inducer. In some embodiments, the amount of protein expressed in the presence of the inducer is at least 70%, at least 71%, at least 72%, at least 73%, at least 74%, at least 75%, at least 76%, at least 77%, at least 78%, at least 79%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, at least 100%, at least 105%, at least 110%, at least 115%, at least 120%, at least 125%, at least 130%, at least 135%, at least 140%, at least 145%, at least 150%, at least 155%, at least 160%, at least 165%, at least 170%, at least 175%, at least 180%, at least 185%, at least 190%, at least 195%, at least 200%, at least 300%, at least 400%, at least 500%, at least 600%, at least 700%, at least 800%, at least 900%, or at least 1000% an amount of expressed transgene produced by a comparable nucleic acid having a strong (e.g., T7) promoter in the presence of the inducer.

Such expression can be carried in a variety of prokaryotic host cells. Indeed, while most expression is carried out in T7 lysogenized *E. coli* DE3 strains, the nucleic acid constructs

of the present disclosure having a modified inducible promoter can be utilized in prokaryotic cells in the absence of T7 lysogenization.

The host cell for expression of a nucleic acid provided herein (e.g., a transgene) can comprise a prokaryotic cell or a eukaryotic cell. In some embodiments, the prokaryotic cell is a bacterial cell. In some embodiments, a prokaryotic cell can include any Gram-positive strain bacterial cell. In some embodiments, a prokaryotic cell can include any Gram-negative strain bacterial cell. Examples of bacterium that can be used include but are not limited to: non-T7 lysogenized *E. coli* strains such as DH5 α , DH10 β , or W3110, *P. putida*, *P. aeruginosa*, *H. influenzae*, *C. trachomatis*, *P. mirabilis*, *P. vulgaris*, *C. pneumoniae*, *K. pneumoniae*, *N. gonorrhoeae*, *H. pylori*, *A. cholera*, *S. aureus*, *S. enterica*, *C. jejuni*, *B. fragilis*, *L. pneumophila*, *V. parahaemolyticus*, and *V. natriegens*.

The host cell for expression can be a eukaryotic cells, e.g., a mammalian cell, an insect cell, a yeast cell, a fungal cell, and the like. A nucleic acid construct provided herein can be regulated in a cell-specific or tissue-specific manner such that it is only active in transcribing the associated coding region of a given transgene in a specific tissue type(s). Exemplary Biosynthesis Pathways for Producing Gene Products

Provided herein are methods of producing a protein, an organic compound, or a molecule using the inducible promoters provided herein, a cell expressing an inducible promoter provided herein, a non-naturally occurring organism provided herein, or kits provided herein. In some embodiments, the non-naturally occurring organism provided herein comprises a nucleic acid construct comprising one or more inducible promoters provided herein and one or more sequence encoding a transgene. In some embodiments, the one or more transgene comprises a sequence encoding a protein in a biosynthesis pathway. In some embodiments, the nucleic acid construct provided herein comprise a biosynthesis pathway transgene. In some embodiments, the biosynthesis pathway is a polyketide synthesis pathway, a terpene synthesis pathway, a non-ribosomal peptide biosynthesis pathway, or a carotenoid biosynthesis pathway. In some embodiments, the cell expressing an inducible promoter provided herein or the a non-naturally occurring organism provided herein are cultured for a period of time. In some embodiments, the period of time is for at least 4 hours, 6 hours, 10 hours, 12 hours, or more. In some embodiments, the period of time is for at least 24 hours. In some embodiments, the period of time is for at least 96 hours.

In some embodiments, the biosynthesis pathway is a carotenoid synthesis pathway. Industrially useful carotenoids are generally produced by chemical synthesis processes for which possibility of undesired actions such as contamination of synthesis auxiliary materials is a major concern for the quality of the product. In addition, tastes of consumers tend to lean toward naturally-occurring carotenoids. However, there is a limit to extraction of carotenoids from plants and natural products, and an effective industrial process is not entirely established. As a production method of naturally-occurring carotenoids, microbial fermentation methods have been used. However, none of such cases enable production of carotenoids in an amount which is enough for economical industrial production. In many cases, through classical mutation and breeding, wild-type of carotenoid producing microorganisms do not generate enough carotenoid product for large-scale manufacturing. carotenoid biosynthesis pathway is made up of various enzymes,

and genes encoding such enzymes have been analyzed by many researches. In a typical pathway, for example, carotenoid is synthesized in its early stage by an isoprenoid biosynthesis pathway which is shared by steroid and terpenoid, starting from mevalonic acid which is a basic metabolite. Farnesyl pyrophosphate having 15 carbons (C15) generating through the isoprenoid basic synthesis system is condensed with isopentenyl diphosphate (IPP) (C5), to give geranylgeranyl diphosphate (GGPP) (C20). Then through condensation of two molecules of GGPP, colorless phytoene which is the first carotenoid is synthesized. The phytoene is then converted into lycopene through a series of unsaturation reactions, and then the lycopene is converted into β -carotene through a cyclization reaction. Then, a hydroxyl group and a keto group are introduced into the (3-carotene, which leads synthesis of various xanthophylls represented by astaxanthin.

Provided herein are methods for producing a carotenoid, the method comprises culturing the non-naturally occurring organism provided herein or a plurality of non-naturally occurring organisms under conditions and for a sufficient period of time and contacting the non-naturally occurring organism with an inducer, thereby producing the carotenoid. In some embodiments, the cells or non-naturally occurring organisms provided herein are cultured in a cell culture medium (e.g., a lysogeny broth, also called LB broth or Luria Broth). In some embodiments, the cell or non-naturally occurring organisms are cultured in a bioreactor, a spinning flask, or a vessel suitable for cell growth and survival.

In some embodiments, the transgene encodes a polypeptide having such an enzymatic activity that converts a methylene group at 4 position in β -ionone ring into a keto group. In some embodiments the transgene comprises a crtW gene. In some embodiments, the transgene encodes a polypeptide having such an enzymatic activity that adds one hydroxyl group to a carbon at 3-position of 4-keto- β -ionone ring and/or at 3-position of β -ionone ring. In some embodiments, the transgene comprises a crtZ gene sequence. In some embodiments, the transgene encodes for a polypeptide having such an enzymatic activity that converts lycopene into β -carotene. In some embodiments, the transgene comprises a crtY gene sequence. In some embodiments, the transgene encodes for a polypeptide having such an enzymatic activity that converts phytoene into lycopene. In some embodiments, the transgene comprises a crtI gene sequence. In some embodiments, the transgene encodes for a polypeptide having prephytoene synthase activity. In some embodiments, the transgene comprises a crtB gene sequence. In some embodiments, the transgene encodes for a polypeptide having geranylgeranyl diphosphate synthase activity. In some embodiments, the transgene comprises a crtE gene sequence. In some embodiments, the transgene encodes for a polypeptide having a lycopene elongase/hydratase activity. In some embodiments, the transgene comprises a crtEB gene sequence. In some embodiments, the transgene comprises a crtEBI gene sequence. In some embodiments, the transgene comprises a crtEBIY gene sequence. In some embodiments, the transgene comprises a crtEBIYZ gene sequence. In some embodiments, the transgene comprises a crtEBI-YZW gene sequence. In some embodiments, the transgene comprises an ABA1 gene sequence. In some embodiments, the transgene comprises an ABA2 gene sequence. In some embodiments, the transgene comprises a sequence that is at least 85% identical to any one of SEQ ID NOS: 69-73, 76-79. In some embodiments, the transgene comprises a sequence that is at least 90% identical to any one of SEQ ID NOS: 69-73, 76-79. In some

embodiments, the transgene comprises a sequence that is at least 95% identical to any one of SEQ ID NOS: 69-73, 76-79. In some embodiments, the transgene comprises a sequence that is at least 99% identical to any one of SEQ ID NOS: 69-73, 76-79. In some embodiments, the transgene comprises any one of SEQ ID NOS: 69-73, 76-79.

Further provided herein are methods of producing benzoic acid. In some embodiments, the biosynthetic pathway gene is a CocE gene. In some embodiments, the transgene comprises a sequence that is at least 85% identical to SEQ ID NO: 74. In some embodiments, the transgene comprises a sequence that is at least 90% identical to SEQ ID NO: 74. In some embodiments, the transgene comprises a sequence that is at least 95% identical to SEQ ID NO: 74. In some embodiments, the transgene comprises a sequence that is at least 99% identical to SEQ ID NO: 74. In some embodiments, the transgene comprises SEQ ID NO: 74.

Exemplary Embodiments:

Provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TXTXXTGT sequence at position -10, relative to a transcriptional start site of the promoter; (b) a TXGXCX sequence at position -35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence, wherein X is any nucleobase. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TATAAT sequence at position -10, relative to a transcriptional start site of the promoter; (b) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprises a transgene. Further provided herein are nucleic acid constructs, wherein the modified inducible promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is inserted into a prokaryotic cell in the presence of an inducer; and wherein: (i) the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct; and (ii) the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise: a modified inducible promoter, wherein the modified inducible promoter comprises: (a) a TATAATGT-sequence at position -10, relative to a transcriptional start site of the promoter; (b) a TTGACA-sequence at position -35, relative to a transcriptional start site of the promoter; and (c) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise a transgene. Further provided herein are nucleic acid constructs, wherein the transgene is selected from the group consisting of: a crtW gene, a crtE gene, a crtY gene, a crtI gene, a crtZ gene, a crtEB gene, a crtEBI gene, a crtEBIY gene, a crtEBIYZ gene, a crtEBI-YZW gene, an ABA1 gene, an ABA2 gene, and a CocE gene. Further provided herein are nucleic acid constructs, wherein the modified inducible promoter, when operatively linked to a transgene, facilitates

expression of the transgene when the nucleic acid construct is inserted into a cell in the presence of an inducer. Further provided herein are nucleic acid constructs, wherein the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein a strong promoter increases the amount of expression of a transgene provided herein relative to a comparable inducible promoter that does not comprise TATAAT, TTGACA, TATAATGT, TTXXTGT, or TXGXCX. Further provided herein are nucleic acid constructs, wherein the strong promoter increases the amount of expression of a transgene provided herein by at least 10% relative to the expression of a transgene expressed by a comparable inducible promoter that does not comprise TATAAT, TTGACA, TATAATGT, TTXXTGT, or TXGXCX. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise a sequence encoding a reporter gene. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise one or more regulatory element. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise a terminator sequence.

Provided herein are nucleic acid constructs comprising: a sequence that is at least 85% identical to a sequence comprising any one of: SEQ ID NOS: 1-14, 39-73. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise a sequence that is at least 90% identical to a sequence comprising any one of: SEQ ID NOS: 1-14, 39-73. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise a sequence that is at least 95% identical to a sequence comprising any one of: SEQ ID NOS: 1-14, 39-73. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise a sequence that is at least 99% identical to a sequence comprising any one of: SEQ ID NOS: 1-14, 39-73. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs comprise a sequence comprising any one of: SEQ ID NOS: 1-14, 39-73.

Provided herein are nucleic acid constructs comprising: (a) a first modified inducible promoter sequence provided herein; and (b) a second modified inducible promoter sequence, wherein the second modified inducible promoter sequence comprises: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the second modified inducible promoter sequence; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the second modified inducible promoter sequence; and (iii) a nucleic acid sequence that encodes a second bacterial ribosome binding sequence. Further provided herein are nucleic acid constructs, wherein the nucleic acid construct further comprises a transgene. Further provided herein are nucleic acid constructs, wherein the modified inducible promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is in the presence of an inducer. Further provided herein are nucleic acid constructs, wherein the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a

comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise a terminator sequence. Further provided herein are nucleic acid constructs, wherein the nucleic acid constructs further comprise one or more regulatory elements. Further provided herein are nucleic acid constructs, wherein the transgene comprises a carotenoid biosynthesis pathway gene. Further provided herein are nucleic acid constructs, wherein the transgene is selected from the group consisting of: a crtW gene, a crtE gene, a crtY gene, a crtI gene, a crtZ gene, a crtEB gene, a crtEBI gene, a crtEBIY gene, a crtEBIYZ gene, a crtEBI-YZW gene, an ABA1 gene, an ABA2 gene, and a CocE gene.

Provided herein are compositions, wherein the compositions comprise two or more nucleic acid constructs provided herein.

Provided herein are cells comprising a nucleic acid construct or a composition provided herein. Further provided herein are cells, wherein the cells are prokaryotic cells. Further provided herein are cells, wherein the cells are bacterial cells. Further provided herein are cells, wherein the cells are Gram-negative bacterial cells. Further provided herein are cells, wherein the cells are *P. putida* bacterial cells or *V. natriegens* bacterial cells. Further provided herein are cells, wherein the cells are eukaryotic cells. Further provided herein are cells, wherein the cells are mammalian cells.

Provided herein are cell-free systems comprising a nucleic acid construct provided herein, a composition provided herein and/or an RNA polymerase. Further provided herein are cell-free systems, wherein the cell-free systems further comprise ribosomes, aminoacyl transfer RNAs, translation factors, and a buffer.

Provided herein are methods of expressing a protein encoded by a transgene in a cell, the methods comprising: (a) transforming a cell with a nucleic acid construct, wherein the nucleic acid construct comprises a modified inducible promoter that comprises: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; (iii) a nucleic acid that encodes a bacterial ribosome binding sequence; and a transgene; and then (b) contacting the cell with an inducer, thereby expressing the protein encoded by the transgene. Further provided herein are methods, wherein the transgene when cloned into a comparable nucleic acid construct that comprises a strong promoter inhibits growth of the cell prior to the contacting with the inducer, thereby preventing expression of the transgene via the comparable nucleic acid. Further provided herein are methods, wherein the cell is a prokaryotic cell. Further provided herein are methods, wherein the prokaryotic cell is a bacterium. Further provided herein are methods, wherein the cell does not comprise a T7 promoter or a T7 polymerase. Further provided herein are methods, wherein when in the absence of the inducer, the expression of the protein encoded by the transgene is lower relative to the expression of the protein in a comparable cell comprising the transgene operably linked to a T7 promoter. Further provided herein are methods, wherein when in the presence of the inducer the expression of the protein encoded by the transgene is greater relative to the expression of the protein in a comparable cell comprising the transgene operably linked to a T7 promoter.

Provided herein are non-naturally occurring organisms comprising a nucleic acid construct comprising: (a) a first inducible promoter sequence comprising: (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (iii) a nucleic acid sequence that encodes a bacterial ribosome binding sequence; (b) one or more of a biosynthesis pathway transgene; and (c) a second modified inducible promoter sequence comprising: (i) a TATAATGT consensus sequence at position -10, relative to a transcriptional start site of the promoter; (ii) a TTGACA sequence at position -35, relative to a transcriptional start site of the promoter; and (iii) a nucleic acid sequence that encodes a bacterial ribosome binding sequence. Further provided herein are non-naturally occurring organisms wherein the one or more biosynthesis pathway transgene is selected from the group consisting of: a crtW gene, a crtE gene, a crtY gene, a crtI gene, a crtZ gene, a crtEB gene, a crtEBI gene, a crtEBIY gene, a crtEBIYZ gene, a crtEBI-YZW gene, and a CocE gene. Further provided herein are non-naturally occurring organisms, wherein the non-naturally occurring organism comprises a bacterium. Further provided herein are non-naturally occurring organisms wherein the bacterium is a Gram-negative bacterium. Further provided herein are non-naturally occurring organisms, wherein the Gram-negative bacterium is a *P. putida* bacterium or a *V. natriegens* bacterium. Further provided herein are non-naturally occurring organisms, wherein the bacterium is a Gram-positive bacterium. Further provided herein are non-naturally occurring organisms, wherein the nucleic acid construct comprises a terminator sequence. Further provided herein are non-naturally occurring organisms, wherein the nucleic acid construct comprises one or more regulatory element. Further provided herein are non-naturally occurring organisms, wherein the nucleic acid construct comprises a reporter gene.

Provided herein are compositions, wherein the compositions comprise: the non-naturally occurring organism provided herein; and an inducer. Further provided herein are compositions, wherein the inducer comprises: anhydrotetracycline (aTc), isopropyl β -d-1-thiogalactopyranoside, cocaine, metallothionine, ecdysone, an antibiotic agent, galactose, a steroid, or a divalent cation.

Provided herein are methods for producing a carotenoid, the methods comprising: culturing the non-naturally occurring organism provided herein; and contacting the non-naturally occurring organism with an inducer, thereby producing the carotenoid. Further provided herein are methods, wherein the non-naturally occurring organism comprises a nucleic acid comprising a gene selected from the group consisting of: a crtW gene, a crtE gene, a crtY gene, a crtI gene, a crtZ gene, a crtEB gene, a crtEBI gene, a crtEBIY gene, a crtEBIYZ gene, and a crtEBI-YZW gene.

Provided herein is a carotenoid produced by the methods provided herein, wherein the carotenoid is lycopene, beta-carotene, zeaxanthin, canthaxanthin, or astaxanthine.

Provided herein are methods for producing benzoic acid, the methods comprising: culturing the cell provided herein; or the non-naturally occurring organism of provided herein; and contacting the cell or the non-naturally occurring organism with an inducer, thereby producing benzoic acid. Further provided herein are methods, wherein the cell or the non-naturally occurring organism comprises a nucleic acid comprising a CocE gene.

Provided herein are compositions comprising benzoic acid, wherein the benzoic acid is produced by the methods provided herein.

Provided herein are kits comprising the nucleic acid construct provided herein, packaging, buffers, and materials therefor.

Provided herein are kits comprising the cell provided herein, packaging, culture medium, buffers, and materials therefor.

Provided herein are kits comprising the non-naturally occurring organism provided herein, packaging, culture medium, buffers, and materials therefor.

Provided herein are nucleic acid constructs comprising a modified inducible promoter, wherein the modified inducible promoter comprises: a TATAAT consensus sequence at position -10, relative to a transcriptional start site of the promoter; a TTGACA sequence at the -35 position, relative to a transcriptional start site of the promoter; a bacterial ribosome binding sequence; wherein the modified inducible promoter, when operatively linked to a transgene, facilitates expression of the transgene when the nucleic acid construct is inserted into a cell in the presence of an inducer; and wherein: the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a weak promoter in a comparable nucleic acid construct; and the expression of the transgene in the presence of the inducer is at least equal to an amount of expression in the presence of the inducer of the transgene operatively coupled to the strong promoter in the comparable nucleic acid construct.

Provided herein are isolated prokaryotic cells comprising a nucleic acid construct provided herein.

Provided herein are methods of expressing a transgene toxic to a prokaryotic cell, the methods comprising: transforming a nucleic acid construct into the prokaryotic cell, wherein the nucleic acid construct comprises a modified inducible promoter comprises: a TATAAT consensus sequence at position -10, relative to a transcriptional start site of the promoter; a TTGACA sequence at the -35 position, relative to a transcriptional start site of the promoter; a bacterial ribosome binding sequence; and contacting the prokaryotic cell with an inducer, thereby expressing the transgene; wherein the transgene when cloned into a comparable nucleic acid construct that comprises a strong promoter inhibits growth of the prokaryotic cell prior to the contacting with the inducer, thereby preventing expression of the transgene via the comparable nucleic acid.

Provided herein are methods of expressing a transgene in a prokaryotic cell that does not comprise a T7 RNA polymerase, the methods comprising: transforming a nucleic acid construct into the prokaryotic cell, wherein the nucleic acid construct comprises a modified inducible promoter comprises: a TATAAT consensus sequence at position -10, relative to a transcriptional start site of the promoter; a TTGACA sequence at the -35 position, relative to a transcriptional start site of the promoter; a bacterial ribosome binding sequence; and contacting the prokaryotic cell with an inducer, thereby expressing the transgene; wherein the expression of the transgene in the absence of the inducer is less than an amount of expression in the absence of the inducer of the transgene operatively coupled to a T7 promoter in a comparable nucleic acid construct; and the expression of the transgene in the presence of the inducer is greater than an amount of expression in the presence of the inducer of the transgene operatively coupled to the T7 promoter in the comparable nucleic acid construct.

Provided herein are nucleic acids, nucleic acid constructs, compositions, cells, non-naturally occurring organisms, cell-free systems, kit, or method provided herein.

For a better understanding of the present disclosure and of its many advantages, the following examples are given by way of illustration and without limiting the scope of this disclosure.

EXAMPLES

Example 1: Inducible Promotor Architecture

In exemplary embodiments, the -10 and -35 boxes of the original lac promoter (FIG. 1Ai), which are targeted by sigma 70 (σ^{70}), were modified to generate improved variants such as the lacUV5 (FIG. 1Aii) and tacI (FIG. 1Aiii) promoters. In the lacUV5 promoter, the -10 box of the original lac promoter was replaced by the Pribnow box (TATAAT). Later the lacUV5 and trp promoters were combined to create the tacI promoter, which increased transcription 11-fold compared to its predecessor with the incorporation of the -35 consensus box TTGACA. The tet promoter (FIG. 1Bi) shares the highly conserved -35 hexamer with tac promoter, however it does not contain the Pribnow box.

Provided herein are exemplary lac and tet expression systems with improved strength, control, and portability.

The genetic architecture of the lac and tet expression systems were modified in three ways:

- (1) addition of the consensus -10 and -35 sequence boxes to be strongly targeted by σ^{70} ;
- (2) incorporation of a strong ribosome binding site recognized by a broad spectrum of Gram-negative bacteria; and
- (3) independent control of the transcriptional regulators by appropriately-tuned constitutive promoters.

The results were validated with the reporter protein mCardinal, which significantly improves the dynamic range of promoter measurements over more commonly used green fluorescent proteins. The improvement seen in the mCardinal dynamic range is due to intrinsic fluorescence of many bacterial species that interferes with measurements in the green wavelengths, and the bacteria provided herein have reduced autofluorescence in the far red spectrum. Addition-

ally, the inducible promoters provided herein were compared with the pET system with the production of the cocaine esterase CocE, a thermosensitive enzyme capable of metabolizing cocaine into benzoic acid. CocE, which is prone to form inclusion bodies in leaky *E. coli* expression systems, is expressed as an inducible and soluble protein using the promoters provided herein. The results and assays provided in Example 2 further support that the expression system provided herein is a significant improvement over available expression systems in providing tight OFF state control while achieving high yields of recombinant protein, and with the advantage of direct portability to alternative host species.

Example 2: Generation of Inducible Promoters in Gram Negative Bacteria

Bacterial Strains

E. coli DH10B, *P. putida* JE90 derivative of KT2440 with BxB1int-attB, and *V. natriegens* Vmax X2 (Codex DNA, Inc.) were used to evaluate the synthetic lac and tet promoters. The plasmid pET21a-mCardinal was evaluated in *E. coli* BL21 strain. Selective markers kanamycin (50 $\mu\text{g/mL}$ for *E. coli* and *P. putida*, and 400 $\mu\text{g/mL}$ for *V. natriegens*), ampicillin (100 $\mu\text{g/mL}$ for *E. coli*) and spectinomycin (60 $\mu\text{g/mL}$ for *E. coli* and *P. putida*, and 250 $\mu\text{g/mL}$ for *V. natriegens*) were supplemented to LB medium when required. *E. coli* strains were transformed by electroporation and *V. natriegens* by chemical transformation. Integration of plasmids into *P. putida* chromosome was performed by electroporation following the protocol as described, for example, in Elmore et al. *Metab Eng Commun*, 5:1-8 (2017), the contents of which is incorporated herein by reference in its entirety.

Plasmid information is listed in Table 1 below. Any plasmid containing any version of the σ^{70} -based promoters as shown in FIG. 2A and FIG. 2B has been labeled “ σ^{70} ” followed by information about the construct and the specific promoter. “Lac” and “Tc” constructs are constitutive and do not express a repressor protein; “LacI” and “TcR” constructs are inducible and include expression of LacI or TetR, respectively. Numbers in parentheses indicate a vector backbone other than pJH0204 ((13) for pJH0228 with the pCloDF13 origin of replication, and (19) for pUC19).

TABLE 1

List of Plasmids and Vectors.				
Name	Backbone	Origin of replication	Selective marker	Construct
pJH0204	pJH0204	pColE1	Kanamycin	
pJH0228	pJH0228	pCloDF13	Spectinomycin/Streptomycin	
pJH0204F	pJH0204	pColE1	Kanamycin	
pFCtacI-GFP	pJH0204	pColE1	Kanamycin	tacI-sfGFP
pFCtacI-cardi	pJH0204	pColE1	Kanamycin	tacI-mCardinal
σ^{70} V1Lac-cardi	pJH0204	pColE1	Kanamycin	V1lac-mCardinal
σ^{70} V2Lac-cardi13	pJH0228	pCloDF13	Spectinomycin/Streptomycin	V2lac-mCardinal
σ^{70} V3Lac-cardi	pJH0204	pColE1	Kanamycin	V3lac-mCardinal
σ^{70} V4Lac-cardi	pJH0204	pColE1	Kanamycin	V4lac-mCardinal
σ^{70} V1LacI-cardi	pJH0204	pColE1	Kanamycin	V1lac/lacI-mCardinal
σ^{70} V2LacI-cardi	pJH0204	pColE1	Kanamycin	V2lac/lacI-mCardinal
σ^{70} V2Lac-cardi13	pJH0204	pCloDF13	Spectinomycin/Streptomycin	V2lac/lacI-mCardinal
σ^{70} V3LacI-cardi	pJH0204	pColE1	Kanamycin	V3lac/lacI-mCardinal
σ^{70} V4LacI-cardi	pJH0204	pColE1	Kanamycin	V4lac/lacI-mCardinal
σ^{70} V1Tc-cardi	pJH0204	pColE1	Kanamycin	V1Tc-mCardinal
σ^{70} V2Tc-cardi	pJH0204	pColE1	Kanamycin	V2Tc-mCardinal
σ^{70} V3Tc-cardi	pJH0204	pColE1	Kanamycin	V3Tc-mCardinal
σ^{70} V4Tc-cardi	pJH0204	pColE1	Kanamycin	V4Tc-mCardinal
σ^{70} V1TcR-cardi	pJH0204	pColE1	Kanamycin	V1Tc/tetR-mCardinal
σ^{70} V2TcR-cardi	pJH0204	pColE1	Kanamycin	V2Tc/tetR-mCardinal

TABLE 1-continued

List of Plasmids and Vectors.				
Name	Backbone	Origin of replication	Selective marker	Construct
po ⁷⁰ V3TcR-cardi	pJH0204	pColE1	Kanamycin	V3Tc/tetR-mCardinal
po ⁷⁰ V4TcR-cardi	pJH0204	pColE1	Kanamycin	V4Tc/tetR-mCardinal
po ⁷⁰ V1TcR-cardi19	pUC19	pMB1	Ampicillin	V2Tc/tetR-mCardinal
pET21a-cardi	pET21a	pET	Ampicillin	pET

Molecular Cloning:

A schematic diagram of plasmid construction are shown in FIGS. 1A-1B. Polymerase chain reactions (PCR) were carried out with Phusion DNA polymerase (ThermoFisher Scientific, USA). Digestion of DNA was performed with fast digest restriction enzymes and DNA fragments were join by T4 DNA ligase or Gibson assembly kit (ThermoFisher Scientific, USA). Oligonucleotides and DNA synthesis were ordered to IDT (IDT, USA). DNA sequencing was performed (QuintaraBio, USA). Shuttle vectors pJH0204 and pJH0228 were used. The synthetic Vlac and VTc promoters were synthesized as gBlocks and incorporated into the MCS of pJH0204 using BamHI and XhoI. Further, sfGFP and mCardinal ORF were codon optimized for *P. putida* and inserted downstream the synthetic promoters using NdeI and HindIII. The transcriptional regulator tetR gene was synthesized with the PJ23119 promoter in a single gBlock and tetR was codon optimized for *P. putida* using the IDT DNA codon optimization tool (available on the world wide web at

<https://www.idtdna.com/pages/tools/codon-optimization-tool>) and inserted after mCardinal with EcoRI and XhoI. The lacI gene was PCR amplified from pET21a vector and placed under the control of PJ23119 using NcoI and XhoI, but this combination resulted toxic for *E. coli*, therefore lacI was PCR amplified together with its native promoter and ligated after mCardinal using EcoRI and XhoI. pJH0228-V2lac-mCardinal and pJH0228-V2lac/lacI-mCardinal vectors were produced using the same cloning strategy. mCardinal was inserted into pET21a using NdeI and XhoI. The transcriptional unit V2Tc/tetR-mCardinal was inserted into the pUC19 vector by Gibson removing the NdeI RE nucleotide sequence in the plasmid and incorporating terminators insulating the transcriptional unit. DNA sequence of CocE was codon optimized for *P. putida* using the IDT DNA codon optimization tool and cloned into the pET21a and V2Tc-tetR expression system using NdeI and HindIII. A complete list of plasmids generated in this study is shown in Table 1. Primers used in this study are listed in Table 2.

TABLE 2

List of Primers.		
Primer name	SEQ ID NO:	Sequence 5' to 3'
pUC19 modified FW XhoI	15	attgcCTCGAGACTGATTTTTAAGCGGACTGATGAGTCGCCCTTTTTTTTGTCTAGCTAACTCACATTAATTGCGTTGCGCT
pUC19 modified FW XhoI	16	gcaatGGATCCagtcacaaagcctccgaccggaggcttttgactAGTACAATCTGCTCTGATGCCGCATAGTT
204F1_fwd	17	Ggcgcgtactcctcaaaaggatctaggtgaagatc
204F1_rev	18	gagttcttctgattagacaaaaaaaggcgac
204F2_fwd	19	tttttttgtctaatacagaagaactcgtcaagaagg
204F2_rev	20	gccccagcggcaggatctcgtcgtgacgcacg
204F3_fwd	21	cacgacgagatcctcgcgcgtcgggcatccgc
204F3_rev	22	gtctagactgcaggaattcaagcttcatatgggatccggacaaaacgAAA
204F4_fwd	23	aagcttgaattcctgcagctctagaccatggctcgaggacgaacaataaggc
204F4_rev	24	Cctagatccttttggagtacgcgccccggga
placI upstream	25	TACTGGTTTCACATTACCCAC
lacI downstream	26	GTGGTGAATGTGAAACAGTAA
lacI upstream	27	TTCCAGTCGGGAAACCT
lacI downstream2	28	AGGTTTCCCGACTGGAAA
tetR upstream	29	CCAATACAACGTGGGTGCT
FwplacI	30	AACCGGAATTCccccgtggccggg

TABLE 2-continued

List of Primers.		
Primer name	SEQ ID NO:	Sequence 5' to 3'
LXhoI RV	31	AACCGCTCGAGTCACTGCCCCGCTTTCC
Fw PstI-lacI	32	attgcCTGCAGTCACTGCCCCGCTTTCCAGTCG
Rv XbaI-lacI	33	gcaatTCTAGAAgtcaaaagcctccgacggaggettttgactCCCCGTGGC CGGGGGACTG
XbaI-V3Lac- NcoI fwd	34	gcctccggtcggaggettttgactt CTAGAGAGCTGTTGACACTTTATG
XbaI-V2Lac- NcoI rev	35	taatgagctcctcacccttactcacCATGGGTACATTCCTCTCTTACTGC
XbaI-V3Lac- NcoI_rev	36	taatgagctcctcacccttactcac
NcoI_rev	37	CATGGGATTCTGTTTCCTGTGTG
Rv cardi XhoI	38	TCAGTCTCGAGTTACTTGTGTCATCATCCTTAT

Primers were used as sequencing primers (e.g., placI upstream, tetR upstream). 204F primers were used to incorporate BamHI, NdeI, HindIII, KpnI, EcorI, PstI, XbaI, NcoI, XhoI RE into the MCS of pJH0204 and remove NcoI from kanamycin cassette. The pUC19 modified FW XhoI primers were used to incorporate terminators into pUC19 vector and BamHI/XhoI sites.

Sequences of cocE, tetR, and mCardinal, codon optimized for *P. putida*, and the PJ23119 promoter sequence, are provided below.

DNA sequence of mCardinal codon optimized for *P. putida*:

(SEQ ID NO: 39)

ATGGTGAGTAAGGGTGAAGAGCTCATTAAAGGAGAATGCACATGAAGC

TGTATATGGAGGGCACCGTAACAACCACTTCAAGTGTAACACCGA

GGGTGAAGGTAAACCTACGAGGGGACGAGACCAACGCATCAAGGTC

GTGGAGGGCGGCCGCTGCCTTTCGATTGACATTCTGGCGACCTGTT

TTATGTACGGCTCAAGACCTTCATCAACCACACCAAGGCATCCCGGA

CTTCTTCAAGCAGAGCTTCCTGAGGGCTTCACTGGGAGCGGTCACC

ACGTATGAAGACGGTGGGTGCTCACCGTGACCCAGGACACGAGCTTGC

AGGATGGCTGCTTGATTACAACGTCAAGCTGCGCGGGTGAACTTCCC

TAGCAACGGGCCAGTGATGCAGAAAAGACGCTGGGTGGGAGGCCACC

ACCGAGACCTGTACCGGCCGACGGGGGCTGGAAGGGCGGTGCGATA

TGGCCCTGAAATTGGTCGGCGCGGTCAATTGCACTGCAATCTCAAGAC

CACGTACCGCTCCAAGAAACCGCCAAAACTGAAGATGCTGCTGTT

TATTTTGTGACCGCGCTGAGCGCATCAAGGAAGCGGACAATGAGA

CGTACGTGGAACAGCAGCAAGTGGCCGTGGCTCGTTATTGCGATCTGCC

GTCGAAGCTGGGTACAACTGAACGGCATGGATGAGCTGTACAAGAT

TATAAGGATGATGACGACAAGTAA

DNA sequence of sfGFP codon optimized for *P. putida*:

(SEQ ID NO: 40)

ATGTCCAAAGGTGAAGAGCTGTTTACCGGCGTCGTGCCATTCTGGTGG

AGCTGGATGGCGACGTCAACGGGCACAAGTTAGCGTCCGTGGCGAAGG

TGAGGGCGACGCCACGAACGGTAAGCTGACGCTGAAATTCAATTGCACC

ACCGGCAAAATTGCCTGTACCTGGCCCCACCTGGTGACCACGCTCACCT

ACGGCGTACAGTGCTTACCGGTTACCGGACCACATGAAGCGTCACGA

CTTCTTCAAAAGCGCATGCCGAGGGTTACGTGCAGGAGCGTACGATT

AGTTTCAAGGACGACGGCACCTATAAGACCCGTGCCGAAGTGAAGTTGC

AAGGCGATACGTGGTGAACCGTATCGAGTTGAAGGGTATCGACTTTAA

GGAAGACGGCAACATCCTGGGCCATAAGCTGGAGTACAATTTCAACAGC

CATAACGTTTACATCACCGCGATAAACAAGAAGACGGCATTAAAGCCA

ACTTTAAGATCCGCCACAACGTCAAGACGGCTCGGTGCAGCTGGCCGA

CCATTATCAGCAAAACACCCCATCGGTGATGGGCCGTGCTGCTGCCG

GATAACCATTTCTGAGCACGAGTCGGTGCTCAGCAAGGACCTAACG

AAAAGCGGATCACATGGTGTGCTGGAGTTCGTACGGCGCGGGGAT

CACCCATGGGATGGACGAGCTCTACAAAGACTATAAAGATGACGATGAC

AAGTAAA

DNA sequence of tetR codon optimized for *P. putida*:

(SEQ ID NO: 41)

ATGTCCCGCTGGATAAATCGAAAGTGATTAACCTCGGCCCTCGAATTGC

TGAATGAAGTCGGTATCGAGGGGCTGACGACCCGTAATTTGGCACAAAA

GTTGGGGGTGGAGCAACCCACGTTGTATTGGCAGCTCAAAATAAGCGG

GCATTGCTGGATGCCCTCGCTATTGAAATGTTGGATGCCACCATACCC

ATTTCTGTCCACTGGAGGGCGAGTCTTGGCAGGACTTTCTCCGCAACAA

CGCGAAATCCTTTGCTGTGCACTCTTGTCCTCATCGGACGGTGCTAAG

GTGCACTTGGGCACCCGTCCACCGAAAAACAATACGAAACCTTGGA

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-continued

ATCAATTGGCGTTTTTGTGCCAGCAAGGGTTAGCTTGGAGAATGCTCT
 CTATGCGCTCTCGGCTGTCCGGACATTTACGTTGGGGTGCCTGTTGGAG
 GACCAGGAGCATCAAGTCGCAAAAGAGGAGCGTGAAACCCCAACCACGG
 ACTCGATGCCACCTCTGCTCCGCCAAGCTATCGAACTCTTCGATCATCA
 GGGCGCGGAGCCAGCCTTCTCTTTGGGCTGGAGCTGATTATCTGCGGT
 TTGGAAAAACAACCTCAAGTGTGAAAGCGGGTCCTAA

DNA sequence of cocE codon optimized for *P. putida*:

(SEQ ID NO: 42)

CATATGGTGGACGGTAATTATTCGCTAGCGTCCAACGTTATGGTGGCGATGCGCGACGGGGTGCCTTGGCTGTA
 GATCTGTACCGCCCGGACGCGAGATGGCCCTGTACCGGTCTGCTGGTCCGCAACCCCTACGACAAATTCGACGTG
 TTCGCTTGGAGTACGAGAGCAGCAACTGGCTGGAATTTGTGCGCGATGGGTACGCCGTCGTCATCCAAGACACC
 CGGGGCTCTTTGCATCCGAAGGTGAGTTCTGTTCCACATGTTGATGACGAGGCGGATGCGGAAGACACGCTGAGC
 TGGATCTTGAACAAGCATGGTGCAGCGCAATGTGGGTATGTTTCGGTGTAAGCTACCTGGGCGTTACGCAGTGG
 CAAGCTGCTGTAGCGGTGTGGGTGGTTTGAAGGCAATCGCCCCGAGCATGGCGAGCGCGGATCTGTACCGTGCC
 CCCTGGTACGGTCTTGGCGGCGCTTGAAGCAGTCTGGGCTGGAGCGCATTGATCGGTACGGGCTG
 ATTACAGCCGTAGCGATGCCGCCCGGAAGACGCGAGCCGACTTCGTACAGCTGGCAGCCATCCTGAACGATGTG
 GCCGGTGCAGCAAGCGTGACCCCTCTGGCCGAACAGCCCTTGTGGGCGCCTGATCCCTTGGGTGATCGACCAG
 GTGGTGGACCATCCAGACAACGACGAGTCGTGGCAGAGCATCTCGCTCTTTGAACGTTTGGGTGGGCTCGCTACC
 CCGGCTTGTATTACCGCCGGGTGGTACGATGGCTTCGTGGCGAGAGCCTCCGTACCTTCGTAGCTGTGAAGGAC
 AACCGGATGCGCGTCTGGTGGTGGGCGCTGGAGCCACAGCAATCTGACCGCCGTAATGCCGACCGTAAGTTT
 GGGATCGCCGCGACCTACCCCATCCAGGAGCGACGACCATGCACAAGGCTTTTTCGACCGGCACCTCCGTGGC
 GAGACCGATGCCCTGGCAGGGGTGCCAAGGTGCGCCTCTTCGTAATGGGTATCGATGAGTGGCGCGACGAGACC
 GACTGGCCATTGCCAGATACCGCTTACACGCTTTTACCTCGGGGCTCCGGTGCGCCAACACGAGCACGGGT
 GGTGGGACCTGTGACCTCGATCAGCGGCACGGAGTCGGCGGACACCTACCTGTATGATCCTGCCGACCCCGTG
 CCAAGTCTGGGCGGCACCTCCTCTTCCATAATGGGGACAACGGTCCAGCTGACCAGCGCCGATTACGATCGC
 GACGACGTGCTGTGCTACTCCACCGAGGTGTGACCGACCCCGTGAAGTAACGGGGACGGTTTCGGCTCGCCTG
 TTCGTGCTCTCGTCGGCGGTGATACCGATTTTACCGCAAGTTGGTGCAGCTGTTCCCGATGGTTCGGCAATC
 GCTCTCTGCGACGGCATCGTGCATGCGCTACCGGGAGACCTTGGTAACTCTACGCTCATTGAGGCCGGTGAG
 ATTTACGAGGTGGCTATTGATATGCTGGCCACAGCAACGTGTTTTTGCCGGGCCACCGCATCATGGTGCAAGTT
 AGCAGCTCGAACTTCCCGAAGTACGACCGCAACTCCAACACCGCGGCGTCATCGCTCGCGAGCAACTGGAGGAA
 ATGTGCACCGCCGTAAACCGCATTACCGCGGCCCGGAACCCGTCATATCGTGTGCGGATCATTAAAGCGC
 GACTATAAGGACGACGACGATAAGTGAAAGCTT

DNA sequence of pJ23119 promoter:

(SEQ ID NO: 43)

TTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGCCGAGTAAGAGAGG
 AATGTACAC

SDS-PAGE

SDS-PAGE was carried out on a 4-12% Bis-tris Midi Protein Gel in an XCell4 SureLock Midi system (Invitrogen, USA). *E. coli*, *P. putida* and *V. natriegens* cell extracts were obtained from diluting an overnight culture 1:50 in 10 mL of fresh LB media and grown at 37° C. for *E. coli* and 30° C. for *P. putida* and *V. natriegens* at 220 rpm until the culture

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reached an optical density (OD) at 600 nm (OD₆₀₀) of 0.7 measured spectrophotometrically, then induced with 0.2 mM IPTG or 0.1 µg/mL aTc accordingly. Induced cultures were grown for 5 hours, and 2 mL culture were spun down at 14,000 rpm and 4° C. for 10 minutes and frozen at -20° C. for further analysis. Cells were lysed with Complete Bacterial Protein Extraction Reagent (ThermoFisher Scientific), and crude extracts were analyzed by SDS-PAGE. Total protein concentration was estimated with Thermo Scientific NanoDrop one and ~10 mg total protein of each cell extract was loaded into the SDS-gel.

(SEQ ID NO: 42)

Expression of CocE and Measurement of Benzoic Acid Production

Recombinant *E. coli* BL21 strain carrying the pET21a-cocE and DH10B containing either pσ⁷⁰ V2TcR-cocE or pσ⁷⁰ V2TcR-cocE19 plasmids were cultivated overnight at 30° C. at 220 rpm and fresh cultures were started with 5% of the ON culture and induced at ~OD₆₀₀ 0.7 with 0.2 mM IPTG or 0.1 µg/mL aTc accordingly. 1 mL samples were collected each hour for 6 hours by centrifugation at 4° C. and 14000 rpm and stored at -80° C. Cell pellets were disrupted using 150 µL of B-PE Complete Bacterial Protein Extraction Reagent (ThermoFisher Scientific) for 25 minutes at room temperature and soluble fractions were collected after centrifugation for 25 minutes at 4° C. and 14000 rpm. ~3

mg/mL of soluble fractions were incubated with 0.015 mg of cocaine for 20 minutes at 28° C. and benzoic acid production was estimated using the benzoic acid detection kit for feed (Attogene, EZ2013-03).

Example 3: Design of Synthetic lac and tet Promoters

A schematic diagram of plasmid construction is shown in FIGS. 1A-1B. All plasmids used in the assay are listed in Table 1. The original lac promoter (FIG. 2Ai) lacks the -35 conserved box recognized by the σ^{70} subunit of RNA polymerase. Original lac and tet (FIG. 2Bi) promoters lack the Pribnow (-10) box. Three synthetic variants of each promoter were constructed containing the highly conserved σ^{70} consensus hexamers at positions -10 and -35. Version 1 (V1lac, FIG. 2Aiv, and V1Tc, FIG. 2Bii) of each promoter resembles the original lac and tet promoters, with the incorporation of a strong RBS from *P. putida*. The version 2 (V2lac, FIG. 2Av, and V2Tc, FIG. 2Biii) replaces the -35 and -10 boxes of Version 1 for the highly conserved σ^{70} consensus hexamers TTGACA and TATAAT, respectively. Version 3 (V3lac, FIG. 2Avi, and V3Tc, FIG. 2Biv) incorporates an additional lacO operator and the RBS of the tacI promoter. Version 4 (V4lac, FIG. 2Avii, and V4Tc, FIG. 2Bv) displaces the location of the -10 and -35 boxes (V4lac) and incorporates an additional -35 box in V4Tc. The synthetic promoters were cloned into the MCS of the shuttle vector pJH0204 containing the origin of replication pColE1 with 25-30 copies per cell in *E. coli*. The design allowed the incorporation of the transcriptional regulators lacI and tetR after a terminator L3S2P21 and under the control of the constitutive promoter PJ23119 (FIG. 2C).

No aberrant phenotypes in the *E. coli* strains carrying the PJ23119-tetR construct were observed. However, *E. coli* failed to maintain the PJ23119-lacI construct. The plasmids containing the lacI repressor gene in this configuration produced slow-growing colonies and yielded plasmids with aberrant restriction patterns, indicating gross plasmid rearrangements. Therefore, the strong PJ23119 promoter was replaced for the native lacI promoter and observed no further toxicity.

The reporter protein mCardinal was incorporated downstream of the synthetic promoters and the plasmids were transformed into *E. coli* DH10B, *P. putida* JE90 derivative KT2440, and *V. natriegens* Vmax X2 strains. The V2lac-mCardinal constructs could not be maintained in *E. coli*, likely due the high strength of the V2lac promoter. To avoid this abnormal phenotype, the genetic circuit V2lac-mCardinal was produced in the pJH0228 vector containing the CloDF13 origin of replication at about 10 copies per cell, allowing stable maintenance of the V2lac promoter.

Example 4: Selection of the Fluorescent Reporter System

GFP and other green fluorescent protein derivatives are widely used as reporters in bacterial expression systems. However, the intrinsic green fluorescence produced by endogenous molecules, such as proteins containing aromatic amino acids, negatively impact the interpretation of the exogenous green fluorescence generated by the reporter system. The autofluorescence is exacerbated in *P. putida*, a member of the fluorescent *Pseudomonas* species. Under iron limited conditions *P. putida* secretes the siderophore pyoverdine, a soluble fluorescent yellow green pigment. Far-red fluorescent proteins were explored, specifically mCardinal, a

monomeric far red shifted derivative of mKate, as wavelengths between 600 and 900 nm are not absorbed by cells and organic molecules, thus reducing the noise of endogenous autofluorescence.

To quantify the impact of autofluorescence on the selected Gram-negatives strains, the endogenous fluorescence of these strains cultivated in LB over time were measured. All three species emit fluorescence in the green spectrum, as expected (FIG. 3A, right panel). This fluorescence is more or less constant over time with *E. coli*, however *P. putida* and *V. natriegens* show variable production of molecules that absorb green light at different cell densities. In contrast, when measured in the far-red spectrum specific for mCardinal, there was approximately 400 times less detection of autofluorescence, in arbitrary units on the same instrument, in all strains as compared to the measurements in the green spectrum (FIG. 3A, left panel). Again, *E. coli* displayed rather consistent far red autofluorescence over time whereas *P. putida* and *V. natriegens* autofluorescence varies with cell density.

To further validate the benefit of using mCardinal instead of sfGFP, the inherent noise of each reporter system was quantified by measuring the fluorescence signal-to-background ratio of both reporter systems expressed under the control of the constitutive tacI promoter in the three species. After 16 hours of growth, the recombinant *E. coli*, *P. putida* and *V. natriegens* expressing sfGFP emitted 64, 11 and 5 times more green fluorescence than their respective wild type strains (not expressing a fluorescent protein). Meanwhile, mCardinal-expressing strains displayed 294, 34 and 13-fold higher red-light emission in *E. coli*, *P. putida* and *V. natriegens* compared to their wild type controls (FIGS. 3B-3D). Overall, the use of mCardinal rather than sfGFP significantly improves the dynamic range and facilitates measurement of promoter strength, as there is little endogenous autofluorescence in the far-red spectrum.

Example 5: Strength and Regulation of Synthetic lac Promoters in *Escherichia coli*

The lac promoter, and its derivatives, are constitutively active in the absence of its transcriptional regulator lacI. The strength of the synthetic lac promoters were evaluated without lacI and compared these against the strong tacI promoter, which drives high levels of transcription and can result in recombinant protein production of up to 30% of total protein. The V1Lac promoter was 10-fold weaker than the tacI promoter (FIG. 4A), consistent with previous data observing an 11-fold difference between these two promoters. The V3lac promoter matched the tacI promoter strength, while the V4lac promoter showed 5 times less mCardinal than tacI (FIG. 4A). The V2lac promoter had to be tested in the low copy plasmid pCloDF13 because the medium copy pColE1 derived plasmid could not be maintained in *E. coli*. Even in the low copy configuration, the V2Lac promoter surpassed by ~1.4-fold the tacI promoter and was therefore the strongest constitutive promoter despite its expression in a low copy plasmid. The result suggests that maintenance of a promoter of this strength within a medium copy plasmid exceeds the sustainable metabolic burden of *E. coli* (FIG. 4A). These results indicate that the incorporation of σ^{70} consensus sequences at positions -35 and -10 efficiently increase the strength of the lac promoter.

Next, the transcriptional regulator lacI was incorporated into these circuits to quantify the efficiency of the OFF state. The V1Lac and V4lac constructs containing the lacI repressor produced a red fluorescence signal comparable to wild

type *E. coli* (FIG. 4A), indicating tight transcriptional repression. The V3lac promoter could not be totally repressed by lacI, showing a slight tendency to leak, even with the presence of two lacO operators. The V2lac promoter with lacI was stably maintained by *E. coli* in both pColE1 and CloDF13-derived vectors. However, the repressed state of the V2lac promoter in the low and medium copy plasmids showed rather a constitutive behavior. LacI repressed ~9-fold the V2Lac promoter in the CloDF13 derived plasmid as compared to its constitutive construct (FIG. 4A), while the medium copy version of the V2lac-lacI system (which was not sustainable as a constitutive circuit) exhibited a strong constitutive expression ~1.3-fold higher than the tacI promoter, thus showing that one lacO operator region is not sufficient to block transcription of V2Lac containing both σ^{70} consensus sequences by LacI, but incorporation of a second lacO reduces the leakage as observed in the V3Lac promoter (FIG. 4A).

To verify the inducibility of the synthetic lac promoters, *E. coli* was exposed to IPTG at the beginning of exponential phase. The V1lac and V4lac promoters showed minimal induction of mCardinal (FIG. 4A). The V3lac promoter, which mimics the V2lac promoter with an additional lacO operator, demonstrated a dynamic range of ~17-fold versus the uninduced state, but could only produce 24% of its full constitutive potential (FIG. 4A). The V2lac promoter in the low copy plasmid produced ~2-fold more mCardinal upon induction; similar behavior was observed in the medium copy version, where production of mCardinal reached the maximum values of all promoters tested at ~2.6-fold stronger than the tacI promoter (FIG. 4A). Overall, the results demonstrate that incorporation of σ^{70} consensus sequences to the lac promoter improve its strength (V2Lac), producing more mCardinal than the strong tacI promoter in both, the constitutive LacI-less version in the low copy plasmid and the repressed and induced states in the medium copy plasmid. However, LacI could not repress the transcription of the synthetic V2Lac promoter. Thus, additional lacO operators are necessary to turn the lac promoter OFF containing the perfect σ^{70} boxes, as observed in V3Lac, though full induction then becomes impossible in this architecture.

Example 6: Strength and Regulation of Synthetic tet Promoters in *Escherichia coli*

The tet promoter also drives transcription constitutively without TetR. The expression profile of the constitutive synthetic tet promoters were evaluated and compared against the tacI promoter. As observed for the lac promoter, replacement of the -10 and -35 sequences with the σ^{70} consensus boxes in the tet promoter also increased the constitutive efficiency of the V2Tc promoter over the original V1Tc, in this case by 20%, and reached the yields obtained with tacI promoter (FIG. 4B). The V4Tc promoter showed similar results to the V2Tc, while the V3Tc lost any capability to initiate transcription (FIG. 4B). Further, incorporation of the tetR repressor to each of the four circuits under the control of the strong constitutive PJ23119 promoter completely silenced mCardinal production from all the synthetic tet promoters, even in the strong V2Tc promoter which harbors the optimal combination of consensus sequences recognized by the σ^{70} (FIG. 4B).

To confirm the induction of each promoter and determine its functional dynamic range, each circuit was induced with anhydrotetracycline (aTc). The original tet promoter (V1Tc) only achieved ~37% of its full constitutive potential, reaching only 30% of mCardinal compared to tacI promoter

confirming the middle level strength of the tet promoter (FIG. 4B). The V2Tc promoter achieved maximal induction reaching the full potential dynamic range and produced similar yields as the strong tacI promoter (FIG. 4B). Interestingly, the V4Tc promoter could not be de-repressed by aTc and showed a weak induction of only 5%, and as expected, the V3Tc promoter showed no activity (FIG. 4B). Together these results confirm that in *E. coli* the incorporation of the consensus boxes targeted by the σ^{70} is key to boost the expression profile of promoters recognized by this transcription initiation factor and achieve the maximal transcriptional capacity of these promoters upon induction.

Example 7: Direct Comparison of Synthetic lac and tet Expression Systems to pET in *Escherichia coli*

In *E. coli*, the pET series of expression plasmids are the most popular systems for recombinant protein production. The efficiency of the synthetic lac and tet promoters were compared directly against pET21a. The pET expression system in the BL21 strain can produce more than 50% of the target gene as the total protein per cell, mCardinal production was measured in BL21. As expected, the pET system induced by IPTG produced ~1.5-fold more mCardinal than the strong constitutive tacI promoter, which is expected to accumulate up to 30% of total cell protein. Despite the massive production of mCardinal by the pET system, its OFF or uninduced state can be considered as a medium high constitutive expression system, yielding 30% of the constitutive tacI promoter as evidenced by mCardinal production (FIG. 4B, FIG. 4C). This transcriptional leakage in the pET system is well known in the scientific community.

Among the synthetic lac and tet promoters, all tested in *E. coli* DH10B, only the V2Lac promoter surpassed the pET system in the medium copy plasmid pColE1. The IPTG induced V2Lac expression system produced ~1.8 times more recombinant protein than the pET system, however, the leakage of V2Lac promoter equals the induced state of the pET system (FIG. 4A and FIG. 8), therefore cancelling the advantages of an inducible expression system. The V3Lac expression system showed a tighter control of the OFF state, therefore, to increase its strength the V3Lac-lacI construct containing mCardinal was inserted into the pUC19 vector, thus increasing the copy number from 30 to 250 copies per cell. *E. coli* failed to maintain the V3lac-lacI construct in the pUC19 vector, likely due to the toxicity of LacI. The V2Tc-tetR expression system was evaluated in the pUC19 vector. *E. coli* stably maintained the V2Tc-tetR construct and no phenotypic or genotypic abnormalities were observed as in pUC19-V3Lac-lacI. mCardinal production by pUC19-V2Tc-tetR matched the pET21a expression system, with one exceptional difference, the pUC19-V2Tc-tetR maintained complete repression in the uninduced after 12 hours (FIG. 4B and FIG. 8). Thus, pUC19-V2Tc-tetR shows significant improvement in transcriptional control and dynamic range over pET21a (FIG. 4B, C). Further, equivalent protein production and tighter transcriptional control were achieved without the obligatory use of the BL21(DE3) strain, suggesting the system provided herein is readily portable to other strains and species.

Example 8: Expression of the Cocaine Esterase CocE and Production of Benzoic Acid

To further validate the advantage of the expression system provided herein over the pET expression system, the production of a functional protein product, the cocaine esterase

CocE, was assayed. This enzyme hydrolyses cocaine into benzoic acid and could expand the use of the narcotic compound as raw source in the production of the carboxylic acid widely used as precursor and preservative in the food and pharmaceutical industries. Expression of CocE has previously been shown to be a difficult and laborious task with the pET expression system, because CocE forms inclusion bodies; consequently, long incubation methods at low temperatures are required to isolate sufficient yields of functional CocE. *E. coli* could not support CocE expression in the repressor-less variants of the synthetic promoters, thus confirming this is a toxic protein for *E. coli*. Next, CocE in the V2Tc-tetR expression system was assessed in both the medium and high copy plasmid backbones, pJH0204 and pUC19 respectively (p σ 70 V2TcR-cocE and p σ 70 V2TcR-cocE, see Table 1), and compared against the pET expression system for the production of benzoic acid. Benzoic acid production is an indication that CocE was correctly folded and properly hydrolyzing cocaine.

The soluble fraction recovered from the recombinant *E. coli* strain containing the pET21a-cocE expression system showed equivalent benzoic acid production in the presence of cocaine in both the uninduced and induced states (FIG. 5B), with a clear tendency to form inclusion bodies, as evidenced by the accumulation of protein in the insoluble fraction upon activation by IPTG (FIG. 5A). Thus, while induction of CocE by IPTG in the pET system does result in the additional production of protein, the additional protein made is insoluble and non-functional protein. Functionally speaking, the expression of CocE in pET21 a is not inducible, but instead behaves constitutively, as there is no difference in cocaine hydrolysis in the uninduced and induced states (FIG. 5B).

The soluble fractions from the recombinant *E. coli* V2Tc-tetR-cocE expression system, in both the pJH0204 and pUC19 plasmids, showed no benzoic acid production in the absence of the inducer. The addition of aTc triggered production of CocE with protein production and function roughly equivalent to the soluble pET21a-cocE plasmid (FIG. 5A, FIG. 5B). In both plasmids, benzoic acid production was observed after only one hour of aTc induction, and reached maximal production by 3 hours (FIG. 5B). These results confirmed the advantage of the incorporation of the σ^{70} boxes to the tetracycline promoter, keeping the toxic enzyme CocE OFF in the uninduced state and producing sufficient amounts of mature CocE after 3 hours of induction. This is significantly shorter than induction of 16 hours previously recorded (data not shown). Further, the expression system provided herein produced soluble and functional protein without evidence of massive toxicity or inclusion body formation. This method facilitates the scalability of this bioprocess which has potential as an alternative, environmentally friendly method to obtain benzoic by replacing petroleum-based starting materials.

Example 9: Strength and Regulation of Synthetic lac and tet Promoters in *Pseudomonas putida*

Genetic control in the soil-dwelling species *P. putida* is of great interest for biotechnological applications due to the ability of this microorganism to synthesize complex natural products and metabolize a variety of xenobiotic compounds. The constitutive strength, repression, and inducibility of synthetic promoters in *P. putida* were assayed. The lac-based promoter systems (which include the tacI promoter) are reported to have poor dynamic range and instability in high-copy plasmids when harbored in *P. putida* as replica-

tive plasmids. To address this problem, the genetic circuits were integrated into the *P. putida* chromosome. The constitutive strength of V1lac, V2lac, V3lac and V4lac promoters were measured using mCardinal, and found out that tacI is stronger than all the synthetic lac variants by ~3, 1.5, 10 and 33-fold respectively (FIG. 6A, FIG. 6C). The incorporation of the transcriptional regulator lacI efficiently repressed the V1lac, V3lac and V4lac promoters, but as seen in *E. coli*, the V2lac showed a tendency to leak, though not as egregiously as in *E. coli* (FIG. 6A, FIG. 6C). Induction by IPTG was barely detected in the V1lac, V3lac and V4lac promoters, thus indicating the weakness of the original lac promoter in *P. putida* (FIG. 6A, FIG. 6C). Interestingly and analogously to *E. coli*, the V2lac promoter showed ~27-fold increase in mCardinal production against the uninduced state and surpassed its theoretical dynamic range as compared to the constitutive version, ultimately achieving similar yields as the strong tacI promoter (FIG. 6A, FIG. 6C). These results confirm that the incorporation of the σ^{70} consensus sequences to the lac promoter (V2Lac) significantly improve the strength of this promoter in the *P. putida* host.

The tet promoters in *P. putida* were also evaluated. The original tet promoter with a strong RBS (V1Tc) was the strongest among all synthetic tet promoters in the absence of tetR, followed by V4Tc, V2Tc and V3Tc (FIG. 6B, FIG. 6C). Incorporation of the tetR repressor under the control of the PJ23119 promoter silenced all the synthetic tet promoters, as in *E. coli* (FIG. 6B, 6C). Addition of aTc activated the four tet promoters and remarkably, the induced V2Tc promoter showed the highest mCardinal induction exceeding its constitutive expression by ~10-fold and matching the tacI promoter. The V2Tc promoter was revealed to be ~2 and 3-fold more efficient than the V1Tc and V4Tc promoters after induction respectively, while the V3Tc proved to be inefficient in *P. putida* (FIG. 6B, FIG. 6C). These results highlight the importance of the consensus sequences targeted by σ^{70} to amplify the strength of inducible promoters also in *P. putida*.

Example 10: Strength and Regulation of Synthetic lac and tet Promoters in *Vibrio natriegens*

The circuits were next evaluated in the marine bacterium *V. natriegens* which has gained popularity for routine molecular biology applications due its ability to double in <10 min. In this host, previous studies indicated that the tacI promoter produced GFP upon activation with IPTG, while induction of the tet promoter resulted in low GFP yields. The constitutive and inducible versions of the synthetic lac and tet promoters were evaluated in the pCoIE1 derived plasmid with ~300 copies per cell, except for the constitutive V2Lac, which was evaluated in pCloDF13 derived vector with ~64 copies per cell in this strain. The constitutive V2Lac and V4Tc promoters outperformed tacI by ~16 and ~2-fold respectively, while V3Lac, V1Tc and V2Tc produced similar levels as tacI (FIGS. 7A-7C). V1Lac and V3Lac underperformed tacI by ~2.7-fold, and V3Tc showed no activity (FIGS. 7A-7C). Incorporation of the transcriptional regulators lacI and tetR turned OFF all the synthetic promoters except for V2Lac, which continued to show leakage, as in *E. coli* and *P. putida* (FIG. 7A). Induction of the synthetic lac promoters by IPTG revealed that only the V2Lac promoter was fully activated, surpassing its constitutive version by ~1.3-fold and showed a potential dynamic range of ~25-fold (FIG. 7A). The V3Lac reached 30% of its full potential, and the V1Lac and V4Lac showed no induction (FIG. 7A). While addition of aTc activated all

four synthetic tet promoters, the V3Tc and V4Tc promoters rather showed weak mCardinal production, and V1Tc produced 2-fold more mCardinal than its constitutive variant (FIG. 7B). The V2Tc promoter, which harbors the 2 consensus σ^{70} boxes, demonstrated strong inducibility producing 11-fold more mCardinal than the *lacI* promoter and displaying full potential of its dynamic range producing 100-fold more mCardinal than its OFF version (FIG. 7B, C). These results confirmed that the adaptation of the two consensus boxes recognized by the σ^{70} in the V2lac and V2Tc promoters improved the performance of the inducible *lac* and *tet* promoters in *V. natriegens*.

Example 11: Additional Considerations for Inducible Promoter Design

Production of recombinant proteins is and will continue to be one of the main tools scientists use to understand biological processes and transfer academic results to industrial applications. The pET system is well known to induce the formation of inclusion bodies, a major drawback in the production of soluble proteins. pET requires specific strains that carry the T7 RNA polymerase, it lacks real tuneability, and fails to keep the target gene OFF. Leakiness in the OFF state negates the main advantage of an inducible system, which is intended to permit time- or context-specific control of gene expression.

In contrast to pET, the results provided in Examples 2-9 show that inclusion of the σ^{70} consensus sequences into the *lac* and *tet* inducible promoters improve both repression and induction. The tight transcriptional control does not require any particular strain background, and permits rapid expression of soluble proteins, including toxic proteins such as CocE. The inducible promoters provided herein can be easily optimized to be used in a variety of Gram-negative hosts, including *P. putida* and *V. natriegens*, thus widening the applicability of these tools to a broad spectrum of bacteria. The inducible promoters provided herein offer a significant advance to the biotechnology industry in offering additional platforms for exogenous protein expression and purification.

The V2lac promoter expressed very highly across three Gram-negative species. Indeed, in *V. natriegens* the V2lac promoter is the strongest promoter yet described, as it surpasses the widely used *lacI* promoter by 16-20-fold (FIG. 7). Additional factors can affect promoter performance, for instance, the length and sequence of the spacer between the -10 and -35 elements can have dramatic effects on promoter strength.

The results in FIG. 5 suggest that complete OFF state control is key to avoiding common problems with protein expression of challenging proteins, such as the formation of inclusion bodies. Future promoter configurations could be aided by the combination of additional mechanisms to achieve complete OFF state control, such as the additional of small RNA transcriptional and translational regulators. These approaches have increased appeal over the pET system with pLysE, for example, as they permit facile portability among strains and species. OFF state control of the *lac* promoters was less of an issue in *P. putida* and *V. natriegens*, where better repression of the strong V2lac promoter was achieved (FIGS. 6A-7C). Without being bound by a particular theory, it is likely that species-specific differences in RNA polymerase binding and promoter clearance may affect the behavior of inducible promoters.

The challenges of leakiness were entirely eliminated by the use of *tet*-based promoters in the system provided herein.

An often-cited drawback of *tet* promoters for protein expression was the limitation of protein production as compared to *lac*-based systems. Incorporation of the σ^{70} consensus sequence into the *tet* promoter (V2Tc) significantly increased expression above that of the original *tet* promoter (V1Tc) to the point where the promoter output equaled (and in the case of *V. natriegens*, exceeded) that of the *lacI* promoter. As a complementary approach to inducible promoter manufacturing and design, the sigma factor can be replaced with σ^{70} sequences to assess promoter performance. Promoter performance was quantified herein by using far-red reporters, which significantly decreased background fluorescence and increased dynamic range in the bacterial species assayed.

Different yields of recombinant protein can be achieved based on promoter and host selection. Overall, the incorporation of the consensus -35 sequence and Pribnow (-10) box unlocks the strength of the *lac* and *tet* promoters in Gram-negative bacteria, facilitating the production of any given target gene in different host with the same set of plasmids.

The V2Tc promoter provided herein offers an improvement over the pET-based protein expression systems that remain very widely used. V2Tc is tightly regulated, robustly inducible, and drives protein production comparable to or better than the *lacI* promoter in all three species that were examined. No specific strain backgrounds were necessary, as it does not rely on the presence of the T7 polymerase.

Example 12: Development of the Dual Expression System V2TcR/V2(3)LacI

The σ^{70} adapted expression systems V2TcR and V3LacI have tight regulation and strong induction in *E. coli* in the presence of anhydrotetracycline (aTc) and IPTG respectively, while in *P. putida* and *V. natriegens* the V2TcR and V2LacI showed the best performance as inducible promoters in these host bacteria. A dual expression system was developed using as backbone the pJH0204 vector containing the origin of replication pColE1 and the *attB* sites specific for *bxhI* recombinase. Each transcriptional unit of the dual expression system was insulated by terminators and the transcriptional regulators *tetR* and *lacI* were located in-between and in opposite direction of the inducible promoters V2Tc and V2/3Lac to block undesired transcription of the open reading frames (ORF). Further, 2 different multicloning sites were located after the inducible promoters V2Tc and V2/3Lac to facilitate the incorporation of target genes (FIGS. 9A-9F, and FIG. 14).

The dual expression systems were tested in *E. coli*, *P. putida* and *V. natriegens* to evaluate i) synergy ii) expression of different recombinant proteins and iii) production of the biosynthetic gene cluster (BGC) leading to the biosynthesis of lycopene and B-carotene. Synergy was evaluated with the reporter system mCardinal which was accommodated in both inducible promoters of the dual expression system (FIG. 9B). Expression of different recombinant proteins was performed by adapting the sfGFP under the control of V2TcR and mCardinal controlled by V2/3LacI (FIG. 9C). Finally, production of the terpenoids lycopene and B-carotene by the dual expression system was achieved by the assembly of the *crtEBIY* genes from *Pantoea ananatis*. The *crtEBIY* genes were synthesized with the codon usage for *E. coli*. The *crtEBI* operon was assembled in the pUC19 vector via Gibson adapting the ribosome binding site (RBS) of J23119 9 to *crtB* and the RBS pJL1 to *crtI*, further the synthetic construct was transferred to the V2TcR expression system to evaluate lycopene production (FIG. 9D). B-carotene

tene production was evaluated with the incorporation of the crtY gene with the J23119 RBS to V2TcR-crtEBI yielding V2TcR-crtEBIY (FIG. 9E). The dual control of the crtEBI operon together with the crtY gene to produce B-carotene was achieved by adapting the crtEBI to the V2TcR expression system and the crtY gene to V2(3)LacI expression system present in the dual vector (FIG. 9F). All measurements were performed using LB as growth medium as the scope of this work is not to optimize the production of terpenoids. After transforming the plasmids listed in Table 3 in the three bacterial species no difference in the wild type strains was observed, thereby confirming the stability of the dual expression systems.

TABLE 3

List of Plasmids.				
Name	Expression system 1	ORF1	Expression system 2	ORF2
p σ^{70} V2TcR-mCardinal	V2TcR	mCardinal		
p σ^{70} V2lacI-mCardinal	V2lacI	mCardinal		
p σ^{70} V3LacI-mCardinal	V3LacI	mCardinal		
p σ^{70} V2TcR-sfGFP	V2TcR	sfGFP		
p σ^{70} V2lacI-sfGFP	V2lacI	sfGFP		
p σ^{70} V3LacI-sfGFP	V3LacI	sfGFP		
pJH0204F				
p σ^{70} V2TcR-mCardinal/V2LacI-mCardinal	V2TcR	mCardinal	V2LacI	mCardinal
p σ^{70} V2TcR-mCardinal/V3LacI-mCardinal	V2TcR	mCardinal	V3LacI	mCardinal
p σ^{70} V2TcR-sfGFP/V2LacI-mCardinal	V2TcR	mCardinal	V2LacI	sfGFP
p σ^{70} V2TcR-sfGFP/V3LacI-mCardinal	V2TcR	mCardinal	V3LacI	sfGFP
pUC19-crtEBI	—	crtEBI		
p σ^{70} V2TcR-crtEBI	V2TcR	crtEBI		
p σ^{70} V2TcR-crtEBIY	V2TcR	crtEBIY		
p σ^{70} V2TcR-crtEBI/V2LacI-crtY	V2TcR	crtEBI	V2LacI	crtY
p σ^{70} V2TcR-crtEBI/V3LacI-crtY	V2TcR	crtEBI	V3LacI	crtY

Note:

SEQ ID NOS: are listed in the SEQUENCES section of this paper.

Example 13: Synergy of the Dual Expression System V2TcR/V2(3)LacI

Increasing the copy number in *E. coli* efficiently boosted production of mCardinal by the p σ^{70} V2TcR expression system but had a negative impact in the p σ^{70} V3LacI expression system due the toxicity of the transcriptional regulator LacI. Therefore, the synergistic effect of the dual expression system p σ^{70} V2TcR/V3LacI both controlling mCardinal as different studies demonstrated that combined promoters enhance production of the target protein. The *E. coli* strain containing the duet expression system V2TcR/V3LacI produced 2-fold and 4-fold less red fluorescence than the strains containing the V2TcR and V3LacI when induced with aTc and IPTG respectively (FIG. 10A). Synergy was observed by the addition of IPTG and aTc to the V2TcR/V3LacI strain improving mCardinal production by 1.5-fold against the induction obtained with aTc alone (FIG. 10A). However, the synergy in mCardinal production in the duet system reached equal amounts of mCardinal as the V2TcR expression system (FIG. 10A). These results indicate that the tandem version of the synthetic promoters V2TcR/V3LacI offer an alternative to co-express different genes at high levels using different inducers.

In *P. putida* the V2TcR and V2lacI expression systems showed the best performance activating transcription of heterologous proteins. Consequently, we integrated both promoters controlling expression of mCardinal into the chromosome of this host. Single induction of the dual system by aTc resulted in induction of mCardinal equivalent to V2TcR system; single induction of the dual system with IPTG resulted in a 3-fold decrease relative to V2lacI alone (FIG. 10B). However, the addition of both inducers (aTc+IPTG) increased mCardinal by 1.1-fold over the V2TcR and V2lacI expression system, thus showing a minimal synergistic effect of the dual promoters in *P. putida* (FIG. 10B).

The duet expression system V2TcR/V2LacI decreased substantially the efficiency of the synthetic promoters in *V. natriegens* compared to the single expression system counterparts by 12 and 2-fold for V2LacI and V2TcR after induction with IPTG and aTc, respectively (FIG. 10C). Co-induction with IPTG+aTc did not have a synergistic effect in mCardinal production, but reached similar yields as obtained by induction with aTc alone (FIG. 10C). Together, this information demonstrates that synergy cannot be achieved with the dual expression system V2TcR/V2(3)LacI in medium copy plasmids as is the case for *E. coli* and *V. natriegens*. On the contrary, the duet expression system diminishes the activity of each promoter probably due the constitutive expression of the transcriptional regulators TetR and LacI, and competition of the σ^{70} for both promoters. In *P. putida* the duet expression system was directly integrated into the chromosome and reduced the quantities of the transcriptional regulators TetR and LacI. Surprisingly, in this host the V2TcR expression system gained strength in the dual version, the V2lacI expression system lost efficiency, and synergy was observed, thus indicating that duet expression system gain performance as single copies in the chromosome of the host bacterium.

Example 14: Expression of Different Recombinant Proteins with the Dual Expression System V2TcR/V2(3)LacI

Prior to the expression systems provided herein there was not a universal dual expression system that allows differential expression of distinct set of genes via two independent inducible promoters that is portable among different Gram-negative species. The V2TcR/V2(3)LacI dual expression system demonstrated that it can control the co-expression of mCardinal, although synergy was only observed in *P. putida*. Two different reporter proteins were expressed to evaluate their production in *E. coli*, *P. putida* and *V. natriegens* using the V2TcR/V2(3)LacI expression system. The transcriptional unit V2TcR controlling the expression of sfGFP while the V2(3)LacI unit controlling mCardinal production.

The three bacterial species under study containing V2TcR-sfGFP/V2(3)LacI-mCardinal expression system showed no green fluorescence in the uninduced state, thus confirming the tight regulation of the V2TcR expression system, however, the V2(3)LacI-mCardinal is prone to leak and red fluorescence was observed in the absence of IPTG (FIGS. 11A-11C). Addition of aTc induced the production of sfGFP, and similar result was observed with the addition of IPTG which triggered production of mCardinal. Interestingly, when the cultures were co-induced with aTc+IPTG the green and red fluorescence reached similar levels compared with the cultures induced with only one inducer, thus indicating that both promoters (V2TcR/V2(3)LacI) are unaffected by the activation of each other reaching their maxi-

mum activity either in their sole activation or their independent combined activation (FIGS. 11A-11C).

Example 15: Production of Lycopene and B-Carotene in *E. coli*, *P. putida* And *V. natriegens* with the Expression System V2TcR

To decipher the capability of the dual-expression system V2TcR/V2(3)LacI to produce natural products we first had to test the ability of the V2TcR expression system to control the expression of multiple genes, we therefore decided to evaluate the production of lycopene and B-carotene in *E. coli*, *P. putida* and *V. natriegens*. Lycopene production has already been demonstrated in *E. coli* and *P. putida* via heterologous expression of the lycopene-producing operon (LYC) containing the crtEBI genes from *Pantoea ananatis*. Lycopene is the biosynthetic precursor of B-carotene, thus expression of the LYC operon together with crtY yields B-carotene production, which also has been demonstrated in *E. coli* and *V. natriegens*. The LYC and the crtEBIY operons were accommodated in the V2TcR expression system and lycopene and B-carotene production were measured by UHPLC after 4 hours of induction with aTc. In *E. coli* production of both, lycopene and B-carotene, was achieved but the induced culture produced only 1.7 and 1.2-fold more lycopene and B-carotene respectively than the uninduced culture, thus demonstrating that the tight control of the V2TcR can't totally repress the expression of the crtEBI and crtEBIY genes (FIGS. 12A-12B). Lycopene biosynthesis in *E. coli* using the pET expression system also demonstrated abundant leakage and production of 170 mg/L lycopene after 40 hours of induction, 1.1-fold more than the uninduced culture. Despite the V2TcR expression system produced 94 mg/L, this was achieved after 4 hours of induction, against the 40 hours reported with the pET expression system. This result confirms the advantage of the V2TcR expression system over the pET expression system as described in Examples 2-10. In *P. putida* and *V. natriegens* containing the V2TcR-crtEBI (SEQ ID NO: 70) and V2TcR-crtEBIY (SEQ ID NO: 71) transcriptional units no lycopene or B-carotene production was observed in the uninduced state, and activation of by aTc yielded production of the terpenoids (FIGS. 12A-12B). *P. putida* was reported to produce 1.22 ng/mL of lycopene after 24 hours of growth with the pSEVA421 vector, with the V2TcR expression system lycopene production reached 17 mg/L after 4 hours of cultivation, which is a substantial increment (13.000x), and B-carotene production by *P. putida* reached 840 mg/L, the highest among the three species under study (FIG. 12). *V. natriegens* was reported to produce 2.93 mg/L of B-carotene, while the V2TcR expression system achieved 597 mg/L of B-carotene and 31 mg/L of lycopene. These results demonstrate that the V2TcR can be efficiently programmed for production of biosynthetic gene clusters in Gram-negative bacteria.

Example 16: Coordinated Expression of the Lycopene and B-Carotene Biosynthetic Gene Clusters (BGC) with the Dual Expression System V2TcR/V2(3)LacI

Natural products are encoded by BGC and ability to activate each gene at a different time point with a different inducer represent an advantage when the genes are toxic, or the metabolites produce by the BGC affect the cell growth of the heterologous host. As provided herein, the V2TcR expression system can activate the production of lycopene and B-carotene. In order to control the production of B-carotene with two different inducers, aTc and IPTG were used. The LYC operon was incorporated to the V2TcR expression system and the crtY gene was controlled by the V2(3)LacI expression system. The plasmid p σ^{70} V2TcR-crtEBI/V3LacI-crtY (SEQ ID NO: 73) was transformed into *E. coli* and the plasmid p σ^{70} V2TcR-crtEBI/V2LacI-crtY (SEQ ID NO: 72) was transformed into *P. putida* and *V. natriegens*. In the three bacteria species no lycopene production was observed despite the V2TcR expression system was controlling the crtEBI genes and activation by aTc should trigger production of lycopene (FIG. 13A). This result is explained in *E. coli* by the leakage of the crtEBIY genes under the control of the synthetic promoters V2TcR and V3LacI, demonstrating the high processivity of the B-carotene BGC to metabolize the endogenous IPP isopentenyl diphosphate directly into B-carotene. However, the V2TcR-crtEBI showed no leakage in *P. putida* and *V. natriegens*, but the V2LacI expression system was reported to leak, consequently, the lycopene produced after addition of aTc is directly transformed into B-carotene by the CrtY present in the cells due its minimal expression and high processivity.

B-carotene production in *E. coli* was observed no matter the absence or presence of the inducers aTc and IPTG (FIG. 13B), thus demonstrating that the V2TcR/V3LacI expression system failed to keep downregulated the B-carotene BGC in this host. In *P. putida* and *V. natriegens* B-carotene production was only achieved when the cultures were induced with aTc or with aTc+IPTG (FIG. 13A). IPTG induction alone did not conduce to B-carotene production because the crtEBI genes were not transcribed by the V2TcR expression system. Addition of aTc resulted in B-carotene levels similar to induction with aTc+IPTG (FIG. 13A), thus indicating that the minimal leakage of the V2LacI-crtY is sufficient to metabolize the lycopene produced by the V2TcR-crtEBI into B-carotene.

Importantly, the yields of B-carotene reached by *E. coli* were always lower than the production achieved by *P. putida* and *V. natriegens*. Thus, *P. putida* can be used to produce complex biosynthetic gene clusters. Therefore, a dual expression system developed in this study expands the genetic tool kit of *P. putida* to produce complex proteins. Also, *V. natriegens* outperformed *E. coli*, and the dual expression system allowed higher production of B-carotene compared with the V2TcR by 1.2-fold. Together these results demonstrate the potential of the dual expression system presented here to fully capitalize the potential of heterologous host in industrial applications.

SEQUENCES

FIG. 2A Sequences:

Lac (SEQ ID NO: 3)

AGGCTTTACACTTTATGCTTCGGCTCGTATGTTGTGTGGAATTGTGAGCGGATAACAATTTACACAGGAAACA
GCTATG

- continued

SEQUENCES

lacUV5 (SEQ ID NO: 4)
AGGCTTTACACTTTATGCTTCCGGCTCGTATAATGTGTGGAATTGTGAGCGGATAACAATTTACACAGGAAACA
GCTATG

tacI (SEQ ID NO: 5)
GAGCTGTTGACAAATTAATCATCGGCTCGTATAATGTGTGGAATTGTGAGCGGATAACAATTTACACAGGAAACA
GAATCATATG

Vilac (SEQ ID NO: 6)
AGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAATTGTGAGCGGATAACAACGCAGTAAGAGAGGAA
TGTACATATG

V2lac (SEQ ID NO: 7)
GAGCTGTTGACAACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAACGCAGTAAGAGA
GGAATGTACATATG

V3lac (SEQ ID NO: 8)
GAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAAGTGAATTGTGA
CGGATAACAATTTACACAGGAAACAGAATCATATG

V4lac (SEQ ID NO: 9)
CTTTATGCTTCCGGCTCGTTGACAGTGTGGAATTGTGAGCGGATAACAATATAATGTGTGGAATTGTGAGCGGAT
AACAATTTACACAGGAAACAGAATCATATG

FIG. 2B Sequences:

Tet (SEQ ID NO: 10)
GGATCCTTGACACTCTATCATTGATAGAGTTATTTTACCACTCCCTATCAGTGATAGAGAAAAGTGAAATG

V1Tc (SEQ ID NO: 11)
GGATCCTTGACACTCTATCATTGATAGAGTTATTTTACCACTCCCTATCAGTGATAGAGACGCAGTAAGAGAGGA
ATGTCATATG

V2Tc (SEQ ID NO: 12)
GAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAGAGGAAT
GTACATATG

V3Tc (SEQ ID NO: 13)
GAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGAGTGAATTGTGAGCGGA
TAACAATTTACACAGGAAACAGAATCATATG

V4Tc (SEQ ID NO: 14)
TTGACACTCTATCATTGATAGAGTTTGACATCCCTATCAGTGATAGAGATATAATGTGTGGAATTGTGAGCGGAT
AACAATTTACACAGGAAACAGAATCATATG

SEQ ID NOS: 15-38 are provided in Table 2.

SEQ ID NOS: 39-43 are provided in Example 2.

SEQ ID NOS: 44-53 are provided in the Detailed Description.

FIG. 14 Full-length Sequence:

V2Tc/tetR-V2lac/lacI (SEQ ID NO: 54)
GAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAGAGGAAT
GTACATATGAAGCTTCTCGGTACCAAAATCCAGAAAAGAGGCCCTCCGAAAGGGGGGCTTTTTCGTTTGGTTC
CGAATTCCTGACAGCTAGCTCAGTCCAGGTATAATGCTAGCCGAGTAAGAGAGGAATGTACACATGTCCCGCC
TGGATAAATCGAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGCTGACGACCCGTA
AATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGGCATTGCTGGATG
CCCTCGCTATTGAAATGTTGGATCGCCACCATACCCATTCTGTCCACTGGAGGGCGAGTCTTGGCAGGACTTTC
TCCGCAACCAACGCGAAATCCTTTTCGCTGTGCACTCTTGTCCATCGGGACGGTGCTAAGGTGCACTTGGGCACCC
GTCCCAACGAAAACAATACGAAACCTTGGAATCAATTGGCGTTTTTGTGCCAGCAAGGGTTAGCTTGGAGA
ATGCTCTCTATGCGCTCTCGGCTGTCCGGCACTTACGTTGGGGTGCGTGTGGAGGACAGGAGCATCAAGTCG
CAAAAGAGGAGCGTGAAACCCCAACCAACGAGCTCGATGCCACCTCTGCTCCGCCAAGCTATCGAACTCTTCGATC
ATCAGGGCGCGGAGCCAGCCTTCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGGAAAAACAACCTCAAGTGTG
AAAGCGGGTCTTAACGTGAGTCACTGCCCGCTTTCCAGTCGGGAAACCTGTGTCGACGCTGCATTAATGAATCG
GCCAACGCGCGGGGAGAGCGGTTTTCGCTATTGGGCGCCAGGGTGGTTTTTCTTTTACCAGTGAGACGGCAAC
AGCTGATTGCCCTTACCAGCTTGGCCCTGAGAGAGTTGCAGCAAGCGGTTCCACGCTGGTTTGGCCAGCAGGCGA
AAATCCTGTTGATGGTGGTTAACGGCGGGATATAACATGAGCTGTCTTCGGTATCGTCGTATCCCACTACCGAG
ATATCCGACCAACGCGCAGCCCGGACTCGGTAATGGCGCGCATTTGCCGCCAGCGCCATCTGATCGTTGGCAACC
AGCATCGCAGTGGGAACGATGCCCTTACGCAATTGTCATGGTTTGTGAAAACCGGACATGGCACTCCAGTCG
CCTTCCCGTTCGCTATCGGCTGAATTTGATTGCGAGTGAGATATTTATGCCAGCCAGCCAGACGCAGACGCGCC
GAGACAGAACTTAATGGGCCCCCTAACAGCGCGAATTGCTGGTGACCAATGCGACCGAGATGCTCCAGCCCGAT
CGCGTACCGTCTTCATGGGAGAAAAATAACTGTTGATGGGTGTCGTGGTCAGAGACATCAAGAAATAACGCCGA
ACATTAGTGCAGGCAGCTTCCACAGCAATGGCATCCTGGTCATCCAGCGGATAGTTAATGATCAGCCCACTGACG
CGTTGCGCGAGAGATTGTGCACCGCCGCTTACAGGCTTCGACGCCGCTTCGTTCTACCATCGACACCACACG
CTGGCACCAGTTGATCGGCGCAGATTTAATCGCCGCGACAATTGCGACGGCGCGTGCAGGGCCAGACTGGAG
GTGGCAACGCCAATCAGCAACGACTGTTTGGCCCGCAGTTGTTGTGCCACGCGGTTGGGAATGTAATTCAGCTCC
GCCATCGCGCTTCCACTTTTTCCCGGTTTTTCGCAAGAACGTGGCTGGCTTCCACGCGGGAAACGGTC

- continued

SEQUENCES

TGATAAGAGACACCGGCATACCTCTGCGACATCGTATAACGTTACTGGTTTCACATTACCACCCCTGAATTGACTC
TCTTCCGGGCGGTATCATGCCATACCGCGAAAGGTTTTCGCCCATTCGATGGTGTCCGGGATCTCGACGCTCTCC
CTTATGCGACTCCTGCATTAGGAAGCAGCCAGTAGTAGGTTGAGGCGGTTGAGCACCGCGCCGCAAGGAATGG
TGCATGCAAGGAGATGGCGCCCAACAGTCCCCCGGCCACGGGAgatcaaaagcctccggtcggaggtctttgact
TCTAGAGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAACGCAGT
AAGAGAGGAATGTACCCATGGCCATGGCTCGAGgacgaacaataaggcctccctaacggggggcctttttattg
ataacaaaa

Additional sequences:

V2Tc/tetR-V3lac/lacI (SEQ ID NO: 55)

GAGCTGTTGACAACTCTATCATTGATAGATTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAGAGGAAT
GTACATATGAAGCTTCTCGGTACCAAAATCCAGAAAAGAGGCCCTCCGAAAGGGGGCCTTTTTCGTTTTTGGTC
CGAATTCTTGACAGCTAGCTCAGTCTAGGTATAATGCTAGCCGCGAGTAAGAGAGGAATGTACACATGTCCCGCC
TGGATAAATCGAAAGTGATTAACTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGCTGACGACCCGTA
AATTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGGCATTGCTGGATG
CCCTCGCTATTGAAATGTTGGATCGCCACCATACCCATTCTGTCCATCGAGGGCGAGTCTGGCAGGACTTTC
TCCGCAACAACGCGAATCTTTCCGCTGTGCACTCTTGTCCCATCGGACGGTGCTAAGGTGCACTTGGGCACCC
GTCCACCCGAAAAACAATACGAAACCTTGGAAAAATCAATTGGCGTTTTTGTGCCAGCAAGGGTTAGCTTGGAGA
ATGCTCTCTATGCGCTCTCGGCTGTGGGCACCTTTACGTTGGGGTGCGTGTGGAGGACCAGGAGCATCAAGTCG
CAAAAGAGGAGCGGTGAACCCCAACCAACGACTCGATGCCACCTCTGCTCCGCCAAGCTATCGAACTCTTCGATC
ATCAGGGCGCGGAGCCAGCCTTCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGGAAAAACAATCAAGTGTG
AAAGCGGGTCTTAAGTCAGTCACTGCCCGCTTTCCAGTCGGGAAACCTGTGTCGACGCTGCATTAATGAATCG
GCCAACGCGCGGGGAGAGGCGGTTTTCGCTATTGGGCGCCAGGGTGTTTTTCTTTTACCAGTGAGACGGGCAAC
AGCTGATTGCCCTTACCAGCCTGGCCCTGAGAGAGTTGCAGCAAGCGGTCCACGCTGGTTTGGCCAGCAGGCGA
AAATCCTGTTTGTATGGTGGTTAACGGCGGGATATAACATGAGCTGCTTTCGGTATCGTCGTATCCCATACCGAG
ATATCCGCACCAACGCGCAGCCCGGACTCGTAATGGCGCGCATTGCGCCAGCGCATCTGATCGTTGGCAACC
AGCATCGCAGTGGGAACGATGCCCTCATTGACGATTGTCATGGTTTGTGAAAACCGGACATGGCACTCCAGTCG
CCTTCCCGTTTCGCTATCGGCTGAATTTGATTGCGAGTGAGATATTTATGCCAGCCAGCCAGACGAGACGCGCC
GAGACAGAACTTAATGGGCCCCGCTAACAGCGCGATTGCTGGTGACCAATGCGACAGATGCTCCACGCCCAGT
CGCGTACCGTCTTCATGGGAGAAAAATAACTGTTGATGGGTGCTGGTCAGAGACATCAAGAAATAACGCCGGA
ACATTAGTGCAGGCGAGTTCCACAGCAATGGCATCCTGGTCATCCAGCGGATAGTTAATGATCAGCCCACTGACG
CGTTGCGGAGAGATTGTCGCCCCGCTTACAGGCTTCGACGCGCTTCGTTTACCATCGACACCACGCG
CTGGCACCCAGTTGATCGGCGCGAGATTAATCGCCGCGACAATTTGCGACGCGCGGTGACGCGCCAGCTGGAG
GTGGCAACGCCAATCAGCAACGACTGTTTTCGCCCGCAGTTGTTGTGCCACGCGGTTGGGAATGTAATTGAGCTCC
GCCATCGCCGCTTCCACTTTTTCCCGCGTTTTTCGAGAAACGTGGCTGGCTGGTTACCACGCGGGAAACGGTC
TGATAAGAGACACCGGCATACCTGCGACATCGTATAACGTTACTGGTTTTCACATTACCACCCCTGAATTGACTC
TCTTCCGGGCGGTATCATGCCATACCGCGAAAGGTTTTCGCGCATTCGATGGTGTCCGGGATCTCGACGCTCTCC
CTTATGCGACTCCTGCATTAGGAAGCAGCCAGTAGTAGGTTGAGGCGGTTGAGCACCGCGCCGCAAGGAATGG
TGCATGCAAGGAGATGGCGCCCAACGATGCTCCCGCCACGGGAgatcaaaagcctccggtcggaggtctttgact
TCTAGAGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAAGTGGAA
TTGTGAGCGGATAACAATTTACACAGGAAACAGAATCCCATGGCTCGAGgacgaacaataaggcctccctaacg
gggggctttttatttgataacaaaa

TABLE 3

List of Plasmids

pg⁷⁰ V2TcR-mCardinal (SEQ ID NO: 56)

ggatccGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGGTGAGTAAAGGGTGAGGAGCTCATTAAGGAGAAACATGCACATGAAGCTGTATATGAGGGGC
ACCGTAACAACCACTTCAAGTGTACCACCGAGGGTGAAGGTAACCCCTACGAGGGGACGCAGACCCAAACGC
ATCAAGGTCGTGGAGGGCGGCGCGCTGCTTTCGCAATTCGACATTCTGGCGACTGTTTTATGTACGGCTCGAAG
ACCTTCATCAACCAACCAAGGCTACCGGACTTCTTCAAGCAGAGCTTCCCTGAGGGCTTCACTTGGGAGCGC
GTACCACGTATGAAGACGGTGGGTGCTCACCGTGACCCAGGACACGAGCTTGCAGGATGGCTGCTTGATTAC
AACGTCAAGCTGCGCGGGTGAATTCCTAGCAACGGGCGAGTGATGCAGAAAAGACGCTGGGTGGGAGGCC
ACCACCGAGACCTGTACCCGCGCAGCGGGGGCTGGAAGGGCGGTGCGATATGGCCCTGAAATTGGTCGGCGGC
GGTCATTTGCACTGCAATCTCAAGACCACTACCGCTCCAAGAAACCGCCAAAACCTGAAGATGCTGTGTT
TATTTTGTGACCGCGCGCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTACGTGGAACAGCACGAAGTGGCC
GTGGCTCGTTATTGCGATCTGCCGTCGAAGCTGGGTCAACAACTGAACGGCATGGATGAGCTGTACAAAGATTAT
AAGGATGATGACGACAAGTAAAGCTTCTCGGTACCAAAATCCAGAAAAGAGGCCCTCCGAAAGGGGGCCTTTT
TTCGTTTTTGGTCCGAATCTTGACAGCTAGCTCAGTCTAGGTATAATGCTAGCCGCGAGTAAGAGAGGAATGTAC
ACATGTCCCGCTCGATAAATCGAAAGTGATTAACTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGC
TGACGACCCGTAATTTGGCACAAAAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGG
CATTGCTGGATGCCCTCGCTATTGAAATGTTGGATCGCCACCATACCCATTCTGTCCATGGAGGGCGAGTCCT
GAGGAGACTTCTTCGCAACAACGCGGAAATCCTTTGCTGTGCACTCTTGTCCCATCGGACGGTGCTAAGGTGC
ACTTGGGCACCCGTCCACCGAAAACATAACGAAACCTTGAAAATCAATTGGCGTTTTTGTGCGACGAAGGGT
TTAGCTTGGAGAAATGCTCTATGCGCTCTCGGCTGTGGGCACCTTTACGTTGGGGTGCGTGTGGAGGACCAGG
AGCATCAAGTCGCAAAAGAGGAGCGGTGAACCCCAACCAAGGACTCGATGCCACCTCTGCTCCGCAAGCTATCG
AACTCTTCGATCATAGGGCGCGGAGCCAGCTTCTCTTTGGGCTGGAGCTGATTATCTCGGTTTTGGAAAAAC
AACTCAAGTGTAAAGCGGGTCTTAAGTCAGTCTAGACCATGGctcgag

V2lacI-mCardinal (SEQ ID NO: 57)

GGATCCGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAACGCAGT
AAGAGAGGAATGTACATATGGTGAGTAAGGGTGAGGAGCTCATTAAGGAGAACATGCACATGAAGCTGTATATGG
AGGGCACCGTAACAACCACTTCAAGTGTACCACCGAGGGTGAAGGTAACCCCTACGAGGGGACGCAGACCC

TABLE 3-continued

List of Plasmids

AACGCATCAAGGTCGTGGAGGGCGGCCGCTGCCTTTTCGCATTTCGACATTCTGGCGACCTGTTTTATGTACGGCT
CGAAGACCTTCATCAACACACCCCAAGGCATCCCGGACTTCTTCAAGCAGAGCTTCCTGAGGGCTTCACCTGGG
AGCGCGTCACACGTCATGAAGACGCTGGGGTGCTCACCGTGACCCAGGACACGAGCTTGCAGGATGGCTGCTTGA
TTTACAACGTCAGAGTCGCGGGGGTGAACCTTCCCTAGCAACGGGCCAGTGATGCAGAAAAAGACGCTGGGTTGGG
AGGCCACACCGAGACCTGTATCCCGGCCGACGGGGGGTGGAAAGGGCGGTGCATATGGCCCTGAAATTTGGTCG
GCGGCGGTCAATTTGCACTGCAATCTCAAGACCACGTACCGCTCAAGAAACCCGCCAAAAACCTGAAGATGCCTG
GTGTTTATTTTGTGACCGGCCCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTACGTGGAACAGCACGAAG
TGGCCGTGGCTCGTTATTTCGATCTGCGCTCGAAGCTGGGTACAAACTGAACGGCATGGATGAGCTGTACAAAG
ATTATAAGGATGATGACGACAAGTAAAGCTTCTCGGTACCAATTCAGAAAAAGAGGCTTCCCGAAAGGGGGG
CTTTTTTCGTTTTTGGTCCGAATTTCCCGGTGGCGGGGGACTGTTGGGCGCCATCTCCTTGCATGCACCATTCCT
TCCGCGCGGGGTCTCAACGGGCTCAACCTACTACTGGCTGCTTCCCTAATGCAGGAGTCGCATAAGGGAGAGCG
TCGAGATCCCGACACCATCGAATGGCGCAAAACCTTTTCGCGGTATGGCATGATAGCGCCGGAAGAGAGTCAAT
TCAGGGTGGTGAATGTGAACAGTAACGTTATACGATGTGCGAGAGTATGCCGGTGTCTCTTATCAGACCGTTT
CCCGCGTGGTGAACCGGCTGACCGCTTCTGCGAAACCGCGGAAAAAGTGAAGCGCGCATGGCGGAGCTGA
ATTACATTTCCCAACCGCGTGGCACAACAACTGGCGGGCAACAGTCGTTGCTGATTGGCGTTGCCACCTCCAGTC
TGGCCCTGCACGCGCGCTCGCAAAATGTGCGGGCGATTAAATCTGCGCGCATCAACTGGGTGCCAGCGTGGTGG
TGTCCATGGTAGAACGAAGCGGCGTCAAGGCTGTAAAGCGCGGTGCACAACTCTTTCGCGCAACGCGTCAGTG
GGCTGATCATTAACTATCCGCTGGATGACAGGATGCCATTGCTGTGAAGCTGCCTGCATAATGTTCCGCGCT
TATTTCTTGATGTCTCTGACAGACACCCATCAACAGTATTATTTCTCCCATGAAGACGGTACGCGACTGGGCG
TGAGCATCTGGTGCATTTGGTCAACGCAAACTCGCGCTGTAGCGGGCCCATTAAGTTCTGTCTCGGCGCGTC
TCGCTCTGGCTGGCTGGCATAAATATCTCACTCGCAATCAAATTCAGCCGATAGCGGAACGGGAAGCGCATGGA
GTGCCATGTCCGGTTTTCAACAAACCATGCAAACTGCTGAATGAGGGCATCGTTCCTGCGCATGCTGGTTGCCA
ACGATCAGATGGCGTGGCGCAATGCGCGCCATTACCGAGTCCGGGCTGCGCGTTGGTGGGATATCTCGGTAG
TGGGTACGACGATACCGAAGACAGCTCATGTTATATCCCGCGTTAACCACCATCAACAGGATTTTCGCTGC
TGGGCAAAACAGCGTGGACCGCTTGTGCAACTCTCTCAGGGCCAGCGGTGAAGGGCAATCAGCTGTTGCCCG
TCTCACTGGTGAAGAAAAACCAACCTGGCGCCCAATACGCAAAACCGCTCTCCCGCGCGTTGGCCGATTTCAT
TAATGCAGCTGGCACGACAGGTTTCCGACTGGAAGCGGGCAGTGACTGCAGCTCGAG

po⁷⁰ V3LacI-mCardinal (SEQ ID NO: 60)

GGATCCGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAAGTGGAA
TTGTGAGCGGATAACAATTTACACAGGAAACAGAATCATATGGTGAGTAAGGGTGAGGAGCTCATTAAGGAGAA
ATGACATGAAGCTGTATATGAAGGGGACCGTAACACACACCACTTCAAGTGTAACACCGAGGGTGAAGGTAA
ACCTTACGAGGGGAGCGAGACCAACGTCATCAAGGTCGTGAGGGCGGCCGCTGCTTTTCGCATTTCGACATTCT
GGCGACCTGTTTTATGTACGGCTCGAAGACCTTCATCAACACACCCCAAGGCATCCCGGACTTCTTCAAGCAGAG
CTTCCCTGAGGGCTTCACTGGGAGCGGTCAACACGTCATGAAGACCGTGGGGTGTCTACCGTGACCGAGACAC
GAGCTTGCAGGATGCTGCTTGAATTACAACGTCAGGTCGCGGGGTGAACCTTCCCTAGCAACGGGCGAGTGAT
GCAGAAAAAGACGCTGGGTTGGGAGGCGCACACCGAGACCTGTACCGCGCGCAGCGGGGGCTGGAAGGGCGGTG
CGATATGGCCCTGGAATTTGGTGGCGCGGTCAATTGCACTGCAATCTCAAGACCACTACCGCTCCCAAGAAAC
CGCCAAAAACCTGAAGATGCTTGTGTTTATTTTGTGACCGCGCTGAGCGCATCAAGGAAGCGGACAATGA
GACGTACGTGGAACAGCACGAAGTGGCGGTGCTGTTATGCGATCTGCGGTGCAAGCTGGGTGCAAACTGAA
CGCATGGATGAGCTGTATCAAGGGGTCAGAGATCCCGGACACCATCGAATGGCGCAAAACCTTTCGCGGTATGGCATG
ATAGCGCCCGGAAGAGAGTCAATTCAGGGTGGTGAATGTGAACCAAGTAACGTTATACGATGTGCGAGAGTATGC
CGGTGTCTCTTATCAGACCGTTTCCCGGTGGTGAACAGGCCAGCCACGTTTCTGCGAAAAACCGGGAAAAAGT
GGAAGCGCGCATGAGCGGAGCGTCAGATCCCGGACACCATCGAATGGCGCAAAACCTTTCGCGGTATGGCATG
GATTGGCGTTGCCACCTCCAGTCTGGCCCTGCACGCGCGTGCAGAAATGTGCGGGCGATTAAATCTGCGCGCGA
TCAACTGGGTGCCAGCGTGGTGGTGTGATGGTAGAAGCAAGCGCGTCGAAGCCTGTAAAGCGGCGGTGCACAA
TCTTCTCGCGCAACGCGCTCAGTGGCTGATCAATTAACCTCGCTGGATGACAGGATGCCATTGCTTGGGAAGC
TGCTGCACTAATGTTCTCGGCGTTATTTCTGATGTCTTGACAGACACCATCAACAGTATTATTTCTCCCA
TGAAGACGTCAGCGCTGGGCGTGGAGCATCTGGTGCATTGGGTCAACAGCAATCGCGCTGTTAGCGGGCCC
ATTAAGTTCTGTCTCGGCGGTGCTGCGCTGGCTGGTCAGATAAATATCTCACTCGCAATCAAAATTCAGCCGAT
AGCGGAACGGGAAGCGACTGAGTGCCATGTCGGTTTTCAACAAACCATGCAAACTGCTGAATGAGGGCATCGT
TCCCACTGCGATGCTGGTTGCCAAGCATCAGATGGCGCTGGGCGCAATGCGCGCCATTACCGAGTCCGGGCTGCG
CGTTGGTGGGATATCTTACGTAGTGGGATACGACGATACCGAAGACAGCTCATGTTATATCCCGCGTTAACCAC
CATCAACAGGATTTTCGCTGCTGGGCGAAACAGCGTGGACCGCTTGTGCAACTCTCTCAGGGCCAGGCGGT
GAAGGGCAATCAGCTGTTGCCCGTCTCACTGGTGAAGAAAAACCAACCTGGCGCCCAATACGCAAAACCGCTC
TCCCGCGCGTTGGCCGATTTCATTAATGCAGCTGGCACGACAGGTTTCCCGACTGGAAGCGGGCAGTGAGTGCA
GCTCGAG

po⁷⁰ V2TcR-sfGFP (SEQ ID NO: 61)

GGATCCGAGCTGTTGACAACTTATCATGTAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAAGTGAAG
AGGAATGTACATATGTCCAAAGGTGAAGAGCTGTTTACCGCGCTCGTGCCATTCTGGTGGAGCTGGATGGCGAC
GTCAACGGGCAACAAGTTTACGCTCGGTGGCGAAGGTGAGGGCGACGCCACGAACGGTAAAGCTGACGCTGAAATTC
ATTTGCACACCGGCAAAATGCTGTACCTTGGCCACCTTGGTGACACGCTCACCTACGGCGTACAGTGCTTC
AGCCGTTACCGGACCATGAAGCGTCACGACTTCTTCAAAGCGCCATGCCGAGGGTTACGTGACGAGCGT
ACGATTAGTTTCAAGGACGACCGGACCTTATAAGACCCGTCGCAAGTGAAGTTCAAGGGCGATACGTTGGTGAAC
CGTATCGAGTTGAAGGGTATCGACTTAAAGGAAGACGGCAACATCTGGGCGATAAGCTGGAGTACAATTTCAAC
AGCCATAACGTTTACATCACCGCGATAAACAGAAGAACGGCATTAAGGCCAATTTAAGATCCGCCACAAACGTC
GAAGACGGCTCGGTGCGAGCTGGCGACCATTATCAGCAAAACACCCCATCGGTGATGGGCGCGTGTGCTGCGG
GATAACCATTAATCTGAGCACGAGTCGGTGTCTCAGCAAGGACCTAACGAAAAGCGCGATCACATGGTGTGCTG
GAGTTCGTACGGCGGGGGGATCAACCATGGATGGACGAGCTCTACAAAGACTATAAAGATGACGATGACAAG
TAAAAGCTTCTCGGTACCAAAATTCAGAAAAAGAGGCTCCCGAAAGGGGGGCTTTTTCTGTTTTGGTCCGAAT
CTTGACAGCTAGCTCAGTCTAGGTATAATGCTAGCCGAGTAAGAGAGGAATGTACACATGTCCCGCTGGATA
AATCGAAAGTGATTAACTCGGCCCTCGAATGTGTAAGTGAAGTCGGTATCGAGGGGCTGACGACCCGTAATTTGG
CACAAAAAGTTGGGGGTGGAGCAACCCAGTCGTTGATTGGCACGTCAAAAATAAGCGGGCATTTGCTGGATGCCCTCG
CTATTGAAATGTTGGATGCCACCATACCATTTCTGTCCACTGGAGGGCGAGTCTGGCAGGACTTTCTCCGCA

TABLE 3-continued

List of Plasmids

ACAACGCGAAATCCTTTCGCTGTGCACTCTTGTCCTCATCGGGACGGTGCTAAGGTGCACTTGGGCACCCGTCCCA
CCGAAAAACAATACGAAACCTTGGAAAAATCAATTGGCGTTTGTGTGCCAGCAAGGGTTAGCTTGGAGAAATGCTC
TCTATGCGCTCTCGCTGTGCGGCACCTTACGTTGGGGTGGCTGTGGAGGACCAGGAGCATCAAGTCGCAAAAG
AGGAGCGTGAAACCCCAACCGGACTCGATGCCACCTCTGCTCCGCCAAGCTATCGAACTCTTCGATCATCAGG
GCGCGGAGCGACCTTCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGAAAAACAACCTCAAGTGTGAAAGCG
GTCCTAACTGCACTCGAG

po⁷⁰ V2lacI-sfGFP (SEQ ID NO: 62)

GGATCCGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAACGCAGT
AAGAGAGGAATGTACATATGTCCAAAGGTGAAGAGCTGTTTACCGGCGTCGTGCCCCATTCTGGTGGAGCTGGATG
CGAGCTCAACCGGGCACAGTTTACGCTCCGTCGGCGAAGGTGAGGGCGACGCCACGAAACGGTAAGCTGACGCTGA
AATTCAATTGCAACACCGGCAAAATGGCTGTACCTTGGCCACCTTGGTGACCACGCTCACCTACGCGGTACAGT
GCTTCAGCCGTTACCCGGACCATGAAGCGTCACGACTTCTTCAAAGCGCCATGCGGAGGGTTACGTGACAGG
AGCGTACGATTAGTTTCAAGGACGACGCGCACCTATAAGACCCGTGCCGAAGTGAAGTTTCAAGGCGATACGTTGG
TGAACCGTATCGAGTTGAAGGTTATCGACTTTAAGGAAGACGGCAACATCTTGGGCCATAAGCTGGAGTACAATT
TCAACAGCCATAACGTTTACATCACCGCCGATAAACAAGAAACGGCATTAAGCCAACTTTAAGATCCGCCACA
ACGTCGAAGACGGCTCGGTGCACTGGCCGACCATTAATCAGCAAAACACCCCATCGGTGATGGGCCGTGCTGC
TGCCGGATAACCATTATCTGAGCAGCGAGTGGTCTCAGCAAGGACCTAACGAAAGCGCGATCACATGGTGC
TGCTGGAGTTTCGTACGCGCGCGGGGATCACCCATGGGATGGACGAGCTCTACAAGACTATAAAGATGACGATG
ACAAGTAAAAGTTTTCGATACCAAAATCCAGAAAAGAGGCTCCCGAAAGGGGGCCCTTTTTCGTTTGGTCC
GAATCCCCCGTGGCGGGGACTGTTGGCGCCATCTCTTGCATGCACCATCTCTTGGCGCGCGGTGCTCAA
CGGCTCAACCTACTACTGGGTGCTTCTAATGTCAGGAGTCGCATAAGGGAGAGCGTCGAGATCCCGACACCA
TCGAATGGCGCAAAACCTTTCGCGGTATGGCATGATAGCGCCCGGAAGAGAGTCAATTACGGGTGGTGAATGTGA
AACCAGTAACGTTATACGATGTCGAGAGTATGCCGGTGTCTCTTATCAGACCGTTTCCCGCGTGGTGAACGAGG
CCAGCCACGTTTCTGCGAAAACGCGGGAAGTGAAGCGGCGATGGCGGAGCTGAATTACATTTCCAAACCGCG
TGCGCAACAACCTGCGCGGCAACAGCTGCTGCTGATTGGCGTTGCCACCTCCAGTCTGGCCCTGCACGCGCGT
CGCAAAATGTCGCGCGGATTAATCTCGCGCGATCAACTGGGTGCCAGCGTGGTGGTGGTGGTGGTGGTGGTGGT
GCGGCGTCGAAGCCTGTAAGCGCGCGGTGCAACAATCTCTCGCGCAACGCGTCAGTGGGTGATCATTAACATATC
CGCTGGATGACGAGATGCGCATTTGCTGTGGAAGCTGCGTGCACATAATGTTCCGGCGTTATTTCTGATGCTCTG
ACCAGACACCCATCAACAGTATTATTTCTCCCATGAAGACGGTACGCGATGGGCGTGGAGCATCTGGTGCAT
TGGGTACACAGCAATCGCGCTGTAGCGGGCCATTAAAGTCTGCTCGCGCGCTGCTGCGTCTGGTGGTGGTGGT
ATAAATATCTCTCATCGCAATCAAAATTCAGCCGATAGCGGAACGGGAAGGCGACTGGAGTGCCATGTCGCGTTTC
AACAACCATGCAATGCTGAATGAGGGCATCGTTCCCACTGCGATGCTGTTGCCAACGATCAGATGGCGCTGG
GCGCAATGCGCGCCATTACCGAGTCCGGGCTGCGCGTTGGTGGGATATCTCGGTAGTGGGATACGACGATACCG
AAGACAGCTCACTGTTATATCCGCTTAACCAACCATCAAAACAGGATTTTCGCTGCTGGGGCAACACGAGCTGG
ACCGCTTGTGCAACTCTCTCAGGCGCAGGCGGTGAAGGGCAATCAGCTGTGCGCCGTCTCACTGGTGAAGAA
AAACACCTTGGCGCCCAATACGCAAAACCGCTCTCTCCCGCGCTTGGCCGATTCAATTAATGACAGCTGGCACGAC
AGGTTTCCGACTGGAAGCGGGCAGTGACTGCAGCTCGAG

po⁷⁰ V3LacI-sfGFP (SEQ ID NO: 63)

GGATCCGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGATAACAAGTGGAA
TTGTGAGCGGATAACAATTTACACAGGAACAGAAATCATATGTCCAAAGGTGAAGAGCTGTTTACCGGCGTCGT
GCCCCATTCTGGTGGAGCTGGATGGCGACGTCACACGGGCACAAGTTTAGCGTCCGTGGCGAAGGTGAGGGCGACGC
CACGAACGGTAAGCTTACGCTGAAATTTCAATTGACACCGGCAAAATGGCTGTACCTGGCCACCCCTGGTGC
CACGCTCACCTACGCGGTACAGTGTTCAGCCGTTACCCGGACACATGAAGCGTCAACGACTTCTTCAAAGCGC
CATGCCGAGGGTTACGTGACGAGCGTACGATTAGTTTCAAGGACGACCGCACCTATAAGACCCGTGCCGAAGT
GAAGTTGGAAGCGGATACGTTGGTGAACCGTATCGAGTTGAAGGGTATCGACTTTAAGGAAGACGGCAACATCT
GGGCCATAAGCTGGAGTACAATTTCAACAGCCATAACGTTTACATCACCGCCGATAAACAAGAACGGCATTA
AGCCAACTTTAAGATCCGCCACAACGTCGAAGACGGCTCGGTGACGCTGGCCGACCATTAACAGCAAAACACCCC
CATCGGTGATGGGCCCGTCTGCTGCGGATAACCATTAATCTGAGCAGCGAGTGGTGTGAGCAAGGACCTTA
CGAAAGCGCGATCACATGGTGTCTGCTGGAGTTGCTCAGCGCGCGGGGATCACCATGGGATGGACGAGCTCTA
CAAAGACTATAAAGATGACGATGCAAGTAAAGCTTCTCGGTACCAAAATTCAGAAAAGAGGCTCCCGAAAGG
GGGGCTTTTTCGTTTGGTCCGAATTTCCCGCGTGGCGGGGAGTGTGGGCGCCATCTCTTGCATGCACCA
TTCTTGGCGGGCGGTGCTCAACGGCTCAACCTACTACTGGGTGCTTCTAATGTCAGGAGTGCATAAGGGA
GAGCGTCGAGATCCCGACACCATCGAATGGCGCAAAACCTTTTCGCGTATGGCATGATAGCGCCCGAAGAGAG
TCAATTCAGGTGGTGAATGTAACACAGTAAACGTTATACGATGTCGAGAGTATGCGCGGTGCTCTTATCAGAC
CGTTTCCGCGTGGTGAACACAGGCGAGCCAGCTTTCTGCGAAAACGCGGGAAGTGAAGCGCGATGGCGGA
GCTGAATTACATTTCCAAACCGCGTGGCAACAACCTGGCGGGCAACAGTCTGTGCTGATTGGCGTTGCCACCTC
CAGTCTGGCCCTGACGCGCGCTCGCAAAATGTCGCGCGATTAATCTCGCGCGATCAACTGGGTGCCAGCGT
GGTGGTGTGATGGTAGAACGAAGCGCGTGAAGCTGTAAAGCGCGGTGCAACAATCTTCTCGCGCAACGCGT
CAGTGGGCTGATCATTAACTATCCGCTGGATGACAGGATGCCATTGCTGTGGAAGCTGCTGCACTAATGTTCC
GGCGTTATTTCTGATGCTCTGTAACACAGACACCATCAACAGTATTAATTTCTCCATGAAGACGATACGCGAT
GGGCGTGGAGCATCTGGTGCATTTGGTCAACAGCAAAATCGCGCTGTAGCGGGCCATTAAAGTCTGCTCTCGGC
GCGTCTGCGTCTGGCTGGCTGGCATAAATATCTCACTCGCAATCAAATTCAGCCGATAGCGGAACGGGAAGGCGA
CTGCTGAGTGCATGTCGCTGTAACCAACCATGCAAAATGCTGAATGAGGGCATCGTTCCCATGCGATGCTGGT
TGCCAACGATCAGATGGCGCTGGGCGCAATGCGGCCATTACCGAGTCCGGGCTGCGCGTTGGTGGCGATATCTC
GGTAGTGGGATACGACGATACCGAAGACAGCTCATGTTATATCCCGCGTTAAACCATCAACAGGATTTTCG
CTGCTGGGGCAAAACCGGCGCTGCTGCAACTCTCTCAGGGCAGGCGGTGAAGGGCAATCAGCTGTT
GCCCGTCTCACTGGTGAAGAAAGAAACACCTTGGCGCCCAATACGCAAAACGCGCTCTCCCGCGCGTGGCCGA
TTCATTAATGACAGCTGGCACGACAGGTTTCCGACTGGAAGCGGGCAGTGACTGCAGCTCGAG

pJH0204F (SEQ ID NO: 64)

aaaaggatctaggtgaagatctcttttgataatctcatgacaaaatcccttaacgtgagtttctggtccactga
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acaaaaaacaccgcgtaccagcggtggtttgtttgcccgatcaagagctaccaactctttttccgaaggttaact
ggcttcagcagagcgagataccaaatactgtctctctagtgtagcgtagttagggccaccactcaagaactct
gtagcaccgcctacatacctcgctctgctaactcctgttaccagtggtgctgcccagtgaggataagtcgtgtctt
accgggttgagactcaagacgatagttaccggataaaggcgacggtcgggctgaacggggggttcgtgcacacag

TABLE 3-continued

List of Plasmids
ccagcttggagcggaacgacctacaccgaactgagatacctacagcgtgagctatgagaaagcgccacgcttccc gaagggagaaaagcgcgacaggtatccggttaagcgcgagggtcggaacaggagagcgacgagggagcttccaggg ggaaacgctggtatctttatagtcctgtcgggtttcgccacctctgacttgagcgtcgatttttgtgatgctcg tcagggggcgagcctatggaaaacgcccagcaacgcgcccttttacgggttctggccttttctgctggcctttt gctcacatgtttcttctcgctgtatccctgattctgtggataacgctattaccgcttttgagtgagctgatacc gctcgccgcagcggaacgaccgagcgagcgagctcagtgagcgaggaagcggaagagcgcccaatacgcgaacccg cctctgggagaccagaaaacaaaaaaggccgcttagcggccttcaataatggacctggcctcctaactgatttt taaggcgactgatgagtcgctttttttgtctaatcagaagaactcgtcaagaaggcgatagaaggcgatgcgc tgccaatcgggagcgcgataccgtaaagcacgaggaagcggtcagcccatcgcgcgcaagctctcagcaata tcacgggtagcccaacgctatgtcctgatagcggtcgcccacaccagccgcccacagtcgatgaatccagaaaag cggccattttccaccatgatattcgccgaagcaggcatcgccatgCgtcaagacgagatcctcgccgtcgggcatc cgcgcttgagcctggcgaaacagttcggtggcgagccccgatgctctctgctccagatcatcctgatcgaca agacgggttccatcagagtcagtgctcgctcgatgagatgttctgctgggtggcgaatggcgaggtagccgga tcaagcgtatgcagcgcccgcatgcatcagccatgatggatactttctcgcgaggagcaaggtgagatgcagg agatcctgccccggcacttcgcccataagcagccagtccttcccgttcagtgacaacgctcgagcacagctgcg caaggaaacggcgtcggtggcagccagatagcccgctgctcgtcttgaggttcatcagggcaccggacagg tcggctcttgacaaaagacggggcgccctcgctgacagcggaacacggcgccatcagagcagcgatgtc tgttgtgcccagtcagtcgaatagcctctccaccaagcgccggagaaactcgctgcaatccatcttgttca atcatgcaaacgatcctcatcctgtctcttgatcagatcttgatccctcgcccatcagatccttggcggaag aaagccatccagtttacttttgagggttcccaacctaccagagggcgcccaagctggcaattccggttgcgtt gctgtccataaaacggccagtcctagctatcgccatgtaagccactgcaagctacctgctttctcttgcgctt gctgtttcccttgtccagatagccagtagctgacattcatccgggacgtcggtccccaactgggtaacctttg agttctctcagtcggggatcgatgctcaaaagcctcggtcgaggcttttgactagcacctcggtaccgaatt ccagaaaagaggcctcccgaaagggggcctttttctgttttggtccggatccCATATGAAGCTTGAATTCCTGC AGTCTAGACCATGGctcgaggacgaacaataaggcctccctaaacggggggcctttttatgtataacaaaatcc acaaggaaaaatataaggggagataaaaatccccctttttggttaactcgcgccgctcggtgttgtctgtgca accaccgcggtctcagtggtgtacggtacaaacccgacgctagcaacgcatgagaaagccccggaagatcacc ttccgggggctttttatttgcgctcggggtgcccaggcggtgccccttggggtcccgggcggtactcc ⁷⁰ V2TcR-mCardinal/V2LacI-mCardinal (SEQ ID NO: 65) ggatccGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG AGGAATGTACATATGGTGAGTAAGGTTGAGGAGCTCATTAAGGAGAAACATGCACATGAAGCTGTATATGGAGGGC ACCGTAAACCAACCACCACTTCAAGTGTACCACCGAGGGTGAAGGTAAACCTTACGAGGGGACGCAGACCCCAACGC ATCAAGGTCGTGGAGGGCGGCCCGCTGCCTTTCGCATTTCGACATTCTGGCGACCTGTTTTATGTACGGCTCGAAG ACCTTCATCAACCAACCAACCAAGGCATCCCGGACTTCTTCAAGCAGAGCTTCCCTGAGGGCTTCACTTGGGAGCGC GTCACCACTATGAAGACGGTGGGTGCTCACCCTGACCCAGGACACGAGCTTGAGGATGGCTGCTGTATTTAC AACGTCAAAGTCGCGGGGTGCACTTCCCTAGCAACGGGCGAGTGATGCAGAAAAGACGCTGGGTGGGAGGGC ACCACCGAGACCTTCTCCGCAACCAACCGGAGTGGAGGGGGTGGAAAGGCGGTGCGATATGGCCCTGAAATGGTGGCGCGC GGTCAATTTGCACTGCAATCTCAAGACCACTACCGCTCCAAGAAACCCGCCAAAACCTGAAGATGCCTGGTGT TATTTTGTGCGACCGCGCTTGGAGCGCATCAAGGAAGCGGACAATGAGACGTACGTGGAACAGCACGAAGTGGCC GTGGCTGTTTATTGCGATTCGCGTCAAGCTGGGTGTCACAACTGAACGGCATGGATGAGCTGTACAAAGATTAT AAGGATGATGACGCAAGTAAAGCTTCTCGGTACCAAACTCCAGAAAAGAGGCTCCCGAAAAGGGGGCCTTTT TTCGTTTTGGTCCGAATCTTGACAGCTAGCTCAGTCTAGGTATAATGTAGCCGAGTAAGAGAGGAATGTAC ACATGTCCCGCTCGATAAATCGAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGC TGACGACCCGTAAATTTGGCACAAGAGTTGGGGGTGGAGCAACCCACGTTGTATTGGCACGTCAAAAATAAGCGGG CATTGCTGGATGCCCTGCTATTGAAATGTTGGATCGCCACCATACCCATTCTGTCCACTGGAGGGCGAGTCTT GGCAGGACTTTCTCCGCAACCAACCGGAATCCTTTGCTGTGCACTCTTGTCCCATCGGGACGGTGTCAAGGTGC ACTTGGGCACCCGTCCCACCGAAAACATACGAAACCTTGGAAAATCAATTGGCGTTTGTGTCAGCAAGGGT TTAGCTTGGAGAAATGCTCTTATGCGCTCTCGGCTGTGCGGCACCTTACGTTGGGGTGCCTGTTGGAGGACCAGG AGCATCAAGTCGCAAAAGAGGAGCTGAAACCCCAACCAAGGACTCGATGCCACCTCTGCTCGCCCAAGCTATCG AACTCTTCGATCATCAGGCGCGGAGCCAGCTTCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGGAAAAC AACTCAAGTGTGAAAGCGGGTCTTAAGTGCAGTCACTGCCCGCTTTCAGTCGGGAAACCTGCTGTCAGCTGC ATTAATGAATCGGCTCAACGCGCGGGGAGAGGCGGTTTGCCTATTGGGCGCCAGGCTGGTTTTCTTTTACCAGT GAGACGGGCAACAGCTGATTGCCCTTACCGCCTGGCCCTGAGAGAGTTGACGCAAGCGGTCACGCTGGTTTGC CCCAGCAGCGCAAAATCCTGTTTGTATGGTGGTTAACGGCGGGATATAACATGAGCTGTCTTCGGTATCGTCGTAT CCCACTACCGAGATATCGCCCAACGCGCAGCCCGGACTCGGTAATGGCGCGCATTCGCCCCAGCGCCATCTGA TCGTTGGCAACAGCATCGCAGTGGGAACGATGCCCTCATTGAGCATTTGCATGGTTTGTGAAAACCGGACATG GCACTCCAGTCGCCCTTCCGCTATCGGCTGAATTTGATTGCGAGTGAGATATTTATGCCAGCCAGCCAGA CGCAGACGCGCCGAGACAGAACTTAATGGGCCCGCTAACAGCGCATTTGCTGGTGACCCAAATGCGACCATGATGC TCCAGCGCCAGTCGCTACCGTCTTTCATGGGAGAAAATAATACTGTTGATGGGTGTCTGGTCAGAGACATCAAGA AATAACGCGCGGAACATTAGTGCAGGCGACTTCCACAGCAATGGCATCTGGTCAATCCAGCGGATAGTTAATGATC AGCCCACTGACGCGTCAAGAGGAGTGTGTCACCGCGCTTTACAGGCTTCGACGCGCTTCTGTTTACCATC GACACCAACAGCTGGCACCCAGTTGATCGGCGCGAGATTAACTCGCGCGACAAATTTGCGACGCGCGTGCAGG GCCAGACTGGAGGTGGCAACGCCAATCAGCAACGACTGTTTGCCTCGCAGTTGTTGTGCCACGCGGTGGGAATG TAATTCAGCTCGCCATCGCCCTTCCACTTTTCCGCGCTTTCGAGAAACGTTGGCTGGCTGGTTCACCAAG CGGGAACGGTCTGATAAGAGACACCGGCATACTCTGCGACATCGTATAACGTTACTGGTTTACATTACCAACC CTGAATTGACTCTCTTCCGGCGCTATCATGCCATACCGGAAAGGTTTTCGCCCATTCGATGGTGTCCGGGATC TCGACGCTCTCCCTTATCGATCTCTGCTATTAGGAAGCAGCCAGTAGTAGTTGAGGCGGTGAGACCCGCGC CGCAAGGAATGGTGCATGCAAGGAGATGGCGCCCAACAGTCCCCGGCCACGGGGagtc aaaagcctccggtcgg aggtttttgactTCTAGAGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGTGTGGAATTGTGAGCGGA TAACAACGCAAGTAAGAGAGGAATGTACCATGGTGAGTAAGGGTGAGGAGCTCATTAAGGAGAAACATGCACATGA AGCTGTATATGAGAGGACCGTAAACAACCACTTCAAGTGTACCACCGAGGGTGAAGGTAAACCTACGAGG GGACGCGACCCCAACGCATCAAGGTCGTGGAGGGCGGCCGCTGCCTTTCGATTTCGACATTCTGGCGACCTGTT TTATGTACGGCTCGAAGACCTTCAACCAACCAAGGCATCCCGACTTCTTCAAGCAGAGCTTCCCTGAGG GCTTCACTTGGGAGCGCTCACCAGTATGAAGACGGTGGGTGCTCACCCTGACCCAGGACACGAGCTTCGAGG ATGGCTGCTTGAATTACAACGTCAGGCTGCGCGGGTGAACCTCCCTAGCAACGGGCGAGTGATGCAGAAAAGA CGTGGGTTGGAGGGCCACACCGGAGCCCTGTACCAGCGCGACGGGGGCTGGAAGGGCGGTGCGATATGGCCC TGAAATTGGTGGCGGGCGGTCAATTTGCACTGCAATCTCAAGACCACTGACCGCTCCAAGAAACCCGCCAAAAC

TABLE 3-continued

List of Plasmids

TGAAGATGCCTGGTGTATTATTTGTGACCGGCGCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTACGTGG
AACAGCACGAAGTGCCCGTGGCTCGTTATTGCGATCTGCCGTGGAAGCTGGGTACAAAACCTGAACGGCATGGATG
AGCTGTACAAAGATTATAAGGATGATGACGACAAGTAACCTCGAG

pg⁷⁰ V2TcR-mCardinal/V3LacI-mCardinal (SEQ ID NO: 66)

ggatccGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGGTGAAGGGTGAAGAGCTCATTAAAGGAGAACATGCACATGAAGCTGTATATGGAGGGC
ACCGTAACAACACCACTTCAAGGTATACCACCGAGGGTGAAGGTAAACCTACGAGGGGACGCAGACCCACGC
ATCAAGGTCGTGGAGGGCGGCCGCTGCTTTCGCATTGACATTCTGGCGACCTGTTTATGTACGGCTCGAAG
ACCTTCATCAACCACACCAAGGCATCCCGGACTTCTTCAAGCAGAGCTTCCCTGAGGGCTTACCTGGGAGCGC
GTCACCCAGTATGAAGACGGTGGGGTCTCACCGTGACCCAGGACACGAGCTTGAGGATGGCTGCTTGATTAC
AACGTCAGCTGCGCGGGTGAACCTCCCTAGCAACGGGCCAGTGATGCAGAAAAGACGCTGGGTGGGAGGCC
ACCACCGAGACCTGTACCCGGCCGACGGGGGGTGAAGGGCGGTGCGATATGGCCCTGAAATTGGTCGGCGGC
GGTCATTGTCAGTGCATCTCAAGACCACTACCGCTCCAGAAAACCGCCAAAACCTGAAGATGCTGGTGT
TATTTGTGACCGCGCCCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTACGTGGAACAGCACGAAGTGGCC
GTGGCTCGTTATTGCGATCTGCGCTCGAAGCTGGGTCAAAAACCTGAACGGCATGGATGAGCTGTACAAAGATTAT
AGGATGATGACGACCAAGTAAAGTTTGGCACAAGCTTCTCGGTACCAATTCAGAAAAGAGGCTCCGAAAGGGGGCCTTT
TTCGTTTTGGTCCGAATCTTGACAGCTAGCTCAGTCTTAGGTATAATGTAGCCGACAGTAAGAGAGGAATGTAC
ACATGTCCCGCTGGATAAATCGAAAGTGATTAACTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGC
TGACGACCGTAAATTGGCACAAGTGGGGTGGAGCAACCCACGTTGTATTGGCAGCTCAAAAATAAGCGGG
CATTGCTGGATGCCCTCGTATTGAAATGTTGGATCGCCACCATACCATTTCTGTCCACTGGAGGGCGAGTCTT
GGCAGGACTTTCTCCGCAACACGCGAAATCCTTTCGCTGTGCACTCTTGTCCCATCGGGACGGTGCTAAGGTGC
ACTTGGGACCCGTCACCCGCAAAAACATACGAAACCTTGGAAAATCAATTGGCGTTTTTGTGCGCAGCAAGGGT
TTAGCTTGAGAAATGCTCTCTATGCGCTCTCGGCTGTGCGGCACCTTACGTTGGGGTGCCTGTTGGAGGACAGG
AGCATCAAGTCGAAAAGAGGAGCGTGAACCCCAACACGAGCTCGATGCCACCTCTGCTCCGCCAAGCTATCG
AAGCTTTCGATCATCAGGCGCGGAGCCAGCCTTCTCTTGGGCTGGAGCTGATTATCTGCGGTTTGGAAAAAC
AAGCTCAAGTGTGAAGCGGGTCTTAAGTCAGTCACTGCCGCTTCCAGTCGGGAAACCTGTGCTGCCAGCTGC
ATTAATGAATCGGCCCAACGCGCGGGGAGAGCGGTTTGGCTATTGGGCGCCAGGGTGGTTTTCTTTTCCACAGT
GAGACGGGCAACAGCTGATTGCGCTTACCCGCTGCGCTGAGAGAGTTGACAGCAAGCGGTCACGCTGGTTTTG
CCCAGCAGGCGAAAATCTGTTTGTATGGTGGTTAACGCGGGGATATAACATGAGCTGTCTTCGGTATCGTCGTAT
CCCCTACCGAGATATCCGACCAACGCGCAGCCGCGACTCGGTAATGGCGCGCATTTGCGCCAGCGCCATCTGA
TCGTTGGCAACAGCATCGAGTGGGAACGATGCCCTCATTCAGCATTTGATGGTTTTGTTGAAAACCGGACATG
GCATTCAGTCCGCTTCCGCTTCCGCTATCGGCTGAATTTGATTGCGAGTGAGATATTATGCCAGCCAGCCAGA
CGCAGACGCGCGAGACAGAACTTAATGGGCCCGCTAACAGCGCGATTTGCTGGTGACCCAAATGCGACCCAGATGC
TCACGCGCCAGTCGCTACCGCTGCTTTCATGGGAGAAAATAACTGTTGATGGGTGTCTGGTCAAGACATCAAGA
AATAACGCGCGAATATTAGTGAGGAGCTTCCACAGCAATGGCATCTGTCATCAGCGGATAGTTAATGATC
AGCCCACTGACGCGTTGCGCGAGAAGATTGTGACCCGCGCTTACAGGCTTCGACGCGGCTTCGTTCTACCATC
GACACCCACAGCTCGAGTGCAGTGGGAGGAGCTTAACTCGCGCGAGATTTAATCGCGGACAAATTCGACGCGCGTGACGG
GCCAGACTGGAGGTGGCAACGCAATCAGCAACGACTGTTTGCCTCGCAGTGTGTTGCCACGCGGTTGGGAATG
TAATTCAGCTCCGCCATCGCGCTTCCACTTTTTCCCGCGTTTTTCGAGAAACGTGGCTGGCCTGGTTCCACACG
CGGAAACCGTCTGATAGCAAGGAGTGGCGCCCAACAGTCCCGGCCACGGGAgatcaaaagcctccggtcgg
agggcttttgactTCTAGAGAGCTGTGACACTTTATGCTTCCGGCTCGTATAATGTGTGGAAATTGTGAGCGGA
TAACAAGTGAATTTGAGCGGATAAACAATTTACACAGGAAACAGAAATCCCATGGTGAAGGTGAGGAGCT
ATTAAAGGAGAACTGCACCTGAAGCTGTATATGGAGGGCAGCGTAAACAACCACTTCAAGTGTACCAACCGA
GGGTGAAGGTAAACCTACGAGGGGACGCAGACCCCAACGCATCAAGGTCGTGGAGGGCGGCCGCTGCCTTTCGC
ATTCGACATTTGCGCGCATGTTTTATGTACGGCTCGAAGACCTTCATCAACCACACCAAGGCATCCCGGACTT
CTTAAGCAGAGCTTCCCTGAGGGCTTACCTGGGAGCGCGTACCACGATGAAGACGGTGGGGTGTCTCACCGT
GACCCAGGACACGAGCTTGCAGGATGGCTGCTGATTTACAACTCAAGCTGCGCGGGGTGAACCTTCCCTAGCAA
CGGGCCAGTGTGACAGAAAAGACGCTGGGTGGGAGGCCACACCGAGACCTGTACCCGGCCGACGGGGGGCT
GGAAGGGCGGTGCGATATGAAGCTGAAATTTGGTGGCGCGCGCTCATTGCACTGCAATCTCAAGACACGCTACCG
CTCCAGAAACCCGCCAAAACCTGAAGATGCTGGTGTATTATTTGTGACCGGCGCTGGAGCGCATCAAGGA
ACGGGACAATGAGACGTACGTGGAACAGCACGAAGTGGCCGTGGCTCGTTATTGCGATCTGCCGTCGAAGCTGGG
TCACAACTGAACGCGATGGATGAGCTGTACAAAGATTATAAGGATGATGACGACAAGTAACCTCGAG

pg⁷⁰ V2TcR-sfGFP/V2LacI-mCardinal (SEQ ID NO: 67)

ggatccGAGCTGTTGACAACTCTATCATTGATAGAGTTATAATGTTCCTATCAGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGTCAAAGGTGAAGAGCTGTTACCGGCGCTCGTCCCATCTGGTGAAGCTGGATGGCGAC
GTCAACGGGCACAAGTTTAGCGTCCGTGGCGAAGGTGAGGGCGACGCCACGAACGGTGAAGCTGACGCTGAAATTC
ATTTGCCACACCGGCAAAATGCTGTATCCCTGGGCCACCTGGTGACCAAGCTCACCTACGGCGTACAGCTTCT
AGCCGTTACCCGGACCATGAAAGCTCACGACTTCTTCAAAGCGCCATGCCGAGGGGTACGTGACGAGAGCGT
ACGATTAGTTTCAAGGACGACGGCACCTATAAGACCCGTCGCGAAGTGAAGTTCGAAGGCGATACGTTGGTGAAC
CGTATCGAGTGAAGGGTATGCACTTTAAGGAAGACGGCAACATCTGGGCCATAAGCTGGAGTACAATTTCAAC
AGCCATAACGTTTACATCACCGCGATAAACAGAAGACGGCATTAAAGCCAACTTTAAGATCCGCCACAACGTC
GAAGACGGCTCGGTGACGTGGCCGACCATTTACGCAAAAACACCCCATCGGTGATGGGCGCGTGTCTGCTGCCG
GATAACCTATCTGAGACGACGAGTCGGTGTCTCAGCAAGGACCTTAAAGGAAAGCGCATCACATGGTGGTGGT
GAGTTCGTACGCGCGCGGGGATCACCATGGGATGGACGAGCTTACAAAGACTATAAAGATGACGATGACAAG
TAAAAGCTTCTCGGTACCAATTCAGAAAAGAGGCTCCCGAAAGGGGGGCTTTTTCTGTTTTGGTCCGAATT
CTTGACAGCTAGCTCAGTCTTAGGTATAATGCTAGCGCAGTAAGAGAGGAATGACACATGTCCTCCGCTGGATA
AATCGAAAGTGATTAACCTCGGCCCTCGAATTGCTGAATGAAGTCGGTATCGAGGGGTGACGACCCGTAATGG
CACAAAGTTGGGGGTGGAGCAACCCAGTTGTATTGGCAGCTCAAAAATAAGCGGGCATTTGCTGGATGCCCTCG
CTATTGAAATGTTGATCGCCACCTATCCATTTCTGTCCACTGGAGGGCGAGTCTTGGCAGGACTTTCTCCGCA
ACAACGCGAAATCCCTTTCGCTGTGCACTTCTGTCCCATCGGACGGTGTCTAAGGTGCACTTGGGCACCCGTC
CCGAAAAACAATACGAAACCTTGGAAAATCAATTGGCGTTTTTGTGCCAGCAAGGGTTAGCTTGGAGAATGCTC
TCTATGCGCTCTCGGCTGTGCGGCACTTACGTTGGGGTGCCTGTTGGAGGACAGGAGCATCAAGTCGCAAAAG
AGGAGCGTGAACCCCAACACGAGCTCGATGCCACCTCTGCTCCGCCAAGCTATCGAAGCTTTCGATCATCAGG

List of Plasmids

po⁷⁰ V2TcR-sfGFP/V3LacI-mCardinal (SEQ ID NO: 68)

ggatccGAGCTGTTGACAACTCTATCATGTATAGAGTTTAAATGTTCCCTTACGTGATAGAGACGCAGTAAGAG
AGGAATGTACATATGTCTCAAAGGTGAAGAGCTGTTTACCGGCGCTGTCGCCATCTCTGTTGGAGCTGGATGGCGAC
GTCAACGGCCACAAGTTTATCGCTCCGTGGCAGAGTGGAGGCGCGACGCAAGCCGTAAAGCTGACGTTGGAATCT
ATTTGCAACACCGGCAAAATGCTGTATCCCTGGCCACCTCGTGACCACGCTCACCTACCGCGGTACAGTGCTTC
AGCCGTTTACCCGGACACATGAAGCTGTCAAGCTCTTCTCAAAGCGCCATGCCGAGGGTTACGTGCAGGAGCGGT
ACGATTAGTTTCTAAGGACGACGCGCCTATAAGCAACCGTGCGAAGTGAAGTTTCAAGGCGCATACGTTGGTGAAC
CGTATCGAGTTGAAGGGTATCGACTTTAAGGAAGACGCGCAACATCTCTGGGCCATAAGCTGGAGTACAATTTCAAC
AGGCATAACGTTTACATCACCCCGGATAAACAAGAAAGCGGCATTTAAGCCAACTTGAAGTCCGCCACAACGCT
GAAGACGGCTCGGTGCGAGCTGGCCGACATTATCAGCAAAACACCCCATCGGTGATGGGCGCTGCTGCTGCCG
GATACCAACATTATCTGAGCACCGAGTCGGTGCTCAGCAAGGCACTTAACGAAAGCGCGCATACATGTGTGTCTGTG
GAGTTCGTCAACGCGCGCGGGGATCACCTATGGGATGGACGAGCTCTAAGAAAGCTATAAAGATGACGATGACAAG
TAAAGACTTCTCGTACCAATTTCCAGAAAGAGGCTCCGAAAGGGGGCTTTTTTCGTTTGGTTCGAAAT
CTTGACAGCTAGCTCAGTCTCATAGTATAATGTCTGCGCGAGTAAAGAGGAAATGTACACATGTCTCCCGCTGGATA
AATCGAAAGTGAATTAATCTCGGCCCTCGAATGTCTGAATGAAGTCGGTATCGAGGGCTGACAGCGCTGAATTTGG
CAGAAAAGTTGGGGGGTGAGACACCCAGCTGTGATTGGGACATCAAAAATAGCGGGCATTTGCTGGATGCCCTGC
CTAATGAAATGTGGATCGCCACCATAACCCATTCTGTCTGATCAGTGAGGGCGAGTCTTGGCAGGACTTTCTCCGCA
ACAACCGCGAAATCCTTTTCGCTGTGCATCTGTGCCATCGGACGGTGTCAAGGTGCACTTGGGCACTCCGCTCCA
CCGAAAAACATACGAAACTTGGAAAACTAATTGGCGTTTGTGTGCGACAGGGTTAGCTTGGAGATGCTC
TCTATCGCCTCTCGGCTGTGCGGCACTTTACGTTGGGGTGCGTGTGTGAGGACACAGGAGCATCAAGTCGCAAAAG
AGGAGCTGAAACCCCAACACCGGATCGATGCCACTCTGCTCGCCAAAGCTATCGAATCTTCGATCATCAGG
CGCGGAGCCAGCCTTCTCTTTGGGCTGGAGCTGATTATCTGCGGTTTGGAAAAACAACTCAAGTGTGAAGCG
GGTCTTAAGTCAGCTCACTGCCCGCTTTCCAGTTCGGGAAACCTGTCTGTCGACAGTGCATTAATGAATCGGCCAAC
GCGCGGGGAGAGCGGTTTTCGTTATTTGGGCGCCAGGGTGTGTTTTCTTTCACAGTGAGACGGGCAACAGCTGA
TGTCCCTTACCGCTGGCCCTGAGAGAGTTGCAGCAAGCGTCAAGCTGAGTTTGGCCAGCAGGACGAAATCC
TGTTTGATGTTGTTTAAACGCGGGGATAAATCATGAGCTGTCTTCGATATCGTCTATCCCACATCCGAGATATCC
GCACCAACCGCGCAGCTCGGACTCGGTAATGGCCGCGATTTGCGCCCGACGGCCATCGATCTGTGGCAACAGCATC
CGAGTGGGAACGATGCTCCCTATTGAGCTGTGATGGTTTGTGTAAGAACCGGACATGCGATCAGTCTGCCCTCC
CGTTTCGCTATCGGCTGAAATTCGATTGCGAGTGCAGTATTTATGCCAGCCAGCCAGACGCGAGACGCGCGGAGACA
CGCTTTAATGGGCGCGCTTAACAGCCGGATTTGCTGGTGACCCAGTGCACAGGATGCTCCACGCCCCAGTCGCGTA
GAACTTATCTGGGAGAAAAATAACTGTGTTGATGGGTGTCTGTGACAGACATCAAGAAATAACCGCGGAACATTA
GTGCAAGGACGCTTCCACGCAATTGGCATCTCTGTGATCCAGCGGATAGTTAATGATCAGCCCATGACGCGTTGCG
GCGAGAAGATTGTGCACCGCCGCTTTACAGGCTTCGACGCGCTTCGTTCTACCATCGACACCCACCGCTGGCA
CCGAGTTGATCGGCGGAGATTTAATCGCGCGCAATTTGCGACGGCGCTGTCAGGCGACAGTGGAGTGGCA
ACGCGAATCAGCAACGACTGTGTGCCCGCCAGTTGTTGTCGACCGGTTGGGAATGTAAATTCAGCTCCGCCATC
CGCGCTTCCACTTTTCCGCGCTTTTTCGCAAGAACGTGGCTGGCCCTGTTTACCACCGCGGAAACGCTGTGATAA
GAGACACCGGCATATCTCGCATCGTATAACGTTACTGGTTTCACTTACCTACACCTGAATTGACTCTCTTCC
GGGCGCTATATCGCCATACCGGAAAGGTTTTCGCCCAATTCGATGGTGTCCGGATCTCTGACGCTCTCCCTTATG
CGACTCTCTGATAGGAAGCAGGCCAGTAGTAGTTGAGCGCTGTGAGCACCGCGCCGCAAGGAATGGTGTGGA
CAAGGAGATGGCGCCACAGACTCCCCGCGCACGGGGagTcaaaagctccggtcgagggtcttgactCTAGA
GAGCTGTTTGACACTTTATGCTTCCGCTCGTATAATGTGTGTGGAATTTGTGAGCGGATTAACAGATGGAATTTGTA
GCGGATAACAATTTACACAGGAAACAGAAATCCCATGTGGTAGTAAGGTTGAGGAGCTTAAGGAGAACCATGCA
CATGAAGCTGTATATGGAGGCGACCGTAAACAAACACCACTTCAAGTGTACCAACGAGGGTGAAGTTAAACCTCA
CGAGGGGACGCGAGAACCCACGATCAAGGTCTGTGAGGGCGCGCCGCTGCTTTTCGATTCGCATTTCTGGCGAC
CTGTTTATATGAGCTCTGAAGCACTTCATCAACCAACCAACCAAGCACTCCCGGACTCTCTCAAGCAGAGCTTCC
TGAGGTTCTACCTGGGAGCGCTCACCAGTATGAAGACGCTGGGTTGCTCAGCTGACCCAGGACAGCAGCTCT

TABLE 3-continued

List of Plasmids

GCAGGATGGCTGCTTGATTACAACTCAAGCTGCGCGGGTGAACCTCCCTAGCAACGGGCCAGTGATGCAGAA
 AAAGACGCTGGGTTGGGAGGCCACACCGAGACCTGTACCCGGCCGACGGGGGGCTGGAAGGGCGGTGCGATAT
 GGCCCTGAAATTGGTGGCGGGCGGTCAATTGACATGCAATCTCAAGACCACGTACCGCTCCAAGAAACCCGCCAA
 AAACCTGAAGATGCTGTTGTTATTTTGTGCGACCGGCGCTGGAGCGCATCAAGGAAGCGGACAATGAGACGTA
 CGTGGAACAGCACGAAGTGGCGGTGCTGCTTATTGCGATCTGCCGTGAAAGCTGGGTCAAAAACCTGAACGGCAT
 GGATGAGCTGTACAAAGATTATAAGGATGATGACACAAGTAACTCGAG

pUC19-crteBI (SEQ ID NO: 69)

CATATGTATCCGTTTATAAGGACAGCCCGAATGACGGTCTGCGCAAAAAACACGTTTCTACTCGCGATGCT
 GCGGAGCAGTTACTGGCTGATATTGATCGACGCCCTTGATCAGTTATTGCCCGTGGAGGGAGAACGGGATGTTGTG
 AGTGCCCGCATGCGTGAAGGTGCGTGGCAGCCGGGAAACGTAATTCGCCCATGTTGCTGTTGCTGACCGCCCGC
 GATCTGGGTTGCGCTGTGAGCCATGACGGATTACTGGATTGCGCTGTGCGGTGGAATGGTCCACGCGGCTTCG
 CTGATCCTTGACGATATGCCCTGCATGGACGATGCGAAGCTGCGGCGCGGACGCCCTACCATTCTATTCTATTAC
 GGAGAGCATGTGGCAATACTGGCGCGGTGCTTGTCTGCTGAGTAAAGCCTTTGGCGTAATTGCCGATGCGATGGC
 CTCACGCCCTGGCAAAAATCGGCGGCTTCTGAACGTGCAACGCCCATCGGCATGCAAGGATTGGTTCAGGGT
 CAGTTCAAGGATCTGCTGAAGGGGATAAGCCGCGCAGCGCTGAAGCTATTTTGTGACGAATCACTTTAAACC
 AGCAGCGTGTGTTGTGCTCCATGCGATGGCCTCGATTGTTGCGAATGCGCTCAGCGAAGCGCGTATTGCTG
 CATCGTTTTTCACTTGATCTGGTCAAGCATTTCAACTGCTGGACGATTGACCGATGGCATGACCGACACCGGT
 AAGGATAGCAATCAGGACGCCGGTAAATCGACGCTGGTCAATCTGTTAGGCCCGAGGGCGGTGAAGAACGTCG
 AGACAACATCTTCAGCTTGCCAGTGAGCATCTCTCTGCGGCTGCCAACACGGGACGCCCACTCAACATTTTATT
 CAGGCTCGTTTGAACAAAACCTCGCTGCCGTGATTACGCGAGTAAAGAGGAATGTAGATATGAATAATCCGT
 CGTTACTCAATCATGCGGTGCAACGATGGCAGTTGGCTCGAAAGTTTTCGACAGCCTCAAAGTTATTTGATG
 AAAAAACCGGGCAGCTACTGATGCTTACGCTGCGGCTGGTCCGCCATTTGACGATGTTATTGACGATCAGACGC
 TGGGCTTTGAGCCCGGCGAGCTGCTTACAAACGCCGGAACAACGTCTGATGCAACTGAGATGAAACCGCGCC
 AGGCTATGACGAGATCGCATGACGCAACCGCGCTTTCGCGCTTTTCAGGAAGTGGCTATGGCTCATGATATCG
 CCGCGCTTACCGCTTTGATCATCTGGAAGGCTTCGCCATGGATGACGCGAAGCGCAATACAGCAACTGGATG
 ATACGCTGCGCTATTGCTATCAGTTGACGGCTTGTGCGCTGATGATGGCGCAATCATGGGCGTGGCGGATA
 ACGCCACGCTGGACCGCGCTGTGACCTTGGGCTGGCATTTCAGTTGACCAATATTGCTCGCGATATTGTGGAG
 ATGGCGCATGCGGGCGCTGTTTATCTGCGCGCAAGCTGGCTGGAGCATGAAGGTCTGAACAAAGAGAATTATGCGG
 CACCTGAAACCGCTCAGGCGCTGAGCCGTATCGCCCGTCTGTTGGTGCAGGAAGCAGAACCTTACTATTGCTCTG
 CCACAGCCGCGCTGGCAGGTTGCCCTTGCCTTCGCGCTGGGCAATCGCTACGCGAAGCAGGTTTACCGGAAAA
 TAGGTGTCAAAGTTGAAACGGCCGCTGAGCAAGCTGGGATCAGCGGCGAGTCAACGACACGCGCCGAAAAATTAA
 CGCTGCTGCTGGCCGCTCTGCTCAGGCCCTTACTTCCGGATGCGGGCTCATCTCCCCGCCCTGCGCATCTCT
 GGCAGCGCCCGCTCTGAATAATTTGTTTAACTTAAAGAGGAGATATAATGAACCAACTACGGTAATTTGGTG
 CAGGCTTCGCTGGCTGCGCTGGAATTCGCTTACAAAGCTGCGGGGATTCCTGCTTACTGCTGAACAACTG
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 GTGAGTGGGCGCTGCTGTTTCCGCTGGCGGCGACCGCGCATTAGTTTACGGGATGATAAAGCTGTTTACGATC
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po⁷⁰ V2Tcr-crteBI (SEQ ID NO: 70)

GGATCCGAGCTGTTGACAACTCTATCATGTAGAGTTATAATGTTCCCTATCAGTGATAGAGACGCAGTAAGAG
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 ACCGCCCCGATCTGGGTTGCGCTGTACGCCATGACGGATTACTGGATTGGCCGTGTGCGGTGGAATGGTCCAC
 GCGCTTCGCTGATCTGATCAATGATGCTGCTGCAAGGAGTGAAGCTGCGGCGCGGACGCCCTACCATTCTAT
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TABLE 3-continued

List of Plasmids

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 AACTGGCGAACTGCAGGCATGGAGAAGCGTTTACAGTAAGGTTGCCAGTTACATCGAAGATGAACATCTGCGCC
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po⁷⁰ V2TcR-crtEBIY (SEQ ID NO: 71)

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 ACCGCCCCGATCTGGGTTGCGCTGTCAGCCTATGACGGATTACTGGATTGCGCTGTGCGGTGGAAATGGTCCAC
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TABLE 3-continued

List of Plasmids

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po⁷⁰ V2Ter-crtEBI/V2LacI-crtY (SEQ ID NO: 72)
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TABLE 3-continued

List of Plasmids

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 CGGTGAGCACCAGCGCGCGCAAGGAATGGTGCATGCAAGGAGATGGCGCCCAACAGTCCCCCGGCCACGGGGagt
 caaaagcctccggtcgagagcttttgaactCTAGAGAGCTGTTGACACTTTATGCTTCCGGCTCGTATAATGTGT
 GTGGAATTGTGAGCGGATGTAACACCGCAGTAAGAGAGGAATGTACCATGGAGCGGCTATGCAACCGCATTTATGAT
 CTGATTCTCGTGGGGCTGAGCTCGCAATGGCCTTATCGCCCTCGCTCTCCAGCAGCAGCAACCTGATATGCGT
 ATTTTGCTTATCGACGCGCACCCAGCGGGCGGGAATCATACGTGGTCATTTACCCACGATGATTGACTGAG
 AGCCAACTCTGTTGGATGTAACACCGCTGGTGGTTTCTACTGCGCCGACTATCAGGTACGCTTTCCACACGCGCT
 CGTAAGCTGAACAGCGGCTACTTTTGTATTACTTCTCAGCGTTTCGCTGAGGTTTACAGCGACAGTTTGCGCCG
 CACTTGTGGATGGATACCGCGGTGCGAGAGTTAATGCGGAATCTGTTCCGTTGAAAAGGGTCAGGTTATCGGT
 GCCCGCGCGTGTATTAGCAGGGCGGGTTATGCGGCAAAATCAGCACTGAGCGTGGGCTTCCAGGCGTTTATTGGC
 CAGGAATGGCGATTGAGCCACCCGATGGTTTATCGTCTCCATTATCATGGATGCCACGCTCGATCAGCAAAAT
 GGTATCGCTTCGTGTACAGCTGCGGCTCTCGCGACCCAGATTGTTAATTGAAGATACGCACTATATTGATAAT
 GCGAATTAGATCTGTAATGCGCGGCAAAATATTGCGACTATGCGCGCAACAGGGTTGCGAGCTTCAGACA
 CTGCTGCGAGAAGAACAGGGCGCTTACCCATTACTCTGTGCGGCAATGCCAGCATTCTGGCAGCAGCGCCCC
 CTGGCCTGTAGTGGATTACGTGCCGCTGTCTCCATCTACCCAGCGCTATTACTGCGCGTGGCGGTTGCCGTG
 GCCACCGCTGAGTGCATCTGATGCTCTTTACGTGCGCTCAATTCACCATGCCATTACGCAATTTGCCCGCGAG
 CGCTGCGCAGCAGCGGCTTTTCCGCGATGCTGAATCGCATGCTGTTTTAGCCGAGCCCGCGATTACGCTGG
 CGGGTTATGACAGCGTTTTATGGTTTACCTGAAGATTAAATGCGCGTTTTATGCGGGAAAACCTACGCTGACC
 GATCGGCTACGTATTCTGAGCGCGCAAGCGCTGTTCCGGTATTAGCAGCATTGCAAGCCATTATGACGACTCAT
 CGTTGACTCGAG

po⁷⁰ V2Ter-crtEBI/V3LacI-crtY (SEQ ID NO: 73)
 ggatccgagctgttgacaaactctatcattgtagaggttataatgttccctatcagtgtagagagcgcagtaagag
 aggaatgtacatatgtatccgttttataaggacagccgaatgacggctctgCGCAAAAAACACGTTCTACTCTCACT
 CGCGATGCTGCGGAGCAGTAACTGAGCTGATTTGATCGACGCTTTCGATCAGTTATTGCGGCTGGAGGGAGACCG
 GATGTTGTGGGTGCGCGATGCGTGAAGGTGCGCTGGCACCGGAAAACGTATTGCGCCCATGTTGCTGTTGCTG
 ACCGCCCCGATCTGGGTTGCGCTGTGACGCTGACGCGATTACTGGATTGGGCTGTGCGGTGGAATGGTCCAC
 GCGGCTTCGCTGATCCTTGACGATATGCCCCGATGAGCAGTGCAGAGCTGCGGCGCGGACGCCCTACCATTCAT
 TCTCATTACGAGAGCATGTGGCAATACTGGCGGCGGTTGCCTTGCTGAGTAAAGCCTTTGGCGTAATTGCCGAT
 GCAGATGGCCTCACGCCGCTGGCAAAAATCGGGCGGTTTCTGAAGTGTCAACGCCATCGCGATGCAAGGATTG
 GTTCAGGGTCAAGTCAAGGATCTGCTGAAGGGGATAAGCCGCGCAGCGCTGAAGCTATTTTGATGACGAATCAC
 TTTAAACAGCAGCCTGTTTTGTGCCCTCATGAGATGGCCTCGATTGTTGCGAATGCCCTCAGCGAAGCGCGT
 GATTGCCCTGCATCGTTTTCACTTGATCTTGGTCAGGCATTTCAACTGCTGGACGATTTGACCGATGGCATGACC
 GACACCGGTAAGGATAGCAATCAGGACGCGCGTAAATCGACGCTGGTCAATCTGTTAGGCCGAGGGCGGTTGAA
 GAACGTCTGAGACAACATCTTCAGCTTGCCAGTGAGCATCTCTGCGGCTGCGCAACACGGGACCGCCACTCAA
 CATTTTATTCAGGCTGTTTGGAAAAAACTCGCTGCCGTCAGTTAACGCGAGTAAGAGAGGAATGTAGATATGA
 ATAATCGCTGTTTACTGATCTGCGGTCGAAACGATGGCAGTTGGCTCGAAAAGTTTTCGCGACGCTCAAGT
 TATTGATGCAAAAACCGGCGCAGCGTACTGATGCTTACGCTCGGTGCCGCTATTGTTGACGATGTTATTGACG
 ATCAGACGCTGGGCTTTAGGCCCCGCGAGCTGCTTACAAACGCCGCAACAGCTGTGATGCAACTTGAGATGA
 AAACGCGCGAGGCTATGCGAGGATGCGCGGCTGACCTTGAGGCTGGCATTTCAGTTGACCAATATTGCGCGATA
 ATGATATCGCCCCGCTTACGCGTTTGTATCATCTGGAAGGCTTCGCCATGGATGTACGCGAAGCGCAATACAGCC
 AACTGGATGATACGCTGCGCTATTGCTATCAGTTGCGAGGCTGTCGCGCTGATGATGGCGCAATCATGGGCG
 TGCGGGATAACGCCACGCGCTTGGACCGGCTGACCTTGCGCTGGCATTTCAGTTGACCAATATTGCGCGATA
 TTGTGGAGCATGCGCATGCGGCGCTGTTATCTGCCGCGAAGCTGGCTGGAGCATGAAGGTCTGAACAAAGAGA
 ATTATGGGCACTGAAAACCGCTCAGGCGCTGAGCCGTATCGCCCGTGGTTGGTGCAGGAAGCAGAACCTTACT
 ATTTGCTGCGCACGCGGCTTGGCAGGTTGCCCTGCGCTTCCGCTGGGCAATCGCTACGCGCAAGCAGGTT
 ACCGAAAATAGGTGTCAAGTTGAACAGGCGGTCAGCAAGCTGGGATCAGCGGCGAGTCAACGACACGCCCC
 AAAAATTAACGCTGCTGCTGCGCGCTCTGCTCAGGCGCTTACTTCCCGGATGCGGGCTCATCTCCCCGCGCTG
 CGCATCTCGGACGCGCGCTGCGAATAATTTGTTTAACTTAAAGAGGAGATATAATGAACCAACTACGCG
 TAATTGGTGCAGGCTTCGGTGGCTGGCACTGGCAATTCGCTACAAGCTGCGGGGATTCCTGCTTACTGCTTG

List of Plasmids

cocE DNA sequence (SEQ ID NO: 74)

ATGGTGGACGGTAATTATTCGGTAGCGTCCAACGTTATGGTGCCGATGCGCAGCGGGGTGCGCTTGGCTGTAGAT
CTTACCGCCCGGACGAGATGGCCCTGTCGGCGTCTGCGTGCGCAACCTCAGCAACAAATTCAGACGTGTTG
CGTTGGAGATACGAGAGACGAACTGGTGGAAATTTGTGGCAGTGGGATACCGCGTGTCTATCCAAAGACACCGG
GGCCTCTTTGATCCGAAGGTGAGTTCTGTTCCACATGTTGTAGCAGGCGCGATGCGGAAGACACGCTGAGCTGG
ATCTTGGAAACAGCATGGTGCGACGGCAATGTGGGTATGTTCGGTGTAAAGCTACTTGGCGGCTTACGACGTGGCAA
GCTGCTGTAGCGGTGTGGGTGTTTGAAGGCAATGCCCGAGATGCGCAGCGGATCTGTACGTGCCCC
TGTGATCGGCTCTGGCGGCCCTTGAGCGTGAAGACACTCTGGGCTGAGACGGATGATCGGTACGGCGCTGATT
ACCAGCCGTAGCAGTGGCCCGCGGAAGACAGCCGACTTCGTACAGCTGGCAGGCATCTGAACGATGTGGCC
GGTGGCCGAAGCGTGACCCCTCTGGCCGAACAGCCCTTGTGGCGCGCTGATCCTTGGGTGATGACACAGGTG
GTGGCGATCTCAGACACGACGAGTGTGGGACGAGCATCTCGCTCTTGAACGTTTGGGTGGGCTGGAACCGCC
GCCTTGATTACCGCGGGTGTACATGGCTTCTGGGGCAGAGCCTTCCGATCTTGTAGCTGTGAAGACCAAC
GCGGATCGCGCTCTGTGGTGGGGCGTGAGGACACAGCAATCTGACCGCGCGTAATGCCGACGTAAAGTTGGG

TABLE 3-continued

List of Plasmids

ATCGCCGCGACCTACCCATCCAGGAGGCGACGACCATGCACAAGGCTTTTTTCGACCGGCACCTCCGTGGCGAG
 ACCGATGCCCTGGCAGGGTGCCCAAGGTGCGCCTCTTCGTAATGGGTATCGATGAGTGGCGCGACGAGACCGAC
 TGGCCATTGCCAGATACCGCTTACACGCGCTTTTACCTCGGGGGCTCCGGTGGCGGCCAACACGAGCACGGGTGGT
 GGGACCTGTGCGACCTCGATCAGCGGCACGGAGTCGGCGGACACCTACCTGTATGATCCTGCCGACCCCGTGCCA
 AGTCTGGGCGGCACCTCCTCTTCCATAATGGGACACCGGTCCAGCTGACCGAGCGCCGATTACGATCGCGAC
 GACGTGCTGTGCTACTCCACCAGGTGTTGACCGACCCCGTGAAGTAACGGGGACGGTTTCGGCTCGCCTGTTTC
 GTGTCTCTCGTGGCGGTGGATACCGATTTTACCGCCAAGTTGGTCGACGTGTTCCCGATGGTCGGGCAATCGCT
 CTCTGCGACGGCATCGTGGGTATGCGCTACCGGGAGACCTGGTAAATCCTACGCTCATTGAGGCGCGTGAGATT
 TACGAGGTGGCTATTGATATGCTGCCACCAGCAACGTGTTTTCGCGGGCCACCGCATCATGGTGCAAGTTAGC
 AGCTCGAACTTCCCGAAGTACGACCGCAACTCCAACACCGCGCGCGTCATCGCTCGCGAGCAACTGGAGGAAATG
 TGCACCGCGCTAAACCGCATTACCGCGGCGCCGAACACCCGTCCTATATCGTGTGCGCGATCATTAAGCGCGAC
 TATAAGGACGACGACGATAAGTGA

cocE Amino acid sequence (SEQ ID NO: 75)

MVDGNYSVASNVMPMRDGVRLAVLDLYRPDADGVPVLLVRNPYDKEDVFAWSTQSTNWLEFVRDGYAVVIQDTR
 GLFASEGEFVPHVDDEADAEDTLWILEQAWCDGNVGMFGVSYLGVTTQWQAASVGVGGLKAIAPSMASADLYRAP
 WYGPGGALSVLEALLCGWSALIGTGLITSRSARPEDAADFVQLAAILNDVAGAASVTPLAEQPLLGRLLIPWVIDQV
 VDHPNDSEWQGISLIFERLGLLATPALITAGWYDGFVGESLRTFVAVKDNADARLVVGPWSHSLNLTGRNADRKFG
 IAATYPIQEATTMHKAFDRHLRGETDALAGVVKVRLVFMGIDEWRDETDPWLPDTAYTTPFYLGSGAANTSTGG
 GTLSTSISGTESADTYLYDPAFVPSLGGTLLFHNGDNGPADQRPIDHRDDVLCYSTEVLTDPVEVTGTVSARLF
 VSSSAVDTFDAKLVDVFPDGRAIALCDGIVRMRVRETLVNPTLIEAGEIYEVAIMDLATSNVELPGHRIMVQVS
 SSNPKYDRNSNTGGVIAREQLEEMCTAVNRIHRGPEHPSHIVLP I IKRDYKDDDDK-

crtE (SEQ ID NO: 76)

ATGTATCCGTTTATAAGGACAGCCCGAATGACGGTCTGCGCAAAAAACACGTTTCATCTCACTCGCGATGCTGCG
 GAGCAGTTACTGGCTGATATTGATCGACGCTTGATCAGTTATTGCCCGTGGAGGGAGAACGGGATGTTGTGGGT
 GCCGCGATGCGTGAGAGGTGCGCTGGCACCAGGAAACGATATTGCGCCCATGTTGCTGTGCTGACCGCCCGCGAT
 CTGGGTGCGCTGTGAGCCATGACGGATTACTGGATTGGCCTGTGCGGTGGAATGGTCCACGCGGCTTCGCTG
 ATCCTTGACGATATGCCCTGCATGGACGATGCGAAGCTGCGCGCGGACGCGCTTACCATTCACTCTATTACCGA
 GAGCATGTGGCAATACTGCGCGCGGTTGCCTTGCTGAGTAAAGCCTTTGGCGTAATTGCCGATGAGATGGCCTC
 ACGCCGCTGGCAAAAAATCGGGCGGTTTCTGAACGTGCAACCGCCATCGGCATGCAAGGATTGGTTCAGGGTCAG
 TTCAAGGATCTGCTGAAGGGGATGAACCGCGCAGCGCTGAAGCTATTGATGACGAATCACTTAAACACGAGC
 ACGCTGTTTTGTGCTCCATGAGATGGCCTCGATTGTTGCGAATGCCCTCCAGCGAAGCGGTGATTGCTGCTG
 CGTTTTCTCACTTGATCTTGGTCAGGCATTTCAACTGCTGGACGATTGACCGATGGCATGACCGACACCGGTAAG
 GATAGCAATCAGGACCGCGTAAATCGACGCTGGTCAATCTGTTAGGCCCGAGGCGGTTGAAGAAGCTCTGAGA
 CAACATCTTCAGCTTGCCAGTGAGCATCTCTGCGGCTGCCAACACGGGCACGCGCTCAACATTTTATTCAG
 GCCTGGTTTGACAAAAAATCGCTGCCGTCAGTTAA

crtB (SEQ ID NO: 77)

ATGAATAATCCGTCGTTACTCAATCATGCGGTGCGAAGCATGGCAGTTGGCTCGAAAAGTTTTGCGACAGCCTCA
 AAGTTATTTGATGTCGAAACCGGCGCAGCGTACTGATGCTTACGCTGGTGGCGCATTTGTGACGATGTTATT
 GACGATCAGACGCTGGGCTTTAGGCCGCGCAGCCTGCCTTACAAACCGCCGAACACGCTGATGCAACTTGAG
 ATGAAAACGCGCCAGGCGCTATGCGAGATCGCAGATGACGAAACCGGCGTTTGCGGCTTTTCAGGAAGTGGCTATG
 GCTCATGATATCGGCCCGGCTTACGCGTTGATCATCTGGAAGGCTTCGCCATGGATGTACGCGAAGCGCAATAC
 AGCCAACTGGATGATACGCTGCGCTATTGCTATCAGCTTGACGGCGTTGTCGGCTTGATGATGGCGCAATCATG
 GGCGTGCGGGATAACGCCACGCTGGACCGCGCTGTGACCTTGGGCTGGCATTTCAGTTGACCAATATTGCTCGC
 GATATTGTTGGACGATGCGCATGCGGGCGCTGTTATCTGCGCGCAAGCTGGCTGGAGCATGAAGGTCTGAACAAA
 GAGAATTATGCGGCACCTGAAAACCGTCAGGCGCTGAGCGCTATCGCCCGTCGTTTGGTGAGGAAGCAGAACCT
 TACTATTTGTCTGCGCACAGCCGGCTGGCAGGGTTGCCCTGCGTTCCGCTGGCAATCGCTACGGCGAAGCAG
 GTTTACCGGAAATAGGTGTCAAAGTTGAACAGGCGCGTCAGCAAGCCTGGGATCAGCGGAGTCACAGACACG
 CCCGAAAAATTAACGCTGCTGCTGGCCGCTCTGCTCAGGCCCTTACTTCCCGATGCGGGCTCATCTCCCCGC
 CCTGCGCATCTCTGCGACGCGCCGCTCTAG

crtI (SEQ ID NO: 78)

ATGAAACCACTACGGTAATTGGTGACGGCTTCGGTGGCCTGGCACTGGCAATTCGTCTACAAGCTGCGGGGATT
 CCGCTCTTACTGCTTGAACACGATGATAAACCCGCGGTGCGGCTTATGCTACGAGGATCAGGGGTTTACCTTT
 GATGACGCGCCGACGCTTATCACCAGTCCAGTGCCATTGAAGAACTGTTGCACTGGCAGGAAAACAGTTAAAA
 GAGTATGTGCAACTGCTGCCGTTTACGCCGTTTACCGCTGTGTTGGGAGTCAGGGAAGGCTTTAATTACGAT
 AACGATCAAAACCCGCTCGAAGCGCAGATTGACGAGTTAATCCCGCGATGTCGAAGGTTATCGTCAGTTTCTG
 GACTATTCACGCGCGGTGTTTAAAGAAAGGCTATCTAAAGCTCGGTACTGTCCCTTTTTATCGTTTACGAGACATG
 CTTCCGCGCGCACCTCAACTGGCGAAATGACGAGCATGGAGAAGCGTTTACAGTAAGGTTGCCAGTTACATCGAA
 GATGAACATCTGCGCGCAGCGGTTTCTTTCCTCACTCGCTGTTGGTGGGCGGCAATCCCTTCGCCACCTCATCCATT
 TATACGTTGATACACGCGCTGAGGCGTGAGTGGGCGCTGTTTCCGCGTGGCGGCACCGCGCATTAGTTTCAG
 GGGATGATAAAGCTGTTTCAGGATCTGGGTGGCGAAGTCGTGTTAAACGCCAGAGTCAGCCACATGGAACGACA
 GGAAACAGAGTTGAAGCCGTGCTATTAGAGGACGGTCGAGGTTCTGACGCAAGCGCTCGCGTCAAAATGACAGAT
 GTGGTTTATACCTATTCGCGACCTGTTAAGCAGCACCCCTGCGCGGTTAAGCAGTCCAAACAAATGACAGACTAAG
 CGCATGAGTAACCTCTCTGTTTGTGCTCTATTTTGGTTTGAATCACCATCATGATCAGCTCGCGCATCACACGGTT
 TGTTCGCGCCGCGTATGATGCTCGGCTTACGAGTACGAGTGGTCAACGACCGGATGTTTACGCGCTTGTATTTTCG
 GACCGAGCTTAATGCTATCATGGCTCAGCCTTTTCTGTGGAGCCGTTTCTACCGAGCGCCTGGTTTCGCGCG
 CATAACCGCGATAAAACCACTTACTAATCTCTACCTGGTGGCGCAGGCACGATCCCGCGCAGGCATTCTGGC
 GTATCGGCTCGGCAAAAGCGACACAGGTTTGAATGCTGGAGGATCTGATTGA

crtY (SEQ ID NO: 79)

ATGGGAGCGGCTATGCAACCGCATTATGATCTGATTCTCGTGGGGCTGGACTCGCGAATGGCCTTATCGCCCTG
 CGTCTCCAGCAGCAGCAACCTGATATGCGTATTTGCTTATCGACGCCGACCCAGCGGGCGGGAATCATACG

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SEQUENCE: 7
gagctgttga cactttatgc ttccggctcg tataatgtgt gtggaattgt gagcggataa 60
caacgcagta agagagggaat gtacatatg 89

SEQ ID NO: 8      moltype = DNA length = 112
FEATURE          Location/Qualifiers
source           1..112
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 8
gagctgttga cactttatgc ttccggctcg tataatgtgt gtggaattgt gagcggataa 60
caagtggaaat tgtgagcggga taacaatttc acacaggaaa cagaatcata tg 112

SEQ ID NO: 9      moltype = DNA length = 106
FEATURE          Location/Qualifiers
source           1..106
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 9
ctttatgtct cgggctcggt gacagtgtgg aattgtgagc ggataacaat ataatgtgtg 60
gaattgtgag cggataacaa ttccacacag gaaacagaat catatg 106

SEQ ID NO: 10     moltype = DNA length = 71
FEATURE          Location/Qualifiers
source           1..71
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 10
ggatccttga cactctatca ttgatagagt tattttacca ctccctatca gtgatagaga 60
aaagtgaat g 71

SEQ ID NO: 11     moltype = DNA length = 86
FEATURE          Location/Qualifiers
source           1..86
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 11
ggatccttga cactctatca ttgatagagt tattttacca ctccctatca gtgatagaga 60
cgcagtaaga gaggaatgta catatg 86

SEQ ID NO: 12     moltype = DNA length = 84
FEATURE          Location/Qualifiers
source           1..84
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 12
gagctgttga caactctatc attgatagag ttataatgtt ccctatcagt gatagagacg 60
cagtaagaga ggaatgtaca tatg 84

SEQ ID NO: 13     moltype = DNA length = 107
FEATURE          Location/Qualifiers
source           1..107
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 13
gagctgttga caactctatc attgatagag ttataatgtt ccctatcagt gatagagagt 60
ggaattgtga gcgataacaa atttcacaca ggaaacagaa tcatatg 107

SEQ ID NO: 14     moltype = DNA length = 106
FEATURE          Location/Qualifiers
source           1..106
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 14
ttgacactct atcattgata gagtttgaca tccctatcag tgatagagat ataatgtgtg 60
gaattgtgag cggataacaa ttccacacag gaaacagaat catatg 106

SEQ ID NO: 15     moltype = DNA length = 81
FEATURE          Location/Qualifiers
source           1..81
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 15
attgcctcga gactgatttt taaggcgact gatgagtcgc cttttttttg tctagctaac 60
tcacattaat tgcgttgccc t 81

SEQ ID NO: 16     moltype = DNA length = 72
FEATURE          Location/Qualifiers
source           1..72

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mol_type = other DNA
organism = synthetic construct

SEQUENCE: 16
gcaatggatc cagtcaaaag cctccgaccg gaggttttg actagtacaa tctgctctga 60
tgccgcatag tt 72

SEQ ID NO: 17      moltype = DNA length = 33
FEATURE           Location/Qualifiers
source            1..33
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 17
ggcgcgtagt ccaaaaggat ctaggtgaag atc 33

SEQ ID NO: 18      moltype = DNA length = 32
FEATURE           Location/Qualifiers
source            1..32
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 18
gagttcttct gattagacaa aaaaaaggcg ac 32

SEQ ID NO: 19      moltype = DNA length = 35
FEATURE           Location/Qualifiers
source            1..35
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 19
tttttttgtc taatcagaag aactcgtcaa gaagg 35

SEQ ID NO: 20      moltype = DNA length = 32
FEATURE           Location/Qualifiers
source            1..32
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 20
gcccgcgaggc gaggatctcg tcgtgacgca tg 32

SEQ ID NO: 21      moltype = DNA length = 31
FEATURE           Location/Qualifiers
source            1..31
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 21
cacgcagaga tcctcgccgt cgggcacccg c 31

SEQ ID NO: 22      moltype = DNA length = 51
FEATURE           Location/Qualifiers
source            1..51
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 22
gtctagactg caggaattca agcttcatat gggatccgga ccaaaacgaa a 51

SEQ ID NO: 23      moltype = DNA length = 51
FEATURE           Location/Qualifiers
source            1..51
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 23
aagcttgaat tcctgcagtc tagaccatgg ctcgaggacg aacaataagg c 51

SEQ ID NO: 24      moltype = DNA length = 31
FEATURE           Location/Qualifiers
source            1..31
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 24
cctagatcct tttggagtac gcgccgggg a 31

SEQ ID NO: 25      moltype = DNA length = 21
FEATURE           Location/Qualifiers
source            1..21
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 25
tactggtttc acattcacca c 21

SEQ ID NO: 26      moltype = DNA length = 22

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FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 26		
gtggtgaatg tgaaccagt aa		22
SEQ ID NO: 27	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 27		
tttcagtcg ggaaacct		18
SEQ ID NO: 28	moltype = DNA length = 18	
FEATURE	Location/Qualifiers	
source	1..18	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 28		
aggtttcccg actggaaa		18
SEQ ID NO: 29	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 29		
ccaatacaac gtgggttgct		20
SEQ ID NO: 30	moltype = DNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 30		
aaccggaatt ccccggtggc cggg		24
SEQ ID NO: 31	moltype = DNA length = 27	
FEATURE	Location/Qualifiers	
source	1..27	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 31		
aaccgctcga gtcactgccc gctttcc		27
SEQ ID NO: 32	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 32		
attgcctgca gtcactgccc gctttccagt cg		32
SEQ ID NO: 33	moltype = DNA length = 62	
FEATURE	Location/Qualifiers	
source	1..62	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 33		
gcaattctag aagtaaaaag cctccgaccg gaggttttg actccccgtg gccgggggac	60	
tg	62	
SEQ ID NO: 34	moltype = DNA length = 49	
FEATURE	Location/Qualifiers	
source	1..49	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 34		
gcctccggtc ggaggctttt gacttctaga gagctgttga cactttatg	49	
SEQ ID NO: 35	moltype = DNA length = 50	
FEATURE	Location/Qualifiers	
source	1..50	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 35		
taatgagtc ctcaccctta ctcaccatgg gtacattcct ctcttactgc	50	

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SEQ ID NO: 36      moltype = DNA length = 25
FEATURE           Location/Qualifiers
source            1..25
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 36
taatgagctc ctcaccctta ctcac                                25

SEQ ID NO: 37      moltype = DNA length = 23
FEATURE           Location/Qualifiers
source            1..23
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 37
catgggattc tgtttcctgt gtg                                23

SEQ ID NO: 38      moltype = DNA length = 34
FEATURE           Location/Qualifiers
source            1..34
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 38
tcagtctcga gttacttgtc gtcacatccc ttat                    34

SEQ ID NO: 39      moltype = DNA length = 759
FEATURE           Location/Qualifiers
source            1..759
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 39
atgggtgagta aggggtgagga gctcattaag gagaacatgc acatgaagct gtatatggag 60
ggcacccgtaa acaaccacca cttcaagtggt accaccgagg gtgaaggtaa accctacgag 120
gggacgcgaga cccaacgcac caaggtcgtg gagggcgccc cgctgccttt cgcattcgac 180
attctggcga cctgttttat gtacggctcg aagaccttca tcaaccacac ccaaggcctc 240
ccggacttct tcaagcagag cttccctgag ggcttcacac gggagcgctg caccacgcat 300
gaagacggtg ggggtgctcac cgtgacccag gacacgagct tgcaggatgg ctgcttgatt 360
tacaacgtca agctgcgcgg ggtgaacttc cctagcaacg ggccagtgat gcagaaaaag 420
acgctggggtt gggaggccac caccgagacc ctgtaccggg ccgacggggg gctggaaggg 480
cgggtgcgata tggccctgaa attggtcggc ggcggtcatt tgcactgcaa tctcaagacc 540
acgtaccgct ccaagaaacc cgccaaaaac ctgaagatgc ctggtgttta ttttgtcgac 600
cggcgccctgg agcgcatcaa ggaagcggac aatgagacgt acgtggaaca gcacgaagtg 660
gcgctggctc gttattgcga ttgcccgtcg aagctgggtc acaaaactgaa cggcatggat 720
gagctgtaca aagattataa ggtgatgacg gacaagtaa                    759

SEQ ID NO: 40      moltype = DNA length = 742
FEATURE           Location/Qualifiers
source            1..742
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 40
atgtccaaag gtgaagagct gtttaccggc gtcgtgccca ttctgggtgga gctggatggc 60
gacgtcaacg ggcacaagtt tagcgtccgt ggcgaagggt agggcgacgc cacgaacggt 120
aagctgacgc tgaattcat tgtcaccacc ggcaaatgac ctgtaccctg gccaccctg 180
gtgaccacgc tcacctacgg cgtacagtgc ttcagccgtt acccggaaca catgaagcgt 240
cacgacttct tcaaaagcgc catgcccggg ggttacgtgc aggagcgtac gattagtctc 300
aaggacgacg gcacctataa gacccgtgcc gaagtgaagt tcgaaggcga tacgttggtg 360
aaccgtatcg agttgaaggg tatcgacttt aaggaagacg gcaacatcct gggccataag 420
ctggagtaca atttcaacag ccataacggt tacatcaccg ccgataaaca gaagaacggc 480
attaaagcca actttaagat ccgccacaac gtccgaagac gctcggtgca gctggccgac 540
cattatcagc aaaaacacccc catcggtgat gggcccgtgc tgcgtccgga taaccattat 600
ctgagcacgc agtcggtgct cagcaaggac cctaacgaaa agcgcgatca catggtgctg 660
ctggagttcg tcacggcgcc ggggatcacc catgggatgg acgagctcta caaagactat 720
aaagatgacg atgacaagta aa                    742

SEQ ID NO: 41      moltype = DNA length = 624
FEATURE           Location/Qualifiers
source            1..624
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 41
atgtcccgcc tggataaatc gaaagtgatt aactcggccc tcgaattgct gaatgaagtc 60
ggtatcgagg ggcgtgacgac ccgtaaattg gcacaaaagt tgggggtgga gcaacccacg 120
ttgtattggc acgtcaaaaaa taagcgggca ttgctggatg cctcgctat tgaatgttg 180
gatcgccacc ataccattt ctgtccactg gagggcgagt cctggcagga ctttctccgc 240
aacaacgcga aatcctttcg ctgtgcactc ttgtcccatc gggacggtgc taaggtcgac 300
ttgggcaccc gtcccaccga aaaacaatac gaaaccttgg aaaaatcaatt ggcgtttttg 360
tgccagcaag ggtttagctt ggagaatgct ctctatgcgc tctcggtgtg cgggcacttt 420
acgttggggg gcgtgttgga ggaccaggag catcaagtcg caaaagagga gcgtgaiaacc 480

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ccaaccacgg actcgatgcc acctctgctc cgccaagcta tcgaactctt cgatcatcag 540
ggcgcgaggc cagccttcct ctttgggctg gagctgatta tctgcggttt ggaaaaacaa 600
ctcaagtgtg aaagcgggtc cttaa                                624

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SEQ ID NO: 42      moltype = DNA length = 1758
FEATURE           Location/Qualifiers
source            1..1758
                  mol_type = other DNA
                  organism = synthetic construct

```

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SEQUENCE: 42
catatggtgg acggttaatta ttccgtagcg tccaacgta tgggtccgat gcgcgacggg 60
gtgcgcttgg ctgtagatct gtaccgcccc gacgcagatg gccctgtacc ggctcgtgctg 120
gtccgcaacc cctacgacaa attcgacgtg ttccgttggg gtacgcagag cacgaactgg 180
ctggaaattt tgccgcatgg gtaccgctgc gtcattccaa acaccggggg cctctttgca 240
tcggaaggtg agttcgttcc acatgttgat gacgaggcgg atgcggaaga cacgctgagc 300
tggatcttgg aacaagcatg gtgcgacggc aatgtgggta tggctcgtgt aagctacctg 360
ggcgttacgc agtggcaagc tgctgttagc ggtgtgggtg gtttgaaggc aatcgccccg 420
agcatggcga gcgcggatct gtaccgtgcc ccttggtacg gtctggcggc cgccctgagc 480
gtggaagcac tcctgggctg gagcgcatg atcggtacgg gcctgattac cagccgtagc 540
gatgccccgc cggaagacgc agccgacttc gtacagctgg cagccatcct gaacgatgtg 600
gcccgtgccc caagcgtgac cctcttgccc gaacagccct tggttggcgg cctgatccct 660
tgggtgatcg accaggtggt ggaccatcca gacaacgacg agtcgtggca gagcatctcg 720
ctctttgaac gtttgggtgg gctcgtacc cgggccttga ttaccgcccg gtggtacgat 780
ggcttcgtgg gcgagagcct ccgtaccttc gtagctgtga aggacaaagg ggatgcgcgt 840
ctggtggtgg ggcggtggag ccacagcaat ctgaccggcc gtaatgccga ccgtaagtgt 900
gggatcgccc cgacctaccc catccaggag gcgacgacca tgcacaaggc tttttcgac 960
cggcacctcc gtggcgagac cgatgccctg gcagggggtg ccaaggtgcg cctcttcgta 1020
atgggtatcg atgagtggcg cgacgagacc gactggccat tgccagatac cgcttacacg 1080
cctttttacc tcgggggctc cggtgcggcc aacacgagca cgggtggtgg gaccctgtcg 1140
acctcgatca gcggcacgga gtcggcggac acctacctgt atgacccctg cgaccccgctg 1200
ccaagtctgg gcggcacctc cctcttccat aatggggaca acggtccagc tgaccagcgc 1260
ccgattcacg atcgcgacga cgtgctgtgc tactccaccg aggtgttgac cgaccccgctg 1320
gaagtaacgg ggacggtttc ggctcgctg ttctgttcct cgtcggcctg ggataccgat 1380
tttaccgcca agttggtcga cgtgttcccc gatggtcggg caatcgctct ctgcgacggc 1440
atcgctcgta tgcgtaccg ggagaccttg gtaaatccta cgtcattga ggcgggtgag 1500
atttacgagg tggctattga tatgctggcc accagcaacg tgtttttgcc gggccaccgc 1560
atcatggtgc aagttagcag ctcgaaactc ccgaagtacg accgcaactc caacacgggc 1620
ggcgctcatg ctgcgagaca actggaggaa atgtgcaccg ccgtaaacgg cattcaccgc 1680
ggccccgaac acccgctcca tatcgtgctg ccgatcatta agcgcgacta taaggacgac 1740
gacgataagt gaaagctt                                1758

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SEQ ID NO: 43      moltype = DNA length = 58
FEATURE           Location/Qualifiers
source            1..58
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 43
ttgacagcta gctcagtcct aggtataatg ctaccgacg taagagagga atgtacac 58

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SEQ ID NO: 44      moltype = length =
SEQUENCE: 44
000

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SEQ ID NO: 45      moltype = length =
SEQUENCE: 45
000

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SEQ ID NO: 46      moltype = length =
SEQUENCE: 46
000

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SEQ ID NO: 47      moltype = length =
SEQUENCE: 47
000

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SEQ ID NO: 48      moltype = length =
SEQUENCE: 48
000

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SEQ ID NO: 49      moltype = length =
SEQUENCE: 49
000

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SEQ ID NO: 50      moltype = length =
SEQUENCE: 50
000

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SEQ ID NO: 51      moltype = length =
SEQUENCE: 51

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-continued

000

SEQ ID NO: 52 moltype = length =
 SEQUENCE: 52
 000

SEQ ID NO: 53 moltype = length =
 SEQUENCE: 53
 000

SEQ ID NO: 54 moltype = DNA length = 2334
 FEATURE Location/Qualifiers
 source 1..2334
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 54
 gagctgttga caactctatc attgatagag ttataatggt ccctatcagt gatagagacg 60
 cagtaagaga ggaatgtaca tatgaagcgt ctcggtacca aattccagaa aagaggcctc 120
 ccgaaagggg ggcctttttt cgttttggtc cgaattcttg acagctagct cagtcctagg 180
 tataatgcta gccgcagtaa gagaggaatg tacacatgtc ccgcctggat aaatcgaaag 240
 tgattaactc ggcctcgaa ttgctgaatg aagtcgggat cgaggggctg acgaccctga 300
 aattggcaca aaagtggggg gtggagcaac ccacgttgta ttggcacgtc aaaaataagc 360
 gggcattgct ggaatgcctc gctattgaaa tgttggatcg ccaccatacc catttctgtc 420
 cactggaggg cgagtccttg caggactttc tccgcaacaa cgcgaaatcc ttctcgtgtg 480
 cactcttgct ccacggggac ggtgctaagg tgcacttggg caccgtccc accgaaaaac 540
 aatacgaaac cttggaaaat caattggcgt ttttgtgcca gcaagggttt agcttggaga 600
 atgctctcta tgcgtctctg gctgtcgggc actttacgtt ggggtgctg ttggaggacc 660
 aggagcatca agtcgcaaaa gaggagcgtg aaaccccaac cacggactcg atgccacctc 720
 tgcctcgcca agctatcgaa ctcttcgata atcaggcgcg ggagccagcc ttctctttg 780
 ggctggagct gattatctgc ggtttgga aaacactcaa gtgtgaaagc gggctcctaac 840
 tgcagtcact gcccgcttcc cagtcgggaa acctgtcgtg ccagctgcat taatgaatcg 900
 gccaacgcgc ggggagaggg ggtttgcgta ttgggcgcca ggggtggttt tcttttcacc 960
 agtgagacgg gcaacagctg attgcccttc accgcctggc cctgagagag ttgcagcaag 1020
 cgggtccacg tgggtttgcc cagcaggcga aaatcctgtt tgatggtggt taacggcggg 1080
 atataacatg agctgtcttc ggtatcgtcg tatccacta ccgagatata cgcaccaaacg 1140
 cgcagcccgg actcggtaat ggcgcgcat cgcgccagcg ccactctgatc gttggcaacc 1200
 agcatcgcat tgggaacgat ccctcattc agcatttgca tggtttggttg aaaaccggac 1260
 atggcactcc agtcgcttcc ccgttccgct atcggctgaa tttgattgct agtgagatat 1320
 ttatgccagc cagccagcag cagacgcgcc gagacagaac ttaatgggac cgctaacagc 1380
 tgcgatttgc ggtgaaccaa tgcgaccaga tgcctcacgc ccagtcgctg accgtcttca 1440
 tgggagaaaa taatactggt gatgggtgtc tggtcagaga catcaagaaa taacgccgga 1500
 acattagtgc aggcagcttc cacagcaatg gcatcctggt catccagcgg atagttaatg 1560
 atcagcccaac tgcgcgcttg cgcgagaaga ttgtgcaccg ccgctttaca ggcttcgacg 1620
 ccgcttctgt ctaccatcga caccaccacg ctggcaccca gttgatcgcc gcgagattta 1680
 atcgcgcgca caatttgcga cggcgcgtgc agggccagac tggaggtggc aacgccaatc 1740
 agcaacgact gtttgccttc gcttctgtt gccaacgggt tgggaatgta attcagctcc 1800
 gccatcgccg cttccacttt ttcccgctt ttccgagaaa cgtggctggc ctgggttcacc 1860
 acgcccggaaa cggctctgata agagacaccg gcatactctg cgacatcgta taacgttact 1920
 ggtttcacat tcaccacct gaattgaact tcttcgggac gctatcatgc cataccgca 1980
 aaggttttgc gccattcgat ggtgtccggg atctcgacgc tctcccttat gcgactcctg 2040
 cattaggaag cagcccagta gtagggttag gccgttgagc accgcgcgcg caagggaatg 2100
 tgcattgcaa gagatgcgc ccaacagtc ccgcgcacag gggagtcaaa agcctccggg 2160
 cggaggcttt tgactttctg agagctgttg acactttatg cttccggctc gtataatgtg 2220
 tgttgaattg tgagcggata acaacgcagt aagagaggaa tgtacccatg gccatggctc 2280
 gaggacgaac aataaggcct ccctaaccgg gggccttttt tattgataac aaaa 2334

SEQ ID NO: 55 moltype = DNA length = 2351
 FEATURE Location/Qualifiers
 source 1..2351
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 55
 gagctgttga caactctatc attgatagag ttataatggt ccctatcagt gatagagacg 60
 cagtaagaga ggaatgtaca tatgaagcgt ctcggtacca aattccagaa aagaggcctc 120
 ccgaaagggg ggcctttttt cgttttggtc cgaattcttg acagctagct cagtcctagg 180
 tataatgcta gccgcagtaa gagaggaatg tacacatgtc ccgcctggat aaatcgaaag 240
 tgattaactc ggcctcgaa ttgctgaatg aagtcgggat cgaggggctg acgaccctga 300
 aattggcaca aaagtggggg gtggagcaac ccacgttgta ttggcacgtc aaaaataagc 360
 gggcattgct ggaatgcctc gctattgaaa tgttggatcg ccaccatacc catttctgtc 420
 cactggaggg cgagtccttg caggactttc tccgcaacaa cgcgaaatcc ttctcgtgtg 480
 cactcttgct ccacggggac ggtgctaagg tgcacttggg caccgtccc accgaaaaac 540
 aatacgaaac cttggaaaat caattggcgt ttttgtgcca gcaagggttt agcttggaga 600
 atgctctcta tgcgtctctg gctgtcgggc actttacgtt ggggtgctg ttggaggacc 660
 aggagcatca agtcgcaaaa gaggagcgtg aaaccccaac cacggactcg atgccacctc 720
 tgcctcgcca agctatcgaa ctcttcgata atcaggcgcg ggagccagcc ttctctttg 780
 ggctggagct gattatctgc ggtttgga aaacactcaa gtgtgaaagc gggctcctaac 840
 tgcagtcact gcccgcttcc cagtcgggaa acctgtcgtg ccagctgcat taatgaatcg 900
 gccaacgcgc ggggagaggg ggtttgcgta ttgggcgcca ggggtggttt tcttttcacc 960
 agtgagacgg gcaacagctg attgcccttc accgcctggc cctgagagag ttgcagcaag 1020

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cgggtccacgc tgggtttgccc cagcaggcga aaatcctgtt tgatggtggt taacggcggg 1080
atataacatg agctgtcttc ggtatcgctc tatcccaacta ccgagatata cgcaccaacg 1140
cgcagcccgag actcggtaat ggcgcgcatt gcgccacagc ccatctgatc gttggcaacc 1200
agcatcgagc tgggaacgat gccctcattc agcatttgca tgggttggtg aaaaccggac 1260
atggcactcc agtcgccttc ccgttcogct atcggctgaa tttgattgag agtgagatat 1320
ttatgccagc cagccagacg cagacgcgcc gagacagaac ttaatgggcc cgctaacagc 1380
gcgatttgct ggtgacccaa tgcgaccaga tgetccacgc ccagtgcgct accgtcttca 1440
tgggagaaaa taactactgt gatgggtgtc tggtcagaga catcaagaaa taacgcggga 1500
acattagtag aggagcttc cacagcaatg gcatcctggt catccagcgg atagttaatg 1560
atcagcccaac tgacgcgttg cgcgagaaga ttgtgcaccg ccgctttaca ggcttcgacg 1620
ccgcttcgtt ctaccatcga caccaccacg ctggcaccga gttgatcggc gcgagattta 1680
atcgcccgca caatttgca cggcgcgctg agggccagac tggaggtggc aacgccaatc 1740
agcaacgact gtttgcccg cagttgttgt gccacgcggt tgggaatgta attcagctcc 1800
gcatcgccg ctccactttc ttcccgctt ttcgcagaaa cgtggctggc ctgggtcacc 1860
acgcgggaaa cggctctgata agagacaccg gcatactctg cgacatcgta taacgttact 1920
ggtttcacat tccaccacct gaattgactc tcttcgggac gctatcatgc cataccgcga 1980
aagggtttgc gcatttcgat ggtgtccggg atctcgacgc tctcccttat gcgactcctg 2040
cattaggaag cagcccagta gtagggttag gccgttgagc accgcgcggc caagggaatg 2100
tgcattgcaag gagatggcgc ccaacagtcg cccggccacg gggagtcata agcctccggt 2160
cggaggcttt tgacttctag agagctgttg acactttatg cttccggctc gtataatgtg 2220
tgtggaattg tgagcggata acaagtggaa ttgtgagcgg ataacaattt cacacaggaa 2280
acagaatccc atggctcgag gacgaacaat aaggcctccc taacgggggg ccttttttat 2340
tgataacaaa a                                     2351

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SEQ ID NO: 56      moltype = DNA length = 1625
FEATURE
source             Location/Qualifiers
                   1..1625
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 56
ggatccgagc tgttgacaac tctatcattg atagagttat aatgttccct atcagtata 60
gagacgcagt aagagaggaa tgtacatatg gtgagtaagg gtgaggagct cattaaggag 120
aacattgcaca tgaagctgta tatggagggc accgtaaaca accaccactt caagtgtacc 180
accgaggggtg aaggttaaacc ctacgagggg acgcagagccc aacgcataca ggtcgtggag 240
ggcggcccgcc tgcctttcgc attcgacatt ctggcgacct gttttatgta cggtcgaag 300
accttcacatc accacaccga aggcaccccg gacttcttca agcagagctt ccctgagggc 360
ttcacctggg agcgcgtcac cacgtatgaa gacggtgggg tgcacaccgt gaccaggac 420
acgagcttgc aggatggctg cttgatttac aacgtcaagc tgcgcggggg gaacttccct 480
agcaacggggc cagtgatgca gaaaaagacg ctgggttggtg agggccaccac cgagaccctg 540
taccggccgc acggggggcg ggaaggggcg tgcgatatgg cctgaaattt ggtcggcggc 600
ggtcatttgc actgcaatct caagaccacg taccgctcca agaaaaccgc caaaaacctg 660
aagatgcctg gtgtttatct tgcgaccggc cgcctggagc gcatacaagg agcggacaat 720
gagacgtacg tggaaacagca cgaagtggcc gtggctcgtt attgcgatct gccgtcgaag 780
ctgggtcaca aactgaacgg catggatgag ctgtacaaag attataagga tgatgacgac 840
aagtaaaagc ttctcggtag caaattccag aaaagaggcc tcccgaaagg ggggcctttt 900
ttcgttttgg tccgaattct tgacagctag ctacgtccta ggtataatgc tagccgcagt 960
aagagaggaa tgtacacatg tccgcctcgg ataaatcgaa agtgattaac tcggccctcg 1020
aattgctgaa tgaagtcggt atcgaggggc tgacgaccgc taaattggca caaaagtgtg 1080
gggtggagca acccacgttg tattggcacg tcaaaaataa gcgggcattg ctggatgcc 1140
tcgtatttga aatgttggat cgcaccataa cccatttctg tccactggag ggcgagtcct 1200
ggcaggactt tctccgaac aacgcgaaat ccttctcgtg tgcactcttg tcccatcggg 1260
acggtgctaa ggtgcctcag ggcacccgtc ccaccgaaaa acaatacgaa accttggaaa 1320
atcaattggc gtttttctgc cagcaagggt ttagcttggg gaatgctctc tatgcgctct 1380
cggctgtcgg tcactttacg ttgggtgctg tgttgaggga ccaggagcat caagtgcgaa 1440
aagaggagcg tgaaccccca accacggact cgatgccacc tctgctccgc caagctatcg 1500
aactcttcca tcatcagggc gcggagccag ccttctctct tgggctggag ctgattatct 1560
gcggtttgga aaaacaactc aagtgtgaaa gcgggtccta actgcagctc agaccatggc 1620
tcgag

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SEQ ID NO: 57      moltype = DNA length = 2234
FEATURE
source             Location/Qualifiers
                   1..2234
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 57
ggatccgagc tgttgacact ttatgcttcc ggctcgtata atgtgtgtgg aattgtgagc 60
ggataacaac gcagtaagag aggaatgtac atatggtgag taaggggtgag gagctcatta 120
aggagaacat gcacatgaag ctgtatatgg agggcaccgt aaacaaccac cacttcaagt 180
gtaccaccga gggtagaagt aaaccctacg aggggacgca gaccacaacg atcaaggctc 240
tggagggcgcc cccgctgctt ttgcgattcg acattctggc gacctgtttt atgtacggct 300
cgaagacctt catcaaccac acccaaggca tcccggaact cttcaagcag agcttccctg 360
agggcttcac ctgggagcgc gtcaccacgt atgaagacgg tgggggtgctc accgtgaccc 420
aggacacgag cttgcaggat ggtctgttga tttacaacgt caagctgcgc ggggtgaact 480
tccctagcaa gggccagtg atgcagaaaa agacgctggg tgggagggac accaccgaga 540
ccctgtaccc ggcgacggg gggctggaag ggcggtgcga tatggccctg aaattggctg 600
gcggcggtca tttgcactgc aatctcaaga ccacgtaccg ctccaagaaa cccgccaaaa 660
acctgaagat gctgtgtgtt tattttgtcg accggcgctt ggagcgcata aaggaagcgg 720
acaatgagac gtaactggaa cagcacgaag tggcgtggc tcgttattgc gatctcgct 780
cgaagctggg tcacaaactg aacggcatgg atgagctgta caaagattat aaggatgatg 840

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acgacaagta aaagctttct ggtaccaaat tccagaaaag aggcctcccc aaaggggggc 900
cttttttcgt tttggtcoga attcccccggt ggccggggga ctggtgggcg ccatctcctt 960
gcatgcacca ttcccttcggt cggcggtgct caacggccctc aacctactac tgggctgctt 1020
cctaattgcag gagtcgcata agggagagcg tcgagatccc ggacaccatc gaatggcgca 1080
aaacctttcg cggtatggca tgatagcgcc cggaagagag tcaattcagg gtggtgaatg 1140
tgaaaccagt aacgtttatc gatgtcgcag agtatgcccg tgtctcttat cagaccgttt 1200
cccgctgggt gaaccaggcc agccacgttt ctgcgaaaac gcgggaaaaa gtggaagcgg 1260
cgatggcgga gctgaattac attcccaacc gcgtggcgca acaactggcg ggcaaacagt 1320
cgttgctgat tggcgttgcc acctccagtc tggccctgca cgcgcgctcg caaattgtcg 1380
cggcgattaa atctcgcgcc gatcaactgg gtgccagcgt ggtggtgtcg atggtagaac 1440
gaagcggcgt cgaagcctgt aaagcggcgg tgcacaatct tctcgcgcaa cgcgtcagtg 1500
ggctgatcat taactatccg ctggatgacc aggatgccat tgcgtgtgaa gctgcctgca 1560
ctaagtgtcc ggcggttatt ctgtatgtct ctgaccagac acctcatcac agtattattt 1620
tctcccatga agcggtagc cgactgggcg tggagcatct ggtcgatttg ggtcaccagc 1680
aaatcgcgct gttagcgggc ccattaagtt ctgtctcggc gcgtctcgct ctggtcggtt 1740
ggcataaata tctcactcgc aatcaaatc agccgatagc ggaacgggaa ggcgactgga 1800
gtgccatgtc cgtgtttcaa caaacatgc aaatgctgaa tgagggcac gttccactg 1860
cgatgctggt tgcacaacgat cagatggcgc tgggcgcaat gcgcgcatt accgagtcgg 1920
ggctgcgctg tgggtcggat atctcggtag tgggatacga cgataccgaa gacagctcat 1980
gttatatccc gccgttaacc accatcaaac aggatatttc cctgctgggg caaacccagc 2040
tggaccgctt gctgcaactc tctcagggcc aggcggtgaa gggcaatcag ctgttgcccc 2100
tctcactggt gaaaagaaaa accaccctgg cgcccaatc gcaaacgcgc tctccccgcg 2160
cgttggcgga ttcattaatg cagctggcac gacagggttc ccgactggaa agcgggcagt 2220
gactgcagct cgag                                     2234

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SEQ ID NO: 58          moltype =   length =
SEQUENCE: 58
000

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SEQ ID NO: 59          moltype =   length =
SEQUENCE: 59
000

```

```

SEQ ID NO: 60          moltype = DNA length = 2257
FEATURE
source                 Location/Qualifiers
                        1..2257
                        mol_type = other DNA
                        organism = synthetic construct

```

```

SEQUENCE: 60
ggatccgagc tgttgacact ttatgcttcc ggctcgata atgtgtgtgg aattgtgagc 60
ggataacaag tgggaattgt agcggataac aatttcacac aggaaacaga atcatatggt 120
gagtaagggg gaggagctca ttaaggagaa catgcacatg aagctgtata tggaggggac 180
cgtaaacaaac caccacttca agtgtaccac cgagggtgaa ggtaaacctc acgaggggac 240
gcagacccaa cgcatacaag tcgtggaggg cggcccgcgt cctttcgcat tcgacattct 300
ggcgacctgt tttatgtacg gtcogaagac cttcatcaac cacacccaag gcatcccgga 360
cttcttcaag cagagcttcc ctgagggcct caccctggag cgcgtcacca cgtatgaaga 420
cgggtggggtg ctcacccgtga ccacggacac gagcttgtag gatggctgct tgatttacaa 480
cgtcaagctg cgcgggggtga acttccttag caacggggcca gtgatgcaga aaaagacgct 540
gggttgggag gccaccaccg agacctgtga cccggccgac ggggggctgg aagggcggtg 600
cgatatggcc ctgaatttgg tcggcgggcg tcatttgcac tgcaatctca agaccacgta 660
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FEATURE                Location/Qualifiers
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                        mol_type = other DNA
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 mol_type = other DNA
 organism = synthetic construct

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source             1..1749
                   mol_type = other DNA
                   organism = synthetic construct

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cgcaaccctc acgacaaatt cgacgtgttc gcttggagta cgcagagcac gaactggctg 180
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gaaggtgagt tegtccaca tgttgatgac gaggcggatg cggaagacac gctgagctgg 300
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SEQ ID NO: 75      moltype = AA length = 582
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source             Location/Qualifiers
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                   mol_type = protein
                   organism = synthetic construct

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VTQWQAASVG VGGKATAPAS MASADLYRAP WYGPAGGALS EALLGWSALI GTGLITSRSD 180
ARPEDAADFV QLAAILNDVA GAASVTPLAE QPLLGLRLIFW VIDQVVDHPD NDESWQSISL 240
FERLGLLATP ALITAGWYDG FVGESLRTFV AVKDNADARL VVGPPWSHNL TGRNADRKFQ 300
IAATYPIQEA TTMHKAFFDR HLRGETDALA GVPKVRLEFM GIDEWRDETQ WPLPDTAYTP 360
FYLGGSGAAN TSTGGTLST SSGTESADT YLYDPADPVP SLGGTLLFHN GDNGPADQRP 420
IHDRDDVLCY STEVLTDPVE VTGTVSARLF VSSSAVDTFD TAKLVDFVFPD GRAIALCDGI 480
VRMRYYRETLV NPTLIEAGEI YEVAIDMLAT SNVFLPGHRI MVQVSSNFP KYDRNSNTGG 540
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SEQ ID NO: 76      moltype = DNA length = 936
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source             Location/Qualifiers
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SEQ ID NO: 77      moltype = DNA length = 930
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source             Location/Qualifiers
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gcctgggtgc gccattgtga cgtatgtatt gacgatcaga cgctgggctt tcaggcccg 180
cagcctgcct tacaacggcc cgaacaacgt ctgatgcaac ttgagatgaa aacgcgccag 240
gcctatgcag gatcgagat gcacgaacgg cgtgttgccg cttttcagga agtggtatg 300
gctcatgata tcgccccggc ttacgcgttt gatcatctgg aagccttcgc catggatgta 360

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SEQ ID NO: 78      moltype = DNA length = 1479
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source             1..1479
                   mol_type = other DNA
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SEQ ID NO: 79      moltype = DNA length = 1161
FEATURE            Location/Qualifiers
source             1..1161
                   mol_type = other DNA
                   organism = synthetic construct

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attatgacga ctcacgtgtg a 1161

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What is claimed is:

1. A nucleic acid construct comprising a modified inducible promoter, wherein the modified inducible promoter comprises:

- a. a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter;
- b. a TTGACA sequence at position -35, relative to the transcriptional start site of the promoter; and
- c. a nucleic acid sequence that encodes a bacterial ribosome binding sequence.

2. The nucleic acid construct of claim 1, wherein the nucleic acid construct further comprises a transgene.

3. The nucleic acid construct of claim 2, wherein the transgene is selected from the group consisting of: a crtW gene, a crtE gene, a crtY gene, a crtI gene, a crtZ gene, a crtEB gene, a crtEBI gene, a crtEBIY gene, a crtEBIYZ gene, a crtEBI-YZW gene, an ABA1 gene, an ABA2 gene, and a CocE gene.

4. The nucleic acid construct of claim 1, wherein the nucleic acid construct further comprises a sequence encoding a reporter gene.

5. The nucleic acid construct of claim 1, wherein the nucleic acid construct further comprises a regulatory element.

6. The nucleic acid construct of claim 1, wherein the nucleic acid construct further comprises a terminator sequence.

7. The nucleic acid construct of claim 1, wherein the nucleic acid construct further comprises:

- a transgene; and
- a second modified inducible promoter sequence, wherein the second modified inducible promoter sequence comprises:

- (i) a TATAATGT sequence at position -10, relative to a transcriptional start site of the second modified inducible promoter sequence;
- (ii) a TTGACA sequence at position -35, relative to the transcriptional start site of the second modified inducible promoter sequence; and

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(iii) a nucleic acid sequence that encodes a second bacterial ribosome binding sequence.

8. The nucleic acid construct of claim 1, wherein promoter nucleic acid sequence integrity downstream of the TATAATGT sequence at position -10 is maintained.

9. The nucleic acid construct of claim 7, wherein promoter nucleic acid sequence integrity downstream of the TATAATGT sequence at position -10 is maintained.

10. A composition comprising the nucleic acid construct of claim 1; and an inducer.

11. A kit comprising the nucleic acid construct of claim 1, packaging, buffers, and, optionally, instructions for use.

12. A method of expressing a protein encoded by a transgene in a cell, the method comprising:

- a. transforming a cell with a nucleic acid construct, wherein the nucleic acid construct comprises a modified inducible promoter that comprises:
 - i. a TATAATGT sequence at position -10, relative to a transcriptional start site of the promoter;
 - ii. a TTGACA sequence at -35 position, relative to the transcriptional start site of the promoter;
 - iii. a nucleic acid that encodes a bacterial ribosome binding sequence; and
 - iv. a transgene; and
- b. contacting the cell with an inducer, thereby expressing the protein encoded by the transgene.

13. The method of claim 12, wherein the cell is a prokaryotic cell.

14. The method of claim 13, wherein the prokaryotic cell is a bacterium.

15. The method of claim 12, wherein the cell does not comprise a T7 promoter or a T7 polymerase.

16. The method of claim 12, wherein, when contacting the cell with the inducer, the protein expression is greater than the protein expression in a comparable cell that comprises the transgene operably linked to a T7 promoter.

* * * * *